Informatica® Intelligent Data Management Cloud

October 2026

Runtime Environments

Informatica Intelligent Data Management Cloud Runtime Environments
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CHAPTER 1

Runtime environments

A runtime environment is the execution platform that runs Intelligent Data Management Cloud (IDMC) assets such as tasks and taskflows. You must have at least one runtime environment in each organization so that users in the organization can run tasks.

A runtime environment consists of one or more Secure Agents. A Secure Agent is a lightweight program that runs all tasks and enables secure communication across the firewall between your organization and IDMC.

You can set up runtime environments in the following ways:

Use the Informatica Cloud Hosted Agent.

When you use the Hosted Agent, you run tasks within the Informatica Cloud hosting facility. Informatica maintains the Hosted Agent runtime environment and agents. For more information about the Informatica Cloud Hosted Agent, see Chapter 2, “Hosted Agent” on page 9.

Create one or more Secure Agent groups.

You can install one or more Secure Agents to run within your network or in a cloud computing services environment such as AWS, Google Cloud, Microsoft Azure, or Oracle Cloud Infrastructure. You can install one Secure Agent on each physical or virtual machine.

When you install a Secure Agent, it is added to its own group by default. You can add multiple agents to one Secure Agent group. For more information about Secure Agent groups, see Chapter 3, “Secure Agent groups” on page 11.

Create an elastic runtime environment

An elastic runtime environment uses elastic scaling to provide a runtime environment option that optimizes resource management. Use an elastic runtime environment to handle fluctuating workload demands and manage workload costs. You can deploy an elastic runtime environment on AWS.

For more information about elastic runtime environments, see Chapter 4, “Elastic runtime environments” on page 31.

Configure a serverless runtime environment.

A serverless runtime environment is an advanced serverless deployment solution that doesn't require downloading, installing, configuring, or maintaining a Secure Agent or Secure Agent group. You can deploy a serverless runtime environment on AWS or Microsoft Azure.

For more information about serverless runtime environments, see Chapter 5, “Serverless runtime environments” on page 65.

When you configure a connection or some types of tasks, you specify the runtime environment to use. The runtime environment determines which agent runs the tasks at run time. If the runtime environment is the Hosted Agent, the Hosted Agent runs the tasks. If the runtime environment is a Secure Agent group, any available agent in the group can run the tasks.

To run a mapping in advanced mode, the agent can create a default local cluster on the agent machine so that you can begin developing and running advanced functionality on small data sets to test mapping logic. For more information, see *Advanced Clusters*.

Before you run mappings in advanced mode on a local cluster, make sure that the Secure Agent has enough resources so that it can create a cluster and run jobs successfully, especially if the Secure Agent is already running other jobs. If the Secure Agent doesn't have enough resources, the jobs that are already running on the Secure Agent and the mappings in advanced mode will fail. It's recommended to have at least 8 cores and 32 GB of memory on the Secure Agent machine.

CHAPTER 2

Hosted Agent

The Hosted Agent can run synchronization, mapping, and replication tasks that use certain connectors.

IDMC manages the Hosted Agent runtime environment, so you cannot add, delete, or configure a Hosted Agent.

The Hosted Agent can run synchronization, mapping, and replication tasks that use the following connectors:

- Amazon Athena Connector
- Amazon Aurora Connector
- Amazon Redshift Connector
- Amazon Redshift V2 Connector
- Amazon S3 Connector
- Amazon S3 V2 Connector
- Box Connector
- Box Oauth Connector
- Cloud Integration Hub
- Concur V2 Connector
- Coupa Connector
- Coupa V2 Connector
- Cvent Connector
- Databricks Delta Connector
- DB2 Warehouse on Cloud Connector
- Eloqua Bulk API Connector
- Google Analytics Connector
- Google Big Query Connector
- Google Big Query V2 Connector
- Google Cloud Storage Connector
- Google Cloud Storage V2 Connector
- JIRA Connector
- Marketo V3 Connector
- Microsoft Azure Blob Storage V2 Connector
- Microsoft Azure Blob Storage V3 Connector
- Microsoft Azure Cosmos DB SQL API Connector

- Microsoft Azure Data Lake Storage Gen1 V2 Connector
- Microsoft Azure Data Lake Storage Gen1 V3 Connector
- Microsoft Azure Data Lake Storage Gen2 Connector
- Microsoft Azure SQL Data Warehouse V2 Connector
- Microsoft Azure SQL Data Warehouse V3 Connector
- Microsoft Azure Synapse SQL Connector
- Microsoft CDM Folders V2 Connector
- Microsoft Dynamics 365 for Operations Connector
- Microsoft Dynamics 365 for Sales Connector
- Microsoft Fabric Data Warehouse Connector
- Microsoft Fabric Lakehouse Connector
- Microsoft Fabric OneLake Connector
- Microsoft SQL Server Connector
- MySQL Connector
- NetSuite Connector
- OData Connector
- Oracle Connector
- PostgreSQL Connector
- REST V2 Connector
- Salesforce Connector
- Salesforce Marketing Cloud Connector
- Salesforce Oauth Connector
- ServiceNow Connector
- Snowflake Cloud Data Warehouse V2 Connector
- SuccessFactors ODATA Connector
- UltiPro Connector
- Workday V2 Connector
- Xactly Connector
- Zendesk V2 Connector
- Zuora AQuA Connector

Note: The Hosted Agent support is specific to connectors. For more information, see the help for the relevant connector.

CHAPTER 3

Secure Agent groups

Use a Secure Agent group as the runtime environment when you need to access data on-premises or when you want to access data in a cloud computing services environment without using the Hosted Agent. When you select a Secure Agent group as the runtime environment for a connection or task, a Secure Agent within the group runs the tasks.

Create Secure Agent groups to accomplish the following goals:

Prevent the activities of one department from affecting another department.

To prevent the activities of one department from impacting a different department, create separate Secure Agent groups for each department. For example, users in the sales department run 10 times as many tasks as users in the finance department, but the finance tasks are more time critical. To prevent the sales tasks from impacting the finance tasks, create separate Secure Agent groups for each department. Then assign the sales tasks to one runtime environment and the finance tasks to the other runtime environment.

Separate tasks by environment.

You can create different Secure Agent groups for test and production environments. When you configure a connection, you can associate it with the test or production database by choosing the appropriate Secure Agent group as the runtime environment.

When you create a Secure Agent group, all users in the organization can select the Secure Agent group as the runtime environment.

You can perform the following actions on a Secure Agent group:

- You can add and remove Secure Agents from a group.
- You can add multiple agents to a Secure Agent group.

  Note: If you use the runtime environment to run a mapping task that is based on a mapping in advanced mode, the Secure Agent group must have only one Secure Agent.
- You can share a Secure Agent group with your sub-organizations.

If you need to access output files on the Secure Agent machine, you can view the **All Jobs** page in Monitor or the **My Jobs** page in Data Integration that's associated with the job's organization to determine where a task ran.

Secure Agent groups with multiple agents

When you create a Secure Agent, it is added to its own group by default. You can add multiple agents to one Secure Agent group. All agents within a group must be of the same type, for example, all agents that run within your network or all agents that run on Amazon EC2 machines.

Add multiple agents to a group to achieve the following goals:

Balance the workload across machines.

Add multiple agents to a group to balance the distribution of tasks across machines. When the runtime environment is a Secure Agent group with multiple agents, the group dispatches a task or a background process such as metadata call to an agent with the fewest number of tasks running or queued.

Improve scalability for connections and tasks.

When you create a connection or task, you select the runtime environment to use. If the runtime environment is a Secure Agent group with multiple agents, the tasks can run if any Secure Agent in the group is up and running. You do not need to change connection or task properties when you add or remove an agent or if an agent in the group stops running.

When you add multiple agents to a group, ensure that all of the Secure Agents are of the same type. For example, your organization installs four Secure Agents on physical machines within your network and two Secure Agents on Amazon EC2 machines. You can create a Secure Agent group that contains some or all of the local agents and a different group that contains the EC2 agents. Do not create a group that contains both a local agent and an EC2 agent.

If you need to access output files on the Secure Agent machine, you can view the job details to determine which Secure Agent ran the task. To view job details, open Monitor, select **All Jobs**, and click the job name.

Service and connector assignment for Secure Agent groups

Your organization can enable and disable specific IDMC and connectors that your organization is licensed to use for a Secure Agent group.

You can perform the following actions:

Enable or disable IDMC for a Secure Agent group.

You can enable services such as Data Integration or Application Integration when you want the agents in the group to run the tasks, processes, and product features associated with the service. By default, when you create a Secure Agent group, all services and connectors that your organization is licensed to use are disabled. When you enable a service, the service starts on each agent in the Secure Agent group.

Some services require other services or connectors. If you enable a service that requires other services, IDMC automatically enables the required services. For example, you enable Data Quality on a Secure Agent group. Data Quality requires Data Integration. When you enable Data Quality, IDMC automatically enables Data Integration as well.

Disable services when you do not want the agents in the group to run the tasks, processes, or product features associated with the service. When you disable a service, the service stops on each agent in the Secure Agent group. Any task, process, or product feature that uses the Secure Agent group as the runtime environment no longer runs.

When you enable a service, you also enable the required Secure Agent services. For more information about the Secure Agent services that each service requires, see Secure Agent services.

Enable or disable connectors for a Secure Agent group.

You can enable specific connectors when you want the agents in the group to be able to communicate with cloud and on-premise applications, platforms, databases and flat files. When you enable a connector, all agents in the group download the packages associated with the connector.

Disable connectors when you do not want the agents in the group to download the packages associated with the connectors. When you disable a connector, any connection that uses the Secure Agent group as the runtime environment no longer runs.

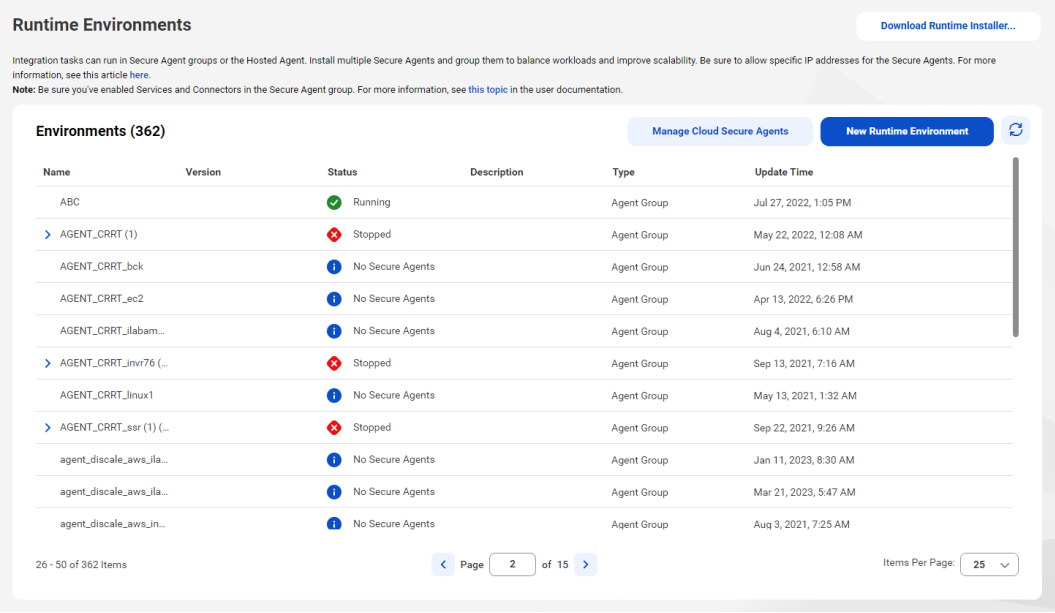
Enable or disable additional services for a Secure Agent group.

You can enable or disable additional services such as Self-Hosted Git Repo or EDC Integration.

You can also enable or disable Secure Agent services for individual Secure Agents within the Secure Agent group. For more information, see "Stopping and starting services on a Secure Agent" on page 116.

After you make service assignments for a Secure Agent group, you might add or remove agents. When you add a Secure Agent to a group, the agent inherits the service assignments of the group that you add it to.

Enable or disable services and connectors for a Secure Agent group on the **Runtime Environments** page:



Service assignment guidelines

Use the following guidelines when you enable and disable services for a Secure Agent group:

- Before you disable a service, verify that no connection, task, or process that uses the group as the runtime environment requires the service.
- Before you disable a service, verify that no feature that uses the group as the runtime environment requires the service.
  If a feature has a Secure Agent group selected as the runtime environment and you disable a required service, the feature cannot be used. For example, the runtime environment for Enterprise Data Catalog integration is set to RuntimeEnv2. If you disable EDC Search Service on RuntimeEnv2, you can no longer perform data catalog discovery.
- When you create a connection, select a runtime environment in which the required services are enabled. For example, you want to create an Advanced SFTP connection for a file ingestion and replication task target. When you create the connection, select a runtime environment in which the Mass Ingestion service is enabled.
- If you configure your organization to store connection properties locally, the Data Integration service must be enabled for the Secure Agent group.

Enabling or disabling services and connectors for a Secure Agent group

You can enable or disable IDMC and connectors for a Secure Agent group. By default, all services and connectors are disabled in newly-created Secure Agent groups. Enable the services and connections you want to run on the group.

1. In Administrator, select **Runtime Environments**.
2. Expand the Actions menu for the Secure Agent group and select **Enable or Disable Services and Connectors**.

   A dialog box listing all the services and connectors for the Secure Agent group is displayed.
3. On the **Services** tab, choose the IDMC services to enable or disable.

   Certain services may be listed in the **Additional Services** tab.
4. On the **Connectors** tab, choose the connectors to enable or disable.
5. On the **Additional Services** tab, select the Secure Agent services to enable or disable.

   For example, if your organization uses source control, and you want to disable it on your Secure Agent group, disable the GitRepoConnectApp service.

   The services that appear in this list vary based on your licenses. You might not see any services listed here.
6. Click **OK**.

   The changes affect every Secure Agent in the group.

Service and connector assignment example

Your organization uses Data Integration and has licenses for mass ingestion and for Enterprise Data Catalog data discovery.

The organization uses the following Secure Agent groups:

- Group 1: Secure Agent 1, Secure Agent 2 , Secure Agent 3
- Group 2: Secure Agent 4
- Group 3: Secure Agent 5

By default, users in your organization can select any group as the runtime environment for any connection or any task, including file ingestion tasks. An administrator can also select any group as the runtime environment for integration with Enterprise Data Catalog.

To balance the load across Secure Agent groups, you want might want to reserve Group 1 for Data Integration tasks except file ingestion tasks, Group 2 for file ingestion tasks, and Group 3 for data catalog discovery.

Therefore, you enable and disable the following Secure Agent services:

Secure Agent Group	Enabled Services	Disabled Services
Group 1	Data Integration Server	Mass Ingestion, EDC Search Agent
Group 2	Mass Ingestion	Data Integration Server, EDC Search Agent
Group 3	EDC Search Agent	Data Integration Server, Mass Ingestion

To avoid task and feature failures, you must also verify the following settings:

- All Data Integration tasks except file ingestion tasks use Group 1 as the runtime environment. All connections that these tasks use also use Group 1 as the runtime environment.
- All file ingestion tasks use Group 2 as the runtime environment. All connections that these tasks use also use Group 2 as the runtime environment.
- On the **Organization** page in Administrator, the Enterprise Data Catalog integration properties use Group 3 as the runtime environment.

Shared Secure Agent groups

If you are the administrator of a parent organization, you can share a Secure Agent group with sub-organizations. When you share a Secure Agent group, all of the sub-organizations can run data integration or ingestion and replication jobs on the Secure Agents within the group.

Note: Share a Secure Agent group when all agents in the group run only one of the following services: the Data Integration Server service (for Data Integration jobs), Mass Ingestion service (for file ingestion and replication jobs), or Database Ingestion agent service (for application ingestion and replication jobs and database ingestion and replication jobs). For other agent services, you cannot run jobs on a shared Secure Agent group.

Share a Secure Agent group to optimize the use of available Secure Agent resources. For example, your organization contains separate sub-organizations for departments in different time zones. Each sub-organization runs data integration tasks at different times of the day. If you create one Secure Agent group for each sub-organization, some Secure Agent groups might be used heavily at certain times of the day while others remain idle. To distribute the tasks more evenly, add the Secure Agents to a Secure Agent group, and share the Secure Agent group with the sub-organizations.

When you share a Secure Agent group, the group appears on the **Runtime Environments** page in all sub-organizations. The sub-organization administrators cannot view the Secure Agents within the group. They cannot perform management tasks on the group such as adding or deleting Secure Agents, renaming, deleting, or unsharing the group, or changing the group permissions.

When a user in the sub-organization creates a connection or task, the user can select the shared Secure Agent group as the runtime environment.

Flat file connections in shared Secure Agent groups

If a shared Secure Agent group contains multiple Secure Agents and the group is used as the runtime environment for a flat file connection, the directory used in the connection must be accessible by all Secure Agents in the group.

If the directory is not accessible by all Secure Agents, tasks that use the connection fail if they are assigned to a Secure Agent that cannot access the directory.

Working with Secure Agent groups

Create Secure Agent groups on the **Runtime Environments** page. After you create a Secure Agent group, you can rename or delete the group, add and remove Secure Agents, and change group permissions. You can also enable services and connectors for the group.

Tip: Click the refresh icon next to **New Runtime Environment** to refresh the page before performing any actions on Secure Agent groups.

You can complete the following tasks:

Create a Secure Agent group.

To create a Secure Agent group, click **New Runtime Environment** and enter a name and optionally a description for the group. After you create a group, you can add Secure Agents to the group.

Note: If you use multi-byte characters in the Secure Agent group name and you create the group in a cloud-hosted environment, verify that the environment also supports these characters.

Edit Secure Agent group properties.

To rename a Secure Agent group or to add or update the description, expand the Actions menu, select **Edit Environment Properties**, and complete the fields in the dialog box. IDMC updates the group name in all services that use the group.

Enable or disable specific IDMC and connectors for a Secure Agent group.

To enable or disable services for a Secure Agent group, expand the Actions menu and select **Enable or Disable Services, Connectors**. On the **Services** tab, select the services to enable or disable. You can enable or disable any service that your organization uses.

Note: Before you disable a service, verify that no connection, task, or process that uses the group as the runtime environment requires the service. If a connection, task, or process has a Secure Agent group selected as the runtime environment and you disable a required service, the task or process cannot run. Similarly, if a feature has a Secure Agent group selected as the runtime environment and you disable a required service, the feature cannot be used.

To enable or disable connectors, expand the Actions menu and select **Enable or Disable Services, Connectors**. On the **Connectors** tab, select the connectors to enable or disable. You can enable or disable any connector that you organization is licensed to use.

To enable or disable additional services such as Self-Hosted Git Repo, expand the Actions menu and select **Enable and Disable Services and Connectors.** On the **Additional Services** tab, select the services to enable or disable. You can enable or disable any service that your organization uses.

Add Secure Agents to a group.

To add Secure Agents to a group, expand the Actions menu and select **Add or Remove Secure Agents**. You can add any agent that is in the Unassigned Agents group on the **Runtime Environments** page.

Alternatively, you can add a new Secure Agent to an existing group by setting the InfaAgent.GroupName property in the infaagent.ini file before you register the agent. When you add a Secure Agent to a Secure Agent group, the Secure Agent inherits the services and connectors that are configured for the Secure Agent groups.

Note: For Application Ingestion and Replication and Database Ingestion and Replication, if you remove a Secure Agent from a Secure Agent group and add it to another Secure Agent group, wait at least 60 minutes before trying to use it to run application or database ingestion and replication jobs. The internal cache of Secure Agent runtime information is refreshed every hour by default.

When you add more than one Secure Agent to a Secure Agent group, all agents must meet the following requirements:

- All of the agents must be of the same type, for example, all local agents or all agents that run on Amazon EC2 machines.
- Each Secure Agent must be configured to connect to the same external systems and have access to files such as libraries, initialization files, and JAR files.
- Each Secure Agent must have access to the files used in tasks. Ensure that all files used in a task are available in a shared location.

Remove Secure Agents from a group.

To remove Secure Agents from a group, expand the Actions menu and select **Add or Remove Secure Agents**. When you remove an agent from a group, IDMC adds it to a group named "Unassigned Agents."

You can remove an agent from a Secure Agent group if the group is not used as the runtime environment for a connection or task. If the group is used, you can remove an agent if it is not the only agent in the group.

Delete a Secure Agent group.

To delete Secure Agent group, expand the Actions menu and select **Delete**. You can delete a Secure Agent group if it does not contain any Secure Agents.

If the Secure Agent group is associated with an advanced configuration and the advanced cluster is running, you must stop the cluster and associate the configuration with a different runtime environment before you can delete the group.

Share or unshare a Secure Agent group.

If you are the administrator of a parent organization, you can share a Secure Agent group so that the sub-organizations can use it. You can unshare a group if it is not used in a connection or task. From the Actions menu associated with the group, choose **Share Secure Agent Group** or **Unshare Secure Agent Group**.

Change permissions for a Secure Agent group.

To change permissions for a Secure Agent group, expand the Actions menu and select **Permissions**. You can define permissions for a Secure Agent group for each user group in your organization.

You can set the following permissions:

Permission	Description
Read	View details about the Secure Agent group and use the Secure Agent group in a task.
Update	Edit the Secure Agent group.
Delete	Delete the Secure Agent group.
Change	Change permissions for the Secure Agent group.

Adding a Secure Agent to a group

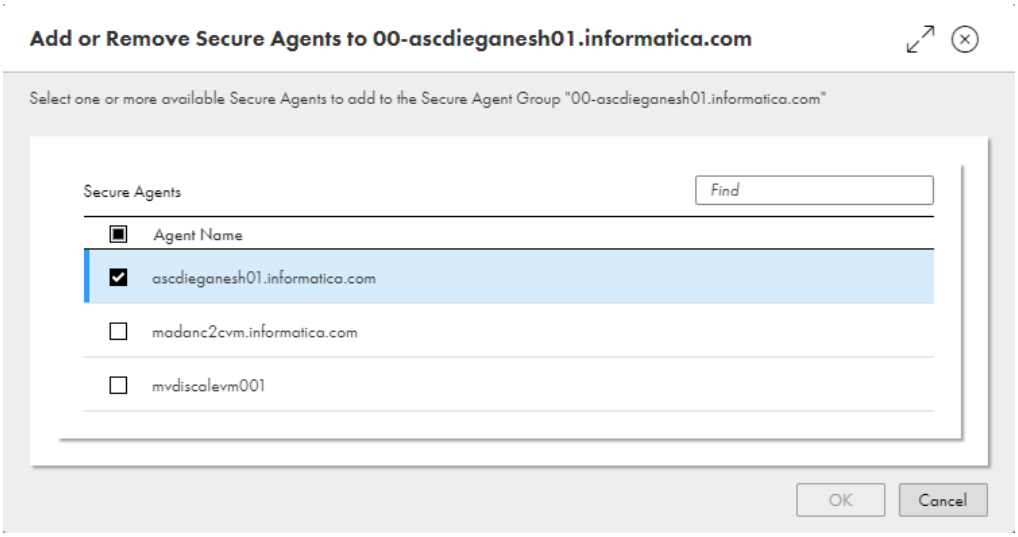
You can add any available Secure Agent to a Secure Agent group. Available agents appear in the "Unassigned Agents" group on the **Runtime Environments** page. You cannot add a Secure Agent to a group if the agent

has already been added to another group. When you add a Secure Agent to a group, the Secure Agent inherits all the services and connectors that are enabled for the group.

1. In Administrator, select **Runtime Environments**.
2. Expand the Actions menu for the Secure Agent group, and select **Add or Remove Secure Agents**.
3. In the **Secure Agents** list, enable the checkbox for the Secure Agents that you want to add to the group.

   If the agent you want isn't listed, that means it is currently assigned to another group. You must remove an agent from a group before you can add it to a different group.

   If there are many agents in the list, use the **Find** box to quickly locate an agent.



4. Click **OK.**

Note: For Application Ingestion and Replication and Database Ingestion and Replication, if you remove a Secure Agent from a Secure Agent group and add it to another Secure Agent group, wait at least 60 minutes before trying to use it to run application or database ingestion and replication jobs. The internal cache of Secure Agent runtime information is refreshed every hour by default.

Adding a new Secure Agent to an existing group

You can add a Secure Agent to an existing Secure Agent group when you install the agent. To add a Secure Agent to an existing group, add the InfaAgent.GroupName property to the infaagent.ini file before you register the agent. When you add a Secure Agent to a group, the Secure Agent inherits all the services and connectors that are enabled for the group.

1. Install the Secure Agent.
2. On Windows, when you are prompted to register the agent, open Windows Services and stop the agent.

   On Linux, when the installation program finishes, do not start the agent.
3. Open `<Secure Agent installation directory>/apps/agentcore/conf/infaagent.ini` in a text editor.
4. Add the following property and save the file:

   ```
   InfaAgent.GroupName=<Secure Agent group name>
   ```
5. Start the agent.

6. Register the agent.

   IDMC adds the Secure Agent to the group you specify in the InfaAgent.GroupName property instead of a new group.

Removing a Secure Agent from a group

You can remove an agent from a Secure Agent group if the group is not used in a connection or task. If the group is used in a connection or task, you can remove an agent if it is not the only agent in the group. When you remove a Secure Agent from a group, IDMC adds it to a group named "Unassigned Agents."

1. In Administrator, select **Runtime Environments**.
2. Expand the Actions menu for the Secure Agent group, and select **Add or Remove Secure Agents**.
3. In the list of **Secure Agents**, clear the check mark next to the agents that you want to remove from the group.
4. Click **OK.**

The Secure Agents that were removed appear in the "Unassigned Agents" group on the **Runtime Environments** page.

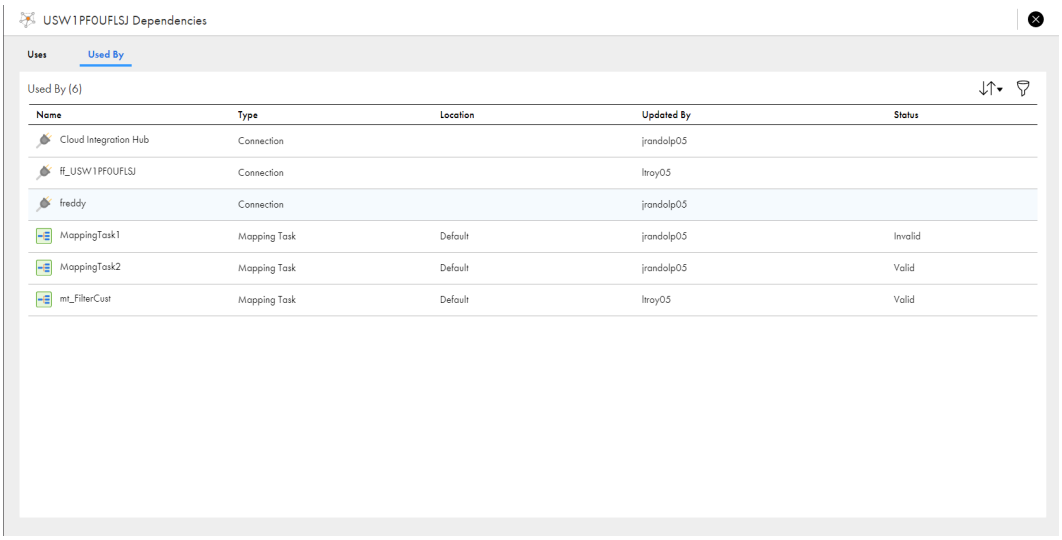
Viewing Secure Agent group dependencies

You can view object dependencies for Secure Agent groups.

When you view dependencies for a Secure Agent group, Administrator lists the connections and assets in each service that use the group as the runtime environment.

To view object dependencies for a Secure Agent Group, expand the Actions menu and select **Show Dependencies**.

The following image shows the **Dependencies** page for a Secure Agent group:



To sort the objects that appear on the page, click the sort icon and select the column name for the property you want to sort by.

To filter the objects that appear on the dependencies page, click the Filter icon. Use filters to find specific objects. To apply a filter, click **Add Field**, select the property to filter by, and then enter the property value. You can specify multiple filters. For example, to find connections with Oracle in the name, add the Type filter and specify Connection. Then add the Name filter and enter "Oracle."

Customer Managed runtime environments

A Customer Managed runtime environment is a Secure Agent group where the Secure Agents run with no dependency on a connection to IDMC.

Customer Managed runtime environments shield their Secure Agents from automatic updates and upgrades. No updates occur unless you specifically initiate them. Customer Managed Secure runtime environments are also ideal in environments with sporadic or unreliable internet connectivity.

Secure Agents in a Customer Managed environment run jobs using definitions that were active after the last synchronization with IDMC. When the job completes, the Secure Agents synchronize the job execution results and agent logs with IDMC, if a connection is available. If not, this synchronization defers until connectivity returns.

Note: A Customer Managed runtime environment should connect to IDMC at least once a month, to avoid billing issues and to install any critical Secure Agent updates.

Is a Customer Managed environment right for you?

While a Customer Managed runtime environment offers advantages such as better control over your Secure Agents and the ability to run without an internet connection, there are additional considerations to keep in mind:

- You need to plan for and maintain additional hardware infrastructure for the Customer Managed Secure Agents and for a PostgreSQL database.
- You are responsible for maintaining high availability and data replication/disaster recovery for the PostgreSQL database.

If you can't or don't want to maintain additional hardware, then a Customer Managed environment might not be the best fit for your organization.

Differences between Customer Managed environments and standard Secure Agent groups

Customer Managed runtime environments and standard Secure Agent groups are similar, with a few key differences.

Note the following key differences:

- Unlike Standard Secure Agent groups, Customer Managed runtime environments don't support all services, transformations, and task types For more information, see “Supported functionality in Customer Managed runtime environments” on page 21.
- You need to provide a PostgreSQL database for Customer Managed runtime environments to store metadata. In standard Secure Agent groups, the metadata is stored within IDMC.

  Important: For PostgreSQL 15 or later, add the following privileges for the DB user (but never to PUBLIC):

  - CONNECT and CREATE on the database
  - USAGE + CREATE on schema public

Next, run the grant as the database owner or a superuser:

```
GRANT USAGE, CREATE ON SCHEMA public TO <db_user>;
```

Verify that the results return `t` (true):

```
SELECT has_schema_privilege('<db_user>', 'public', 'CREATE');
```

PostgreSQL versions prior to 15 have these privileges by default, so these steps aren't necessary.

- In standard Secure Agent groups, the flow of information occurs seamlessly between the agents and IDMC. In a Customer Managed runtime environment, you need to synchronize all changes manually. For more information, see "Synchronize the runtime environment" on page 29.
- In standard Secure Agent groups, the Secure Agents automatically upgrade when a new version is available. In a Customer Managed runtime environment, you can manually check and install upgrades. You are also alerted to available upgrades when you start a synchronization. For more information, see "Working with Customer Managed runtime environments" on page 27.
- You can't use the IDMC UI to start, restart, or pause jobs on Secure Agents within a Customer Managed runtime environment. To perform these tasks, use the REST API for Customer Managed runtime environments. For more information, see *REST API*.

Supported functionality in Customer Managed runtime environments

Customer Managed runtime environments support only a subset of the functionality supported by standard Secure Agent groups.

The following table summarizes the supported services and functionality:

Supported Service	Unsupported functionality
Cloud Application Integration	
Cloud Data Ingestion & Replication	Streaming Ingestion and Replication
Cloud Data Integration	- Advanced Integration - Dynamic tasks - Linear taskflows - Taskflows that span more than one Customer Managed runtime environment (RTE) The taskflow and all of its steps that require a runtime environment must use the same Customer Managed RTE (or RTE group). - Access Policy transformation - Data Masking transformation - Data Services transformation
Cloud Data Integration Connectors	Connections created with OAuth authentication.
Cloud Data Integration for PowerCenter	
Cloud Data Quality	
Industry Solutions for Healthcare	

All services not listed in the above table are not supported in Customer Managed runtime environments.

Steps to set up a Customer Managed runtime environment

You perform a series of setup steps to get your Customer Managed runtime environment ready.

1. Contact Informatica Global Customer Support to license the Customer Managed feature for your organization.
2. Create a Customer Managed runtime environment. For more information, see "Create a Customer Managed runtime environment" on page 22.
3. Configure a PostgreSQL database for your Customer Managed runtime environment. For more information, see "Configure a database for a Customer Managed runtime environment" on page 23.
4. Install Secure Agents into the Customer Managed runtime environment. For more information, see "Install Secure Agents to a Customer Managed runtime environment" on page 23.
5. Push tasks, schedules, and dependencies to the Secure Agents using the **Synchronize Runtime Environment** action. For more information, see "Synchronize the runtime environment" on page 29.
6. Create a password to access the API for the local launch of on-demand jobs for a Customer Managed Secure Agent. For more information, see "Create a password to access the Customer Managed Secure Agent API" on page 27.

The Secure Agents are now fully autonomous and ready to operate in a Customer Managed environment.

Create a Customer Managed runtime environment

Create a Customer Managed runtime environment in Administrator.

Note: You can't convert a Customer Managed runtime environment to a standard Secure Agent group.

1. In Administrator, open the **Runtime Environments** page.
2. Select **New Runtime Environment**.
3. On the **General** page, enter a name for the runtime environment.
4. Select **Customer Managed Runtime Group** as the runtime environment type and optionally add a description.

   Note: If the option isn't visible, verify that the feature is enabled on the **Settings** page. For more information, see "Steps to set up a Customer Managed runtime environment" on page 22.
5. Click **Next** to configure the database.
6. Select **PostgreSQL** as the connection type.
7. Select any connection from the drop-down list to be used temporarily or click **Connections** to create a new one.

   Note: The temporary connection is required to create the Customer Managed runtime environment. After the environment is created, you can use it to configure the actual PostgreSQL connection. Once this connection is available, use the **Database Configuration** tab of the Customer Managed runtime environment to update the connection to the PostgreSQL connection.
8. Click **Next** and then enable the services and components for your runtime environment.
9. Optionally select any additional services to enable.

   Note: You can also enable services and connectors from the Actions menu after you create the Customer Managed runtime environment.
10. Click **Create**.

    The Customer Managed runtime environment appears on the Runtime Environments page.

Configure a database for a Customer Managed runtime environment

A Customer Managed runtime environment uses a database that you supply to maintain its metadata store.

The database stores job metadata, schedules, execution state, service discovery information.

Use the **Database Configuration** tab of a Customer Managed runtime environment's details page to enter or modify connection details.

Connection Details

The following table describes the connection details:

Property	Description
Connection Name	Name of the database connection.
Database Type	PostgreSQL
Host	Domain of the database host, for example, "db-prod.example.com".
Port	Port number for the database host.
Database Name	Name of the database.

Authentication

The following table describes the authentication properties:

Property	Description
Authentication Type	**Username/Password**
Username	User name
Password	Password

After you enter the connection details, click **Test Connection** to verify that the connection is successful.

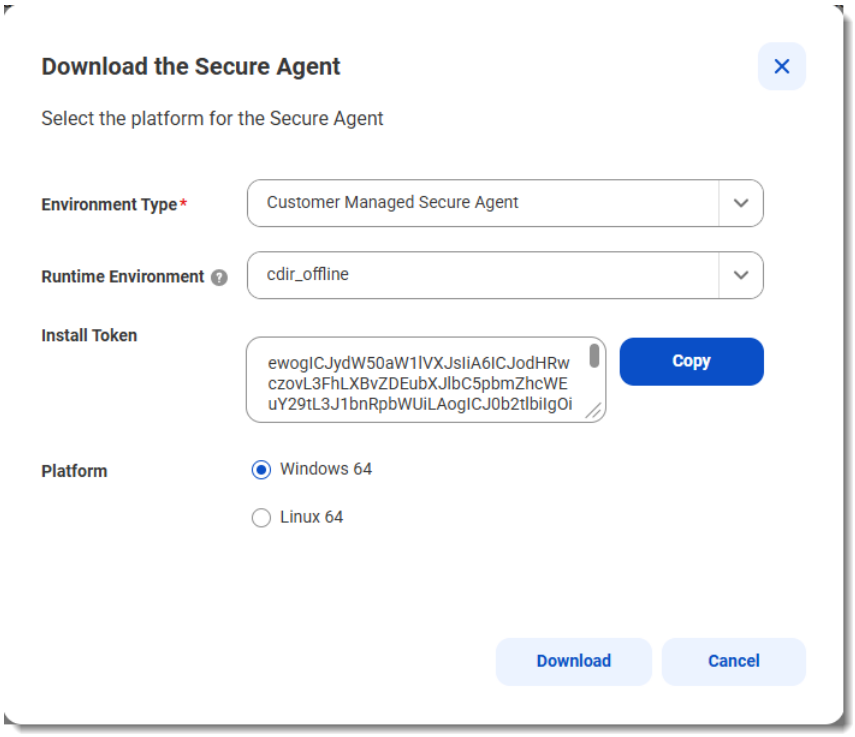
To test a connection, there must be at least one Secure Agent in the Customer Managed runtime environment.

Install Secure Agents to a Customer Managed runtime environment

You can install Secure Agents to a Customer Managed runtime environment. The procedure is virtually identical to installing to a standard Secure Agent group.

For complete installation instructions, see Chapter 7, “Secure Agent installation” on page 130.

The following image shows the **Download Secure Agent** dialog box, where the differences occur:



Perform the following steps:

1. Select **Customer Managed Secure Agent** as the environment type.
2. Select the Customer Managed]runtime environment.
3. Click **Copy** to copy the install token.
   You need this token to install the Secure Agent. The token expires after 24 hours. If the token expires before you install the Secure Agent, repeat steps 1 to 3 to obtain a new token. You don't need to download the installer again.
4. Specify your platform, **Windows 64** or **Linux 64**.
5. Click **Download** and save the installer.
6. Install the Secure Agent for your platform:
   - On Windows, run the installer file and follow the installation wizard.
   - On Linux, run the following shell command in the directory where you saved the installation file:

     ```
     ./agent64_install_ng_ext.bin -i console
     ```

     When the installer completes, navigate to the following directory:

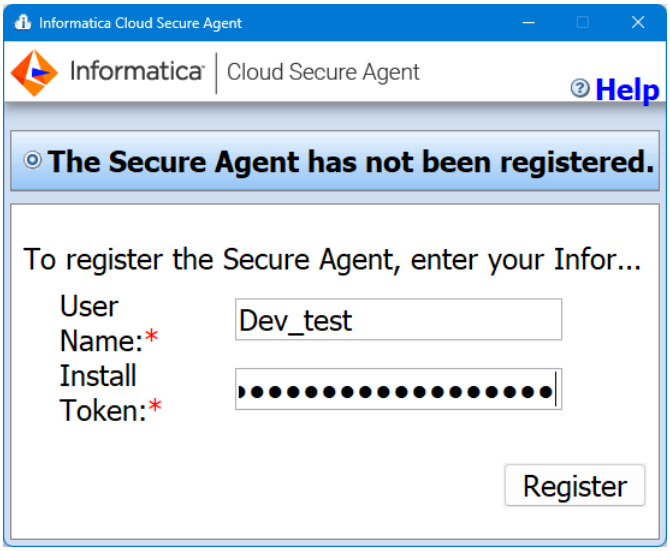
     ```
     <Secure Agent installation directory>/apps/agentcore
     ```

     Start the Secure Agent using the following command:

     ```
     ./infaagent.sh startup
     ```

Registering the Secure Agent on Windows

When the installation is complete, you are prompted to register the Secure Agent. The following image shows the registration dialog box:



1. Enter the user name that you use to access IDMC.
2. Paste the install token that you copied earlier.
3. Click **Register.**

Registering the Secure Agent on Linux

Perform the following steps:

1. Navigate to the following directory:

   ```
   <Secure Agent installation directory>/apps/agentcore
   ```

2. Run the following command:

   ```
   ./consoleAgentManager.sh configureWithEmbeddedToken <token>
   ```

   Where "<token>" is the install token that you copied earlier.

Monitor synchronizations for a Customer Managed runtime environment

The **Monitor** tab on a Customer Managed runtime environment's details page shows the status and history of runtime synchronizations. Knowing the status allows you to take corrective actions as necessary.

Customer Managed runtime environments require that you manually synchronize all changes to the Secure Agents. The **Monitor** tab on the details page of a Customer Managed runtime environment displays the status and history of these synchronizations.

Runtime synchronizations

A runtime synchronization represents a deployment of task metadata and assets that IDMC pushes to the Secure Agents running in the Customer Managed runtime environment. The **Monitor** tab lists all the synchronizations for the environment.

Runtime Syncs table

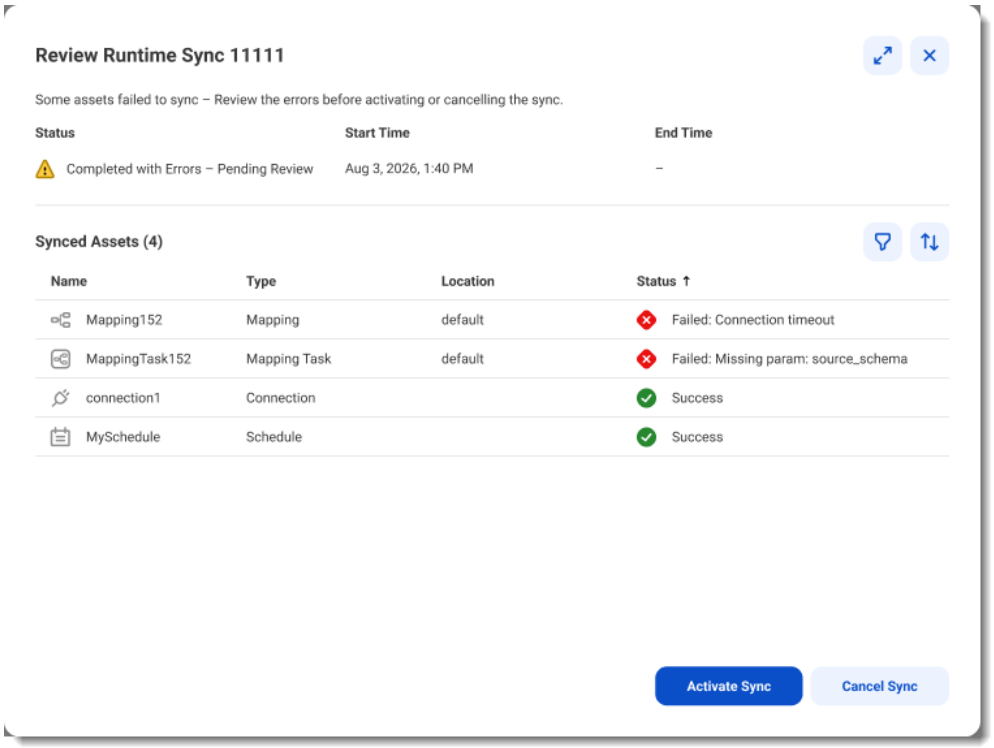
The runtime synchronizations table on the **Monitor** tab displays the following information for each synchronization:

- A sync ID that you can select to view more details about the synchronization.
- The start and end times of the runtime synchronization, based on your user profile time zone.
- The status of the runtime synchronization.
  The following table describes each status:

Status	Description
Activated	Synchronization is complete.
In Progress	Auto synchronization is in progress.
Completed Successfully	Auto synchronization was successful. Moves automatically to Activated.
Completed with Errors	Auto synchronization encountered errors and requires a user response. If the user chooses to continue synchronization despite errors, the status moves to Activated.
Cancelled	User cancelled the synchronization after errors are found.

Synchronization errors

If a synchronization status shows **Completed with Errors**, click **Pending Review** in the job's review pane on the Monitor tab to view the details of the errors. The following image shows the review pane for a job named "Runtime Sync 11111:"



Review the failures and choose one of the following actions:

- To continue with the synchronization despite the errors, click **Activate Sync**. The "Success" assets will receive the latest versions pushed down through this synchronization. Failed assets continue to use the last successful synchronized version. After synchronization, the status moves to "Activated."
- To cancel the synchronization so that you can address the failures, click **Cancel Sync**. You can restart the synchronization later through the **Synchronize the runtime environment** action. For more information, see "Synchronize the runtime environment" on page 29.

Create a password to access the Customer Managed Secure Agent API

The Secure Agent API allows you to perform tasks without using the IDMC user interface. Create a separate password to allow Customer Managed Secure Agents to authenticate API calls. The password doesn't grant access to the IDMC user interface.

1. In Administrator, select **Settings**.
2. Under **Customer Managed Secure Agent**, create a password.

Grant access to Customer Managed Secure Agent API

To allow a user to access the Customer Managed Secure Agent API, assign a role with the appropriate feature enabled.

1. In Administrator, select **User Roles**.
2. Click **Add Role** and enter a role name.
3. Select the **Features** tab and enable **Offline Secure Agent API Access**.
4. Click **Save**.
5. In Administrator, select **Users**.
6. Select the user who needs the API access.
7. In the **Roles** tab, select the role that you just created.
8. Synchronize the runtime environment to send the updated user role to the Customer Managed Secure Agents. For more information, see "Working with Customer Managed runtime environments" on page 27.

Working with Customer Managed runtime environments

After you create a Customer Managed runtime environment, you can edit, synchronize, and delete it. You can also view dependencies, change permissions, and check for upgrades.

You can perform the following tasks from the **Actions** menu:

Change the name or description of the environment

To change the name or description of your Customer Managed runtime environment, select **Edit Environment Properties**.

Enable or disable services and connectors

You can enable a service to allow the Customer Managed runtime environment to run product features associated with the service. You can enable specific connectors to allow the environment to communicate with cloud and on-premises applications, platforms, databases, and flat files.

To enable services and connectors, select **Enable or Disable Services, Connectors**.

Manage configuration

To change the configuration on the **Database Configuration** tab, select **Manage configuration**.

Synchronize the runtime environment

To push changes to Secure Agents in a Customer Managed runtime environment, select **Sync Runtime Environment**.

For more information, see "Synchronize the runtime environment" on page 29.

Check and install upgrades

Check whether upgrades are available for the Secure Agents in the Customer Managed runtime environment and then install the upgrades.

Install any upgrades as soon as possible, as you can't synchronize your Customer Managed runtime environment with outdated Secure Agents.

Tip: The **Sync Runtime Environment** action automatically checks for upgrades.

To check for upgrades, select **Check and install upgrades**.

Delete the runtime environment

You can delete a Customer Managed runtime environment if it is no longer being used.

To delete a Customer Managed runtime environment, select **Delete**.

View dependencies

View dependencies to find the tasks or connections that use the Customer Managed runtime environment or to find the objects that are used by the Customer Managed runtime environment.

To view dependencies, select **Show Dependencies**.

Change permissions

To change permissions for a Customer Managed runtime environment, select **Permissions**. You can define permissions for a Customer Managed runtime environment for each user group in your organization.

The following table describes the permissions that you can set:

Permission	Description
Read	View details about the Customer Managed runtime environment and use the Customer Managed runtime environment in a task.
Update	Edit the Customer Managed runtime environment.
Delete	Delete the Customer Managed runtime environment.
Execute	Run tasks in the Customer Managed runtime environment.
Change Permissions	Change permissions for the Customer Managed runtime environment.

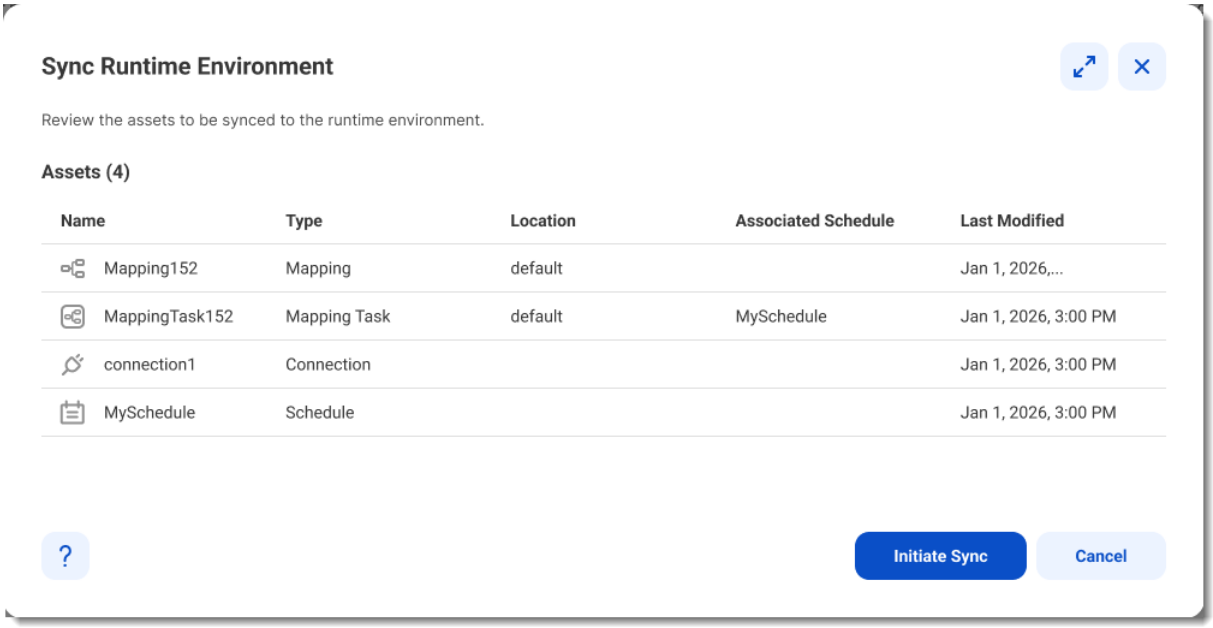
Synchronize the runtime environment

When you use the **Sync Runtime Environment** action of a Customer Managed runtime environment, you manually push changes to tasks, assets, or configurations to the Secure Agents in the environment. Unlike standard runtime environments, these changes don't transmit automatically.

It's important to understand the two types of synchronizations in a Customer Managed runtime environment:

- Automatic synchronization. Secure Agents in a Customer Managed runtime environment automatically return results of jobs back to IDMC when a connection is available. This operation runs in the background, with no user interaction.
- Manual synchronization. Any changes within IDMC such as new tasks, mappings, connections, or schedules require you to manually deploy the changes to the Secure Agents. You do this using the **Sync Runtime Environment** option in the **Actions** menu of a Customer Managed runtime environment.

The following image shows a confirmation dialog listing a set of assets that will be synchronized:



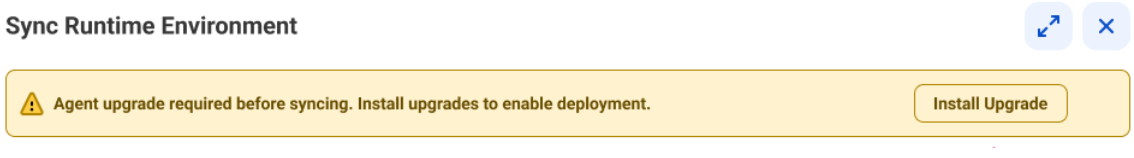
If everything is in order, click **Initiate Sync** to begin the synchronization. This may take several minutes to complete and you will be notified when the process is finished.

If you need to make changes, click **Cancel**, make the corrections and synchronize.

Secure Agent upgrade

If your Secure Agent requires an upgrade, a banner prompts you to upgrade before synchronization is possible.

The following image shows the upgrade banner:



Click **Install Upgrade** to continue and then click **Upgrade** in the confirmation dialog to proceed with the upgrade.

Optionally, you can select to upgrade all available Customer Managed runtime environments. If you don't select this option, only the agents in the current Customer Managed runtime environment are upgraded.

CHAPTER 4

Elastic runtime environments

An elastic runtime environment uses elastic scaling to provide a runtime environment option that optimizes resource management. Use an elastic runtime environment to handle fluctuating workload demands and manage workload costs.

An elastic runtime environment uses Kubernetes to run a Secure Agent and to process data on cluster nodes. Cluster nodes that process data can scale up or down based on workload demands from IDMC.

The following services can use an elastic runtime environment:

- API Center (preview)
- Application Integration (preview)
- Data Governance and Catalog*
- Data Integration

* *Only document classification can run in an elastic runtime environment.*

Note: Effective in the April 2026 release, elastic runtime environments are available for preview in API Center and Application Integration. Preview functionality is supported for evaluation purposes but is unwarranted and is not supported in production environments or any environment that you plan to push to production. Informatica intends to include the preview functionality in an upcoming release for production use, but might choose not to in accordance with changing market or technical circumstances. For more information, contact Informatica Global Customer Support.

To create an elastic runtime environment, complete the following tasks:

1. Verify that you have an AWS account with the correct permissions and that your organization administrator has the correct feature privileges in Administrator.
2. Set up the AWS environment to create and configure the required AWS resources.
3. Create the elastic runtime environment in Administrator.
4. Deploy the elastic runtime environment on AWS.
5. Optionally, upload parameter files and supplementary files to the EFS for system storage to allow tasks to access parameter configurations and custom dependencies.
6. Optionally, upload flat files to the EFS for data storage to use them as sources in mappings and tasks.

Protecting your data

Informatica implements fine-grained authentication and authorization mechanisms to deploy and manage an elastic runtime environment in your VPC, access data sources that you use in tasks, and store output logs.

An elastic runtime environment is isolated at the workload level to increase reliability. A user configures the environment through their organization. Then, the master node auto-scales and manages AWS resources. Images, artifacts, and configurations are stored separately in the IDMC control plane.

Interaction with the elastic runtime environment involves distinct authentication and authorization methods for users and the master node.

Authentication

To configure an elastic runtime environment, a user authenticates to their organization by logging in through a mechanism such as a password or SSO. The master node uses an IAM instance profile to manage AWS resources.

Authorization

An elastic runtime environment uses IDMC and IAM roles to authorize users and instances to access and manage IDMC and AWS resources:

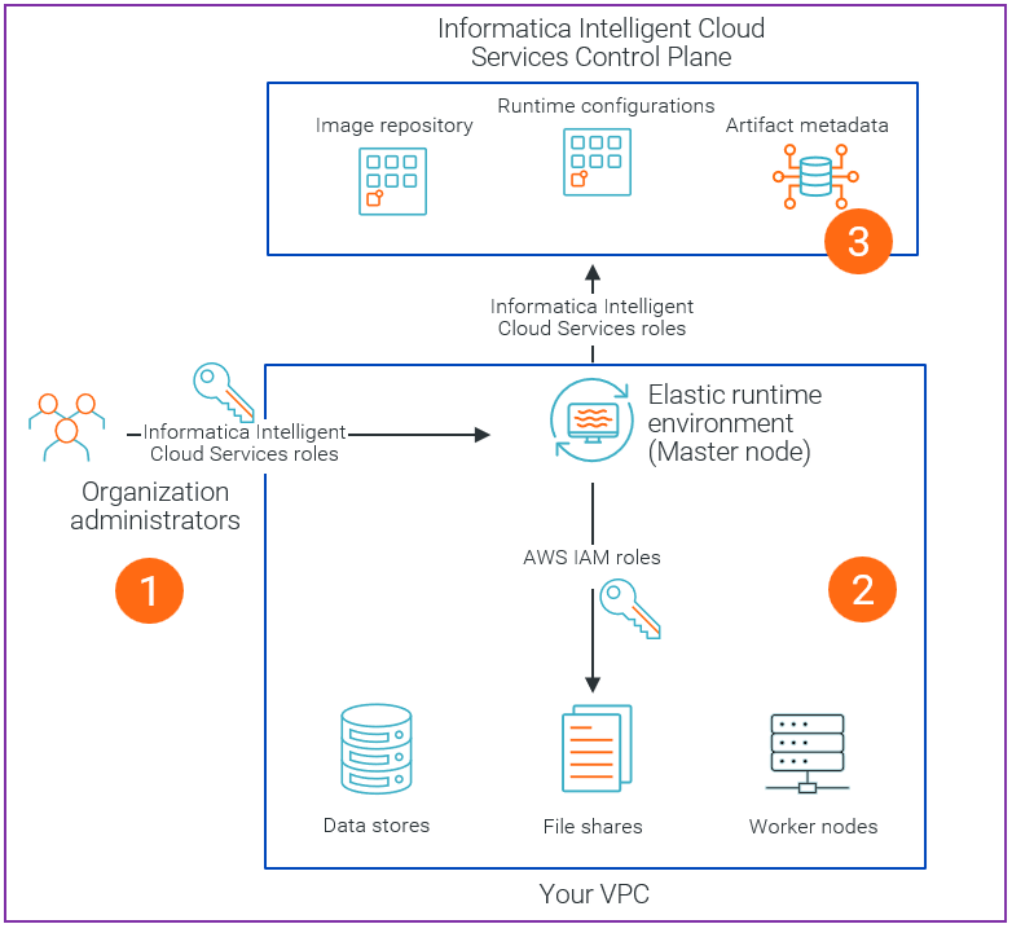
User-based authorization

Users have authorization to access and manage IDMC and an elastic runtime environment through their IDMC role. The role grants a user privileges to create, modify, and deploy the elastic runtime environment and to access the Secure Agent. Roles also isolate the elastic runtime environment for department-level access so that each department is authorized to administer their own elastic runtime environment.

Instance-based authorization

The master node has authorization to access and manage worker nodes in the environment and AWS resources in your VPC. A master node is granted authorization through an IAM role, which is associated with one or more IAM policies and an instance profile. Policies provide granular access to the AWS resources that an elastic runtime environment uses, such as file shares and virtual machines. For example, the master node is authorized to create EC2 instances and change auto-scaling settings.

The following image shows how Informatica uses roles to manage authorization to access and manage IDMC and AWS resources:



1. IDMC roles grant access to configure the elastic runtime environment in your organization.
2. IAM roles grant access to data stores, file shares, and worker nodes in the elastic runtime environment.
3. IDMC roles grant access to resources in the IDMC control plane, including Informatica's image repository, artifactory, and runtime configuration store.

Verify the prerequisites

Before you set up an elastic runtime environment, verify the permissions in your AWS account, verify that your organization administrator has the correct feature privileges in Administrator, and verify the minimum hardware requirements for the master node.

Verify the following prerequisites:

- You have an AWS account with permissions to create resources. If your organization has a dedicated AWS administrator, the AWS administrator can work with the Informatica administrator to create the resources.

- Your IDMC organization administrator has the following Administrator feature privileges:
  - Elastic runtime environment - get images
  - Elastic runtime environment - manage tokens

  For more information about feature privileges, see *User Administration*.
- The master node in the elastic runtime environment has a minimum of 4vCPUs, 16 GB memory, and 60 GB free disk space.

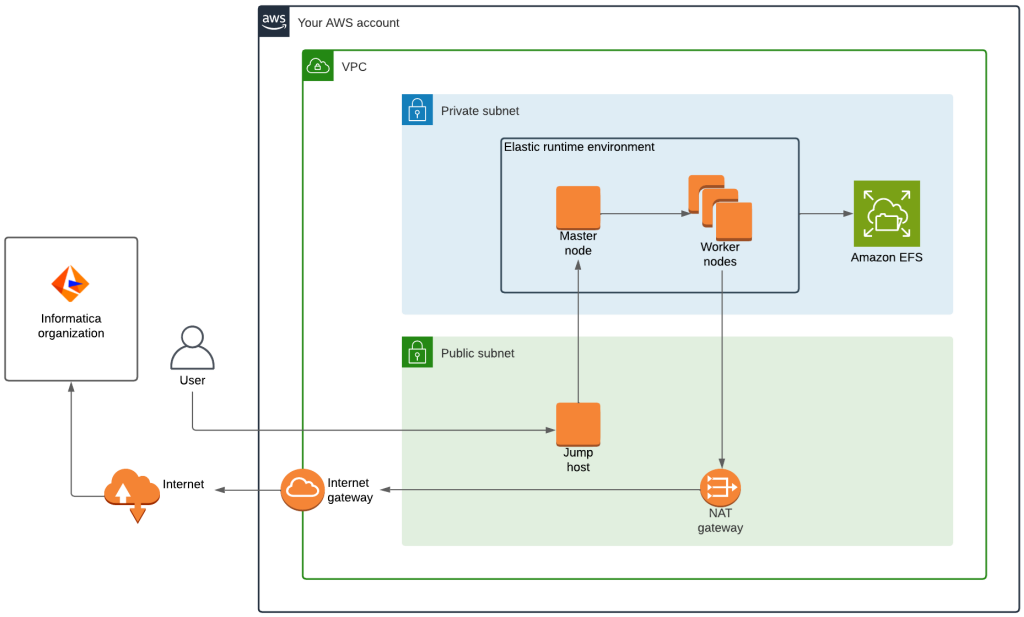
Set up the AWS environment

Set up your AWS environment to create and configure the required AWS resources before deploying an elastic runtime environment.

To set up the AWS environment, complete the following tasks:

1. Create and configure AWS resources that the elastic runtime environment can use to start a Secure Agent and to run jobs.
2. Create a cluster installer policy that contains the permissions that allow the cluster to interact with an elastic runtime environment.
3. Create an IAM role for the cluster installer that allows the cluster installer to deploy an elastic runtime environment.
4. Create a worker policy that contains the permissions that allow worker nodes to process data in an elastic runtime environment.
5. Create an IAM role for the worker nodes that allows the worker nodes to process data in the elastic runtime environment.
6. Optionally, launch an EC2 instance for the jump host to remotely access the elastic runtime environment. If you can access the private subnet through your enterprise network, you don't need to create a jump host.
7. Launch an EC2 instance for the master node.

The following image shows how the AWS environment appears with a deployed elastic runtime environment:



Step 1. Create AWS resources

Create and configure AWS resources that the elastic runtime environment can use to start a Secure Agent and run jobs.

Create and configure the following AWS resources:

VPC

A VPC contains all your AWS resources, including the Kubernetes cluster that hosts the Secure Agent.

Create a VPC in your AWS account. Use IPv4 CIDR manual input and enter the CIDR block for the VPC to use.

Public subnet

A public subnet provides internet access through a NAT gateway. To create the public subnet, use the following guidelines:

- Use any availability zone in the region where you created the VPC.
- To set the IPv4 VPC CIDR block, use the same IPv4 CIDR block that you specified when you created the VPC.
- To set the IPv4 subnet CIDR block, use a CIDR range that's within the VPC CIDR range. Choose a range based on the number of IP addresses that you want to have within the subnet. For example, if you use 10.1.0.0/16 for the VPC, you might use 10.1.0.0/20 for the public subnet.

Private subnet

A private subnet hosts IDMC servers and resources. To create the private subnet, use the following guidelines:

- Use the same availability zone that you used to create the public subnet.
- To set the IPv4 VPC CIDR block, use the same IPv4 CIDR block that you specified when you created the VPC.

- To set the IPv4 subnet CIDR block, use a CIDR range that's within the VPC CIDR range. For example, if you use 10.1.0.0/16 for the VPC, you might use 10.1.240.0/20 for the private subnet.

  Choose a range that has enough available IP addresses to accommodate the maximum number of worker nodes in the elastic runtime environment. For example, if the environment has a minimum of one worker node and a maximum of 10 worker nodes, then at least 10 IP addresses must be available in the private subnet to accommodate the worker nodes.

NAT gateway

A NAT gateway allows outbound traffic to the internet from nodes in the private subnet. The NAT gateway ensures that private nodes are isolated from the public internet.

To create the NAT gateway, use the following guidelines:

- Use the public subnet as the subnet.
- Set the connectivity type to **Public**.
- Allocate an elastic IP address to the NAT gateway.

Internet gateway

An internet gateway is used for internet access. The public subnet and the internet gateway allow the jump host to receive SSH connections from the public internet.

Create an internet gateway in AWS and then attach it to the VPC.

For information about the jump host, see “Step 6. Create the jump host (optional)” on page 44.

Public route table

A public route table routes traffic in the public subnet. To create the public route table, use the following guidelines:

- Use the VPC that you created.
- Add a route using `0.0.0.0/0` as the destination and the internet gateway that you created.
- Edit the subnet association and select the public subnet that you created.

Private route table

A private route table routes traffic in the private subnet. To create the private route table, use the following guidelines:

- Use the VPC that you created.
- Add a route using `0.0.0.0/0` as the destination and the NAT gateway that you created.
- Edit the subnet association and select the private subnet that you created.

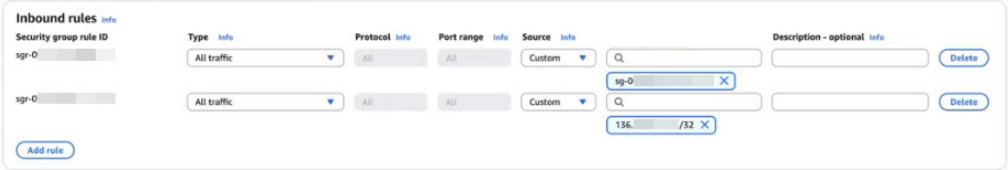
Security group for the elastic runtime environment

A security group allows SSH access to the elastic runtime environment. You specify this security group in the config.txt file. For more information about the config.txt file, see “Deploy an elastic runtime environment” on page 48.

To create the security group, use the following guidelines:

- Use an existing security group or create a new one.
- Use the VPC that you created.
- Add inbound rules that allow the following types of traffic:
  - All traffic from the same security group
  - All traffic from the local machine that you're using to create AWS resources

The following image shows an example of the inbound rules:



If any required inbound rules are missing, the cluster installer populates them.

Security group for the jump host (optional)

The jump host is an EC2 instance in the public subnet that you can use to SSH into nodes in the elastic runtime environment in the private subnet. A security group allows SSH access to the jump host from your local machine. If you can access the private subnet through your enterprise network, you don't need to create a jump host or a security group for the jump host.

To create a security group for the jump host, use the following guidelines:

- Use the VPC that you created.
- Add an inbound rule that allows SSH traffic on port 22 from source `<local machine IP address>/32`.

For information about creating the jump host, see "Step 6. Create the jump host (optional)" on page 44.

EFS file systems for system storage (required) and data storage (optional)

An elastic runtime environment uses EFS file systems for the following storage types:

- System storage is required for Secure Agent operations. You upload parameter files and supplementary files to system storage, such as JDBC drivers, third-party libraries, license files, and custom scripts.
- Data storage stores flat files that the elastic runtime environment can use as source data to run tasks.

Create an EFS file system for system storage and then create an access point for it. Optionally, you can create another EFS file system for data storage and create an access point for it as well.

To create each file system, use the following guidelines:

- For the VPC, use the VPC that you created.
- For the system disk, create at least one mount target in the availability zone where you intend to create the master node and run the cluster installer. When you run the cluster installer on the master node to deploy the elastic runtime environment, the cluster installer will automatically create mount targets for all remaining availability zones.

  For the data disk, specify the private subnet as the mount target.
- On the **Network access** page, customize the file system.

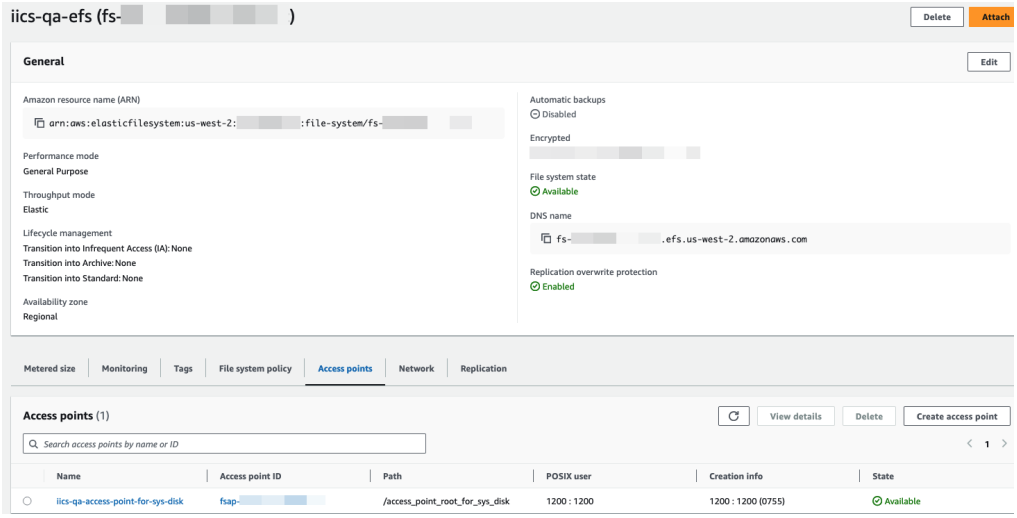
  - Availability zone. Use the name of the availability zone of the private subnet.

  - Security group. Use the name of the security group that you created.
- Create the file system only after you've customized it.

To create each access point, use the following guidelines:

- For the file system, use the file system that you created.
- Enter a root directory, such as `/ert_sysdisk` or `/ert_datadisk`.
- For the POSIX user, use the user ID `1200` and group ID `1200`.
- In the root directory creation permissions, use owner user ID `1200`, owner group ID `1200`, and access point permission `0755`.

The following image shows an example of an EFS file system in the AWS Management Console:



Amazon Machine Image (optional)

An Amazon Machine Image (AMI) is an image that includes the software to start on the master and worker nodes. You can create your own image to manage the infrastructure of the elastic runtime environment and meet compliance and governance needs.

To create the AMI, make sure it meets the following requirements:

- The operating system is Rocky Linux 10.x.
- A root user or a user account with sudo access is available.
- The AMI includes a mounted storage disk.
- The kernel can load the following required modules:

  br_netfilter

  nf_conntrack

  nfs

  nfsv4

  overlay

  vxlan
- IMDSv2 is enabled with the hop limit set to 2.
- The swap service can be stopped and disabled.
- The security tools SELinux and firewalld can be configured to open ports and set custom security policies.
- Third-party software doesn't interfere with Informatica's Kubernetes components or consume Informatica or Kubernetes resources. If you install third-party tools, set up resource monitoring for them.
- The AMI doesn't use automatic patching tools. To update the AMI, create and deploy a new AMI.

Step 2. Create a cluster installer policy

Create a cluster installer policy that contains the permissions that allow the cluster to interact with an elastic runtime environment.

To create the cluster installer policy, use the JSON document in this section and replace the placeholders:

The following table describes each placeholder:

Placeholder	Description
<AWS account ID>	AWS account ID.
<cluster installer role>	Cluster installer role name, such as `cluster_installer_role`. The name must follow the AWS naming convention for roles. **Note:** You can decide on the role name in this step, and then use the same role name when you create the role. For information about creating the cluster installer role, see "Step 3. Create a cluster installer role" on page 41.
<worker role>	Worker role name, such as `worker_node_role`. The name must follow the AWS naming convention for roles. **Note:** You can decide on the role name in this step, and then use the same role name when you create the role. For information about creating the cluster installer role, see "Step 5. Create a worker role" on page 43.

The following JSON document contains the policy content:

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Sid": "VisualEditor0",
            "Effect": "Allow",
            "Action": [
                "autoscaling:CreateAutoScalingGroup",
                "autoscaling:DeleteAutoScalingGroup",
                "autoscaling:TerminateInstanceInAutoScalingGroup",
                "autoscaling:DescribeAutoScalingGroups",
                "autoscaling:DescribeScalingActivities",
                "autoscaling:DescribeTags",
                "autoscaling:UpdateAutoScalingGroup",
                "autoscaling:CreateOrUpdateTags",
                "autoscaling:DeleteTags",
                "autoscaling:DescribeAutoScalingInstances",
                "autoscaling:DescribeLaunchConfigurations",
                "autoscaling:DescribeLifecycleHooks",
                "autoscaling:ResumeProcesses",
                "autoscaling:SetDesiredCapacity",
                "autoscaling:SuspendProcesses"
            ],
            "Resource": "*"
        },
        {
            "Sid": "VisualEditor1",
            "Effect": "Allow",
            "Action": [
                "ec2:DescribeAvailabilityZones",
                "ec2:DescribeImages",
                "ec2:DescribeInstanceAttribute",
                "ec2:DescribeInstances",
                "ec2:DescribeInstanceTypes",
                "ec2:DescribeSecurityGroups",
                "ec2:DescribeLaunchTemplateVersions",
                "ec2:DescribeLaunchTemplates",
                "ec2:DescribeTags",
```

```
            "ec2:AuthorizeSecurityGroupIngress",
            "ec2:DeleteLaunchTemplate",
            "ec2:ModifyInstanceAttribute",
            "ec2:DescribeVpcs"
        ],
        "Resource": "*"
    },
    {
        "Sid": "VisualEditor2",
        "Effect": "Allow",
        "Action": "ec2:RunInstances",
        "Resource": "*"
    },
    {
        "Sid": "VisualEditor3",
        "Effect": "Allow",
        "Action": [
            "ec2:CreateKeyPair",
            "ec2:CreateTags",
            "ec2:CreateLaunchTemplate"
        ],
        "Resource": [
            "arn:aws:ec2:*:<AWS account ID>:security-group/*",
            "arn:aws:ec2:*:<AWS account ID>:network-interface/*",
            "arn:aws:ec2:*:<AWS account ID>:launch-template/*",
            "arn:aws:ec2:*:<AWS account ID>:instance/*",
            "arn:aws:ec2:*:<AWS account ID>:subnet/*",
            "arn:aws:ec2:*:<AWS account ID>:volume/*",
            "arn:aws:ec2::*::image/*",
            "arn:aws:ec2:*:<AWS account ID>:key-pair/*"
        ]
    },
    {
        "Sid": "VisualEditor5",
        "Effect": "Allow",
        "Action": "ec2:CreateTags",
        "Resource": [
            "arn:aws:ec2:*:<AWS account ID>:network-interface/*",
            "arn:aws:ec2:*:<AWS account ID>:key-pair/*",
            "arn:aws:ec2:*:<AWS account ID>:launch-template/*",
            "arn:aws:ec2:*:<AWS account ID>:instance/*",
            "arn:aws:ec2:*:<AWS account ID>:volume/*",
            "arn:aws:ec2:*:<AWS account ID>:subnet/*"
        ],
        "Condition": {
            "StringEquals": {
                "ec2:CreateAction": [
                    "RunInstances",
                    "CreateKeyPair",
                    "CreateLaunchTemplate",
                    "CreateLaunchTemplateVersion",
                    "network-interface",
                    "CreateTags",
                    "CreateAutoScalingGroup"
                ]
            }
        }
    },
    {
        "Sid": "VisualEditor13",
        "Effect": "Allow",
        "Action": "iam:PassRole",
        "Resource": [
            "arn:aws:iam::<AWS account ID>:role/<cluster installer role>",
            "arn:aws:iam::<AWS account ID>:role/<worker role>"
        ]
    },
    {
        "Sid": "VisualEditor21",
        "Effect": "Allow",
        "Action": [
```

```
                "secretsmanager:DescribeSecret",
                "secretsmanager:GetSecretValue",
                "secretsmanager:ListSecrets",
                "secretsmanager:CreateSecret",
                "secretsmanager:TagResource",
                "secretsmanager:DeleteSecret",
                "secretsmanager:UpdateSecret"
            ],
            "Resource": "arn:aws:secretsmanager:*:<AWS account ID>:secret:*"
        },
        {
            "Sid": "VisualEditor10",
            "Effect": "Allow",
            "Action": [
                "elasticfilesystem:CreateMountTarget",
                "elasticfilesystem:DescribeAccessPoints",
                "elasticfilesystem:DescribeFileSystems",
                "elasticfilesystem:DescribeMountTargets"
            ],
            "Resource": "*"
        },
        {
            "Sid": "VisualEditor11",
            "Effect": "Allow",
            "Action": [
                "ec2:DescribeSubnets"
            ],
            "Resource": "*"
        },
        {
            "Sid": "EC2Management",
            "Effect": "Allow",
            "Action": [
                "ec2:AssociateIamInstanceProfile",
                "ec2:CreateKeyPair",
                "ec2:CreateLaunchTemplate",
                "ec2:CreateLaunchTemplateVersion",
                "ec2:CreateTags",
                "ec2:DeleteLaunchTemplate",
                "ec2:DeleteTags",
                "ec2:ModifyLaunchTemplate",
                "ec2:RunInstances",
                "ec2:TerminateInstances",
                "iam:PassRole"
            ],
            "Resource": "*"
        }
    ]
}
```

For more information about the permissions in the cluster installer policy, see "Cluster installer policy statements" on page 59.

Step 3. Create a cluster installer role

Create an IAM role for the cluster installer that allows the cluster installer to deploy an elastic runtime environment.

To create the cluster installer role, use the following guidelines:

- Set the trusted entity to **AWS service** and the use case to **Amazon EC2**.
- Use the same cluster installer role name that you specified in the cluster installer policy.
- Edit the trust relationship and specify the following policy:

```
{
    "Version": "2012-10-17",
    "Statement": [
```

```
            {
                "Effect": "Allow",
                "Principal": {
                    "Service": "ec2.amazonaws.com"
                },
                "Action": "sts:AssumeRole"
            }
        ]
    }
```

- If AWS doesn't automatically create an instance profile for the role, create the instance profile manually.

Step 4. Create a worker policy

Create a worker policy that contains the permissions that allow worker nodes to process data in an elastic runtime environment.

To create the worker policy, use the following JSON document:

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Sid": "DescribeActions",
            "Effect": "Allow",
            "Action": [
                "elasticfilesystem:DescribeMountTargets",
                "autoscaling:DescribeAutoScalingGroups",
                "autoscaling:DescribeAutoScalingInstances",
                "autoscaling:DescribeLaunchConfigurations",
                "autoscaling:DescribeTags",
                "ec2:DescribeInstances",
                "ec2:DescribeLaunchTemplateVersions",
                "ec2:DescribeTags",
                "autoscaling:DescribeLifecycleHooks",
                "autoscaling:DescribeScalingActivities",
                "ec2:DescribeLaunchTemplates",
                "ec2:DescribeInstanceTypes",
                "ec2:DescribeVpcs",
                "secretsmanager:DescribeSecret",
                "secretsmanager:ListSecrets"
            ],
            "Resource": "*"
        },
        {
            "Sid": "AutoscalingCapacity",
            "Effect": "Allow",
            "Action": [
                "autoscaling:SetDesiredCapacity",
                "autoscaling:TerminateInstanceInAutoScalingGroup",
                "autoscaling:CreateAutoScalingGroup",
                "autoscaling:CreateOrUpdateTags",
                "autoscaling:DeleteAutoScalingGroup",
                "autoscaling:DeleteTags",
                "autoscaling:SuspendProcesses",
                "autoscaling:UpdateAutoScalingGroup"
            ],
            "Resource": "*"
        },
        {
            "Sid": "EC2Management",
            "Effect": "Allow",
            "Action": [
                "ec2:AssociateIamInstanceProfile",
                "ec2:CreateKeyPair",
                "ec2:CreateLaunchTemplate",
                "ec2:CreateLaunchTemplateVersion",
                "ec2:CreateTags",
                "ec2:DeleteLaunchTemplate",
```

```
                "ec2:DeleteTags",
                "ec2:ModifyLaunchTemplate",
                "ec2:RunInstances",
                "ec2:TerminateInstances",
                "iam:PassRole"
            ],
            "Resource": "*"
        },
        {
            "Sid": "EFSManagement",
            "Effect": "Allow",
            "Action": [
                "elasticfilesystem:CreateAccessPoint",
                "elasticfilesystem:DeleteAccessPoint"
            ],
            "Resource": "*"
        },
        {
            "Sid": "SecretsManagerAccess",
            "Effect": "Allow",
            "Action": [
                "secretsmanager:GetSecretValue",
                "secretsmanager:UpdateSecret"
            ],
            "Resource": "*"
        }
    ]
}
```

For more information about the permissions in the worker policy, see "Worker policy statements" on page 62.

Step 5. Create a worker role

Create an IAM role for the worker nodes that allows the worker nodes to process data in the elastic runtime environment.

To create the worker role, use the following guidelines:

- Set the trusted entity to **AWS service** and the use case to **Amazon EC2**.
- Use the same worker role name that you specified in the cluster installer policy.
- Edit the trust relationship and specify the following policy:

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Principal": {
                "Service": "ec2.amazonaws.com"
            },
            "Action": "sts:AssumeRole"
        }
    ]
}
```

- If AWS doesn't automatically create an instance profile for the role, create the instance profile manually.

Step 6. Create the jump host (optional)

Optionally, launch an EC2 instance for the jump host to remotely access the elastic runtime environment. If you can access the private subnet through your enterprise network, you don't need to create a jump host.

The jump host is an EC2 instance in the public subnet that you can use to SSH into nodes in the elastic runtime environment in the private subnet. Because the elastic runtime environment is designed to run in a private subnet, a jump host can remotely access the environment.

Using a jump host is a best practice for secure and scalable infrastructure so that the elastic runtime environment isn't exposed to the internet. The jump host greatly reduces the attack surface and prevents unauthorized access to the elastic runtime environment, and access is controlled using SSH authentication and security group rules. If the jump host is compromised, nodes in the elastic runtime environment are protected by a firewall and security groups.

Create a jump host using the launch instance wizard in AWS. Use the following guidelines:

- Use Amazon Linux as the OS image.
- Use instance type `t3.small`.
- Create a new key pair.
- Use the VPC that you created.
- For the subnet, select the public subnet that you created.
- Enable **Auto-assign public IP**.
- Use the security group that you created for the jump host. For more information, see "Step 1. Create AWS resources" on page 35.

Then, launch the instance.

Step 7. Create the master node

Launch an EC2 instance for the master node.

Create the master node using the launch instance wizard in AWS. Use the following guidelines:

- Launch the instance using either Informatica's AMI or the AMI that you created. To find Informatica's AMI, search `infa-elastic-master-node` in the community AMIs and select the latest AMI to launch an instance.
- Use an instance type with at least 2 CPUs and 4 GB of memory.
- Use the key pair that you created when you launched the jump host.
- In the network settings, use the VPC that you created.
- Use the private subnet that you created as the subnet.
- Disable **Auto-assign public IP**.
- Use the security group that you created for the elastic runtime environment. For more information, see "Step 1. Create AWS resources" on page 35.
- To set the IAM instance profile, use the instance profile for the cluster installer role.
- Keep the default metadata settings. Make sure the master node uses IMDSv2 and keep the hop limit set to 2.

Then, launch the instance.

Create an elastic runtime environment

Create an elastic runtime environment in Administrator.

1. In Administrator, open the **Runtime Environments** page.
2. Select **New Runtime Environment**.
3. On the **General** page, enter the name of the elastic runtime environment.
4. Select **Elastic Runtime Environment** as the environment type.
5. Optionally, enter a description.
6. Click **Next**.
7. On the **Enable Components** page, select the services to enable.
8. Select the connectors to enable for the elastic runtime environment.

   For a list of connectors that an elastic runtime environment can use, see "Connectors in an elastic runtime environment" on page 58.
9. Select additional services to enable, such as Secret Manager.

   Note: If you don't enable services and connectors when you create the elastic runtime environment, you can enable them later. For more information, see "Working with elastic runtime environments" on page 55. For instructions about setting up Secret Manager, see *Organization Administration*.
10. Click **Create**.
11. On the **Environment Configuration** tab, configure the elastic runtime environment properties.

Environment configuration

Configure the elastic runtime environment properties to determine how IDMC runs the Kubernetes cluster and processes data in the elastic runtime environment.

Platform

The following table describes the platform properties:

Property	Description
Region	Region in which to create the elastic runtime environment. **Note:** You can't update the region after you deploy the elastic runtime environment.
Worker Instance Type	Instance type to host the worker nodes. To verify that the instance type is supported in your AWS setup, refer to the AWS documentation. **Note:** The worker nodes for the Secure Agent services use the same instance type as the master node.
Worker Instance Profile	Instance profile to attach to worker nodes. The instance profile is the value at the end of the ARN.

Property	Description
Enable High Availability	Enables high availability for the elastic runtime environment. When high availability is enabled, the elastic runtime environment maintains an odd number of master nodes. Because you manually create one original master node as part of the setup, the environment creates an additional, even number of master nodes based on the number of subnets that you provide. For example, if you provide two subnets, the environment will have a total of three master nodes. **Note:** Worker nodes are always highly available and are created across multiple availability zones and subnets regardless of whether high availability is enabled. For more information about high availability, refer to the Kubernetes documentation.
Subnets	Subnet IDs from different regions when high availability is enabled. Enter the subnet ID for at least two subnets. For example: `subnet-01234567890abcdef` A master node is created in each subnet as long as the total number of master nodes is an odd number. Each master node in the environment, including the original master node, must be in a different region. The created master nodes use the same configuration as the original master node but have different IP addresses.
EBS Volume Size	Size of the EBS volume to attach to a worker node for temporary storage during data processing.
AMI ID	Amazon Machine Image ID. For example: `ami-01234567890abcdef`

Note: An elastic runtime environment can become highly available to eliminate a single point of failure when the master node goes down. If you enable high availability and one master node goes down, other master nodes will be available and jobs can continue running. When an elastic runtime environment is highly available, watch out for job failures in the following scenarios:

- If all master nodes go down, jobs will fail.
- If too many master nodes go down, the Kubernetes API server becomes unavailable. The threshold for the number of failures is `(n+1)/2` where `n` is the number of master nodes. For example, if the cluster has 3 master nodes and 2 master nodes go down, the Kubernetes API server becomes unavailable and jobs fail on the cluster.

Job auto-scaling policy

An elastic runtime environment uses the following sets of worker nodes:

- Worker nodes that run jobs. The job auto-scaling policy determines the number of worker nodes that are created to run jobs in the organization.
- Worker nodes that run the Secure Agent services. This set of worker nodes scales automatically to accommodate high job volume.

If no jobs are submitted to the elastic runtime environment by the time it reaches the worker node idle timeout, the number of worker nodes is calculated using the following equation:

```
(min worker nodes in job auto-scaling policy) + (worker nodes for Secure Agent services)
```

Even if the minimum number of worker nodes in the job auto-scaling policy is set to zero, the environment has at least one worker node that runs the Secure Agent services.

The following table describes the properties in the job auto-scaling policy:

Property	Description
Min Worker Nodes	Minimum number of worker nodes.
Max Worker Nodes	Maximum number of worker nodes.
Worker Node Idle Timeout	Amount of idle time before a worker node times out.

System storage

The following table describes the system storage properties:

Property	Description
Type	Type of system storage. Select EFS.
File System ID	File system ID for the EFS file system.
Access Point ID	Access point ID for the EFS file system. The ID is the alphanumeric value at the end of the ARN.
Source File Path	Relative path under the access point where source files physically reside. Exclude the leading forward slash. **Note:** The source file path can't be empty.

Data storage

The following table describes the data storage properties:

Property	Description
Type	Type of data storage. Select EFS.
File System ID	File system ID for the EFS file system.
Access Point ID	Access point ID for the EFS file system. The ID is the alphanumeric value at the end of the ARN.
Source File Path	Relative path under the access point where source files physically reside. Exclude the leading forward slash. For example: `data` **Note:** The source file path can't be empty.
Projected File Path	Path that includes the file system mount point and the access point where files are projected in the runtime environment. For flat files, use the first part of the directory that you enter in the flat file connection. Exclude the leading forward slash. For example, a flat file connection accesses files at `/mnt/shared_data/california/sales/data` where `/mnt/shared_data` is the file system mount point, `/california/sales` is the access point, and `data` is the source file path. For the projected file path, enter `mnt/shared_data/california/sales`.

Note: For more information about how data storage paths map to flat file connections, see "Upload files to data storage" on page 52.

Advanced configuration

The following table describes the advanced properties:

Property	Description
Cloud Ecosystem Tags	Tags for cloud resources. Add at least one tag. The NAME and Name tags are reserved for system use. **Note:** You can't update the tags after you deploy the elastic runtime environment.
Runtime Properties	Properties to customize the elastic runtime environment and the jobs that run in the environment.

Deploy an elastic runtime environment

Deploy the elastic runtime environment on AWS to make the Kubernetes cluster available to process data from IDMC.

Complete the following tasks:

1. SSH into the master node so that you can download and run the cluster installer.
2. If you're redeploying an elastic runtime environment, run the cluster uninstaller to clean up AWS and Kubernetes resources.
3. Download the cluster installer to a directory for the elastic runtime environment on the master node.
4. Optionally, download certificates on the master node for third-party software that your organization requires.
5. Update the configuration file to provide details about the deployment to the cluster installer.
6. Run the cluster installer to deploy the elastic runtime environment.
7. Verify that the elastic runtime environment is running by checking the status of the Secure Agent services and cluster nodes in Administrator.

Step 1. SSH into the master node

SSH into the master node so that you can download and run the cluster installer.

1. Optionally, SSH into the jump host.
   a. Contact your organization's IT team to create a passphrase for the private key on your local machine and configure the jump host to authenticate using the public key.

      To view an example of creating a passphrase and configuring authentication, see the HOW TO: Create a key-pair (Public and Private key) using OpenSSH on UNIX for Support Console Knowledge article.

   b. SSH into the jump host from your local machine by running the following command:

      ```
      ssh -i <private key name>.pem <user name>@<jump host public IP address>
      ```
2. SSH into the master node by running the following command:

   ```
   ssh -i <private key name>.pem cloud-user@<master node private IP address>
   ```

Step 2. Clean up resources

If you're redeploying an elastic runtime environment, run the cluster uninstaller to clean up AWS and Kubernetes resources.

Note: Run the cluster uninstaller only after you have run the cluster installer. If you haven't run the cluster installer, the uninstaller fails.

1. Before you run the uninstaller, verify that there are no jobs running on the elastic runtime environment. Running the uninstaller is irreversible and terminates any running jobs.
2. To run the uninstaller, run the following command on the master node:

   ```
   sudo ./cluster_uninstall.sh
   ```

   The cluster uninstaller deletes resources including Auto Scaling groups, EC2 instances, launch templates, and secrets.
3. In AWS, manually delete the inbound rules that a previous cluster installation created for the security groups.

Step 3. Download the cluster installer

Download the cluster installer to a directory for the elastic runtime environment on the master node.

1. On the master node, create a directory for the elastic runtime environment by running the following command:

   ```
   mkdir ert
   ```
2. Download the cluster installer.
   - To download the cluster installer from Administrator, open the **Runtime Environments** page. Select **Download Runtime Installer**. Then, select **Elastic** as the **Environment Type** and select **Download**. The cluster installer downloads as a ZIP file to the machine.
   - To download the cluster installer using the command line, run the following command on the machine:

     ```
     wget https://global-package.dm.informaticacloud.com/package-manager/files/
     binary/dm-us_informaticacloud/cluster-installer/<version>/linux64/cluster-
     installer.<version>.zip
     ```

     Note: To find the version number to use, download the cluster installer from Administrator and check the version number.
3. Copy the cluster installer to the elastic runtime environment directory that you created.
4. In the directory, unzip the cluster installer by running the following command:

   ```
   unzip cluster-installer.<version>.zip
   ```

Step 4. Download certificates on the master node (optional)

Optionally, download certificates on the master node for third-party software that your organization requires.

For example, you can download certificates for software that inspects network traffic.

1. In the elastic runtime environment directory, create a directory to store certificates by running the following command:

   ```
   mkdir certs
   ```
2. Download the certificates to the certs directory.

Step 5. Update the configuration file

Update the configuration file to provide details about the deployment to the cluster installer.

1. In the elastic runtime environment directory, open the config.txt file by running the following command:

   ```
   vi config.txt
   ```

2. In the config.txt file, read the environment variable descriptions and update the variables.

Note: If your organization uses an outgoing proxy server to connect to the internet, the elastic runtime environment connects to IDMC through the proxy server. You can configure the proxy server settings in the config.txt file. When the cluster installer runs, it prompts you for the proxy server password. The proxy server settings can't be changed after the elastic runtime environment is deployed.

Step 6. Run the cluster installer

Run the cluster installer to deploy the elastic runtime environment.

1. In the elastic runtime environment directory, run the following command:

   ```
   sudo ./cluster-installer.sh
   ```

   If you use the latest version of the installer, you can tail the installation log using the following command:

   ```
   tail -f log.txt
   ```

2. Enter the following details when the cluster installer prompts you to enter them:
   - IDMC user name.
   - IDMC password.
   - Optionally, path to the certificates on the master node, such as `ert/certs`. After you provide the path, the installer validates the certificates and stores them as secrets in AWS Secrets Manager.
   - Proxy password if you configured proxy server settings in the config.txt file.

Warning: Using Ctrl + C terminates the installation.

If cluster installation fails and installs a partial cluster, you need to uninstall the cluster before trying to install it again. To uninstall the cluster, see "Step 2. Clean up resources" on page 49.

Step 7. Verify that the elastic runtime environment is running

Verify that the elastic runtime environment is running by checking the status of the Secure Agent services and cluster nodes in Administrator.

1. In Administrator, open the **Runtime Environments** page.
2. Expand the elastic runtime environment.
3. Verify that the Secure Agent services are running and that one or more instances are running.

   Note: It might take a few minutes for a Secure Agent service to start the instances.

Upload files to system storage

Upload parameter files and supplementary files to the EFS for system storage to allow tasks to access parameter configurations and custom dependencies.

You can upload the following types of files:

- Supplementary files, such as JDBC drivers, third-party libraries, license files, and custom scripts
- Parameter files for mapping tasks

Uploading custom drivers, libraries, and scripts

To access dependencies in mappings and tasks, such as JDBC drivers, third-party libraries, license files, and custom scripts, upload the supplementary files to the EFS for system storage.

You can upload the following supplementary files:

- Database drivers
- JAR files and Java libraries for the Java transformation
- Native libraries and executables
- SAP connectivity drivers and libraries
- SSL/TLS certificates and keystores

To upload the files, perform the following steps:

1. In Administrator, get the elastic runtime environment ID.
   a. Open the **Runtime Environments** page.
   b. Next to the elastic runtime environment, open the **Actions** menu and select **Edit**.
   c. Copy the ID from the URL.

      For example, in the URL `https://usw1.dmr-us.informaticacloud.com/cloudUI/products/administer/main/elastic-agent/KUBERNETES/0141GU25000000000002/overview`, the elastic runtime environment ID is `0141GU25000000000002`.
2. SSH into the master node.
3. Navigate to the file system mount point:

   ```
   cd /mnt/system_disk
   ```

   Note: During cluster deployment, the cluster installer automatically provisions an EFS mount point for system files. The directory `/mnt/system_disk` acts as a logical alias that maps to the following EFS path:

   ```
   /mnt/<file system ID>/<access point>/sys
   ```

   For example:

   ```
   /mnt/fs-06ee607d6bac45c8d/idmc-qa-sysdisk-ap/sys
   ```
4. Navigate to the subfolder that has the same name as the elastic runtime environment ID. For example:

   ```
   cd 0141GU25000000000002
   ```
5. Create subfolders to organize supplementary files according to their type.

   For example, create a jdbc folder to store database drivers:

   ```
   mkdir jdbc
   ```
6. Upload the supplementary files by adding them to their appropriate subfolders.

   For example, place JDBC drivers in `/mnt/system_disk/0141GU25000000000002/jdbc/`.

7. Register the supplementary files by updating the supplementary file specification using the REST API.

   For more information, see Updating the supplementary file specification.

Uploading parameter files for mapping tasks

To use a parameter file to pass different parameter values to mapping tasks, upload the parameter file to the EFS for system storage.

1. In Administrator, get the elastic runtime environment ID.
   a. Open the **Runtime Environments** page.
   b. Next to the elastic runtime environment, open the **Actions** menu and select **Edit**.
   c. Copy the ID from the URL.

      For example, in the URL `https://usw1.dmr-us.informaticacloud.com/cloudUI/products/administer/main/elastic-agent/KUBERNETES/0141GU25000000000002/overview`, the elastic runtime environment ID is `0141GU25000000000002`.
2. SSH into the master node.
3. Navigate to the file system mount point:

   ```
   cd /mnt/system_disk
   ```

   Note: During cluster deployment, the cluster installer automatically provisions an EFS mount point for system files. The directory `/mnt/system_disk` acts as a logical alias that maps to the following EFS path:

   ```
   /mnt/<file system ID>/<access point>/sys
   ```

   For example:

   ```
   /mnt/fs-06ee607d6bac45c8d/idmc-qa-sysdisk-ap/sys
   ```
4. Navigate to the subfolder that has the same name as the elastic runtime environment ID. For example:

   ```
   cd 0141GU25000000000002
   ```
5. Create a folder for parameter files. For example:

   ```
   mkdir parameters
   ```
6. Upload the parameter files to the parameters folder.

Upload files to data storage

Upload flat files to the EFS for data storage to use them as sources and targets in mappings and tasks. By defining access points and data storage configurations in Administrator, you can organize files, isolate data for different teams, and change physical file locations without breaking existing flat file connections.

The following table lists the path layers that an elastic runtime environment uses to process flat files:

Path layer	Origin	Example
EFS access point	Configured in AWS.	`/california/sales`
Source file path	Subfolder structure under the access point where files physically reside. Configured in the data storage properties in Administrator. Excludes the leading forward slash.	`data`

Path layer	Origin	Example
Physical EFS location	Comes from the following formula: `Access point + Source file path`	`/california/sales/data`
File system mount point for the EFS	Local directory where the EFS is attached so that you can interact with it.	`/mnt/shared_data`
Projected file path	Path inside Kubernetes Pods where files are projected for processing. Comes from the following formula: `Mount point + Access point` Configured in the data storage properties in Administrator. Excludes the leading forward slash.	`mnt/shared_data/california/sales`
Flat file connection directory	Configured in the flat file connection.	`/mnt/shared_data/california/sales/data`

For production workloads, upload flat files to the EFS for data storage using a dedicated staging server or your organization's standard file transfer process. If you mount the EFS on a staging node, you can create a file system mount point like `/mnt/efs` to mount the EFS instance, and then upload flat files under the mount point.

Informatica doesn't recommend mounting the EFS for data storage on the master node. Unlike system storage, the cluster installer doesn't create a mount directory for data storage on the master node. The master node rotates during release upgrades and environment configuration changes, which unmounts local connections.

For more information about flat file connections, see the help for the appropriate connector.

Uploading flat files for mappings and tasks

To use a flat file as a source in a mapping or task, upload the flat file to the EFS for data storage. Then, configure the data storage properties in Administrator.

1. Upload your flat files to the EFS for data storage using SMB, NFS, or an external file transport mechanism.

   You can organize flat files using any directory structure. For example, you can upload them to `/data` under the access point.
2. In Administrator, edit the elastic runtime environment.
3. In the **Data Storage** table, add a row and enter the following details:
   - **Type:** Select **EFS**.
   - **File System ID:** Enter the EFS file system ID.
   - **Access Point ID:** Enter the access point ID.
   - **Source File Path:** Enter the relative path under the access point with no leading forward slash. For example: `data`
   - **Projected File Path:** Enter the file system mount point and access point, excluding the leading forward slash. For example, if the file system mount point is `/mnt/shared_data` and the access point is `/california`, enter `mnt/shared_data/california`.

Example: Multi-team access control

Enable secure, isolated, multi-team access to a shared EFS file system by defining team-specific EFS access points, flat file connections, and data storage properties in the elastic runtime environment.

An organization has two sales teams, California Sales and Oregon Sales. Both teams share the same EFS file system but must be strictly isolated to their own directories:

- California Sales data: `/california/sales/data`
- Oregon Sales data: `/oregon/sales/data`

To enforce boundary isolation so that the teams access only their own files, complete the following tasks:

1. In AWS, create access points on the EFS with the following root directories:
   - California access point: `/california/sales`
   - Oregon access point: `/oregon/sales`

   In the AWS Management Console, the root directories appear under the **Path** column for each access point.
2. In IDMC, create flat file connections with the following directories for each regional team:
   - California directory: `/mnt/shared_data/california/sales/data`
   - Oregon directory: `/mnt/shared_data/oregon/sales/data`
3. Edit the elastic runtime environment and add the following rows in the **Data Storage** table:
   - California Sales
     - **Type:** EFS
     - **File System ID:** fs-06ee607d
     - **Access Point ID:** fsap-ca12345
     - **Source File Path:** data
     - **Projected File Path:** mnt/shared_data/california/sales
   - Oregon Sales
     - **Type:** EFS
     - **File System ID:** fs-06ee607d
     - **Access Point ID:** or67890
     - **Source File Path:** data
     - **Projected File Path:** mnt/shared_data/oregon/sales

The elastic runtime environment performs the following actions:

- Reads and writes files at `/california/sales/data` and `/oregon/sales/data` on the EFS.
- Processes files at `/mnt/shared_data/california/sales` and `/mnt/shared_data/oregon/sales` on the Kubernetes Pods.

Because each EFS access point enforces its own permission sets, California Sales can access flat files only under `/california/sales/data`, and Oregon Sales can access flat files only under `/oregon/sales/data`.

Example: Migrate data without modifying connections or tasks

Migrate data to a new secure EFS access point and process the data in an elastic runtime environment without modifying existing connections or tasks.

An organization has an existing flat file connection that reads data from `/mnt/legacy/sales/orders`. The organization wants to migrate the data to a new secured access point, `/ca_region`, without changing existing connections or tasks.

To configure the elastic runtime environment to read data from the new location, complete the following tasks:

1. In Administrator, edit the elastic runtime environment.
2. Add the following row to the **Data Storage** table:
   - **Type:** Select **EFS**.
   - **File System ID:** Enter the EFS file system ID.
   - **Access Point ID:** Enter the access point ID for `/ca_region`.
   - **Source File Path:** orders
   - **Projected File Path:** mnt/legacy/sales

The elastic runtime environment performs the following actions:

- Reads and writes files at `/ca_region/orders` on the EFS.
- Processes files at `/mnt/legacy/sales` on the Kubernetes Pods.

Because the elastic runtime environment projects flat files from `/ca_region/orders` to `/mnt/legacy/sales`, it allows existing connections and tasks to continue running without modification.

Working with elastic runtime environments

After you create an elastic runtime environment, you can edit, monitor, and delete it. You can also view dependencies and change permissions.

You can perform the following tasks:

Change permissions.

To change permissions for an elastic runtime environment, expand the **Actions** menu and select **Permissions**. You can define permissions for an elastic runtime environment for each user group in your organization.

The following table describes the permissions that you can set:

Permission	Description
Read	View details about the elastic runtime environment and use the elastic runtime environment in a task.
Update	Edit the elastic runtime environment.
Delete	Delete the elastic runtime environment.

Permission	Description
Execute	Run the elastic runtime environment.
Change Permissions	Change permissions for the elastic runtime environment.

Change the name or description of the environment.

To change the name or description of your elastic runtime environment, expand the **Actions** menu and select **Edit Environment Properties**.

Configure Secure Agent services.

You can configure Secure Agent services or change service properties to optimize performance or if you are instructed to do so by Informatica Global Customer Support.

Modify Secure Agent service properties on the **Service Configuration** tab or add custom properties for Secure Agent services. Select **Reset** to reset the Secure Agent service properties to their system defaults.

For more information about the Secure Agent services and their properties, see *Secure Agent Services*.

Delete the environment.

You can delete an elastic runtime environment if it has no running instances.

To clean up instances, run the cluster uninstaller. For more information, see "Step 2. Clean up resources" on page 49.

To delete an elastic runtime environment, expand the **Actions** menu and select **Delete**.

Enable or disable services and connectors.

You can enable a service to allow the elastic runtime environment to run product features associated with the service. You can enable specific connectors to allow the environment to communicate with cloud and on-premise applications, platforms, databases, and flat files.

To enable services and connectors, expand the **Actions** menu and select **Enable or Disable Services, Connectors**.

Monitor and troubleshoot cluster upgrades.

Occasionally, the Kubernetes cluster in the elastic runtime environment needs to upgrade. For example, the cluster upgrades to use a new Kubernetes version, to use a new AMI, or to patch a vulnerability. If the cluster upgrades, you'll get a notification in IDMC.

During the cluster upgrade, master nodes upgrade all at once and worker nodes upgrade one at a time so that jobs can continue running. You can't submit new jobs while the master nodes are upgrading, and you can't connect to a node while it's upgrading.

When a node upgrades, a new node instance starts up using a different IP address that you can view in the AWS Management Console. Make sure to pick up the new IP address if you need to reconnect to the cluster using third-party tools.

To troubleshoot an upgrade failure, you can drill down on the elastic runtime environment and open the **Monitor** tab to view the audit log. The audit log shows each step of the upgrade process.

Monitor the audit log, nodes, and jobs.

To monitor an elastic runtime environment, drill down on the elastic runtime environment and open the **Monitor** tab. You can view the following details:

- Audit log that displays events that occur on the Secure Agent services running in the elastic runtime environment.
- Lifecycle graph that shows the number of nodes in the elastic runtime environment over time.
- A list of all jobs that were submitted to the elastic runtime environment. You can stop and restart jobs, and you can download log files.

Update the environment configuration.

To update the environment properties, drill down on the elastic runtime environment and edit the properties on the **Environment Configuration** tab.

After you update the property values and save the configuration, it takes some time for the changes to take effect in the deployed elastic runtime environment. To verify certain changes, you can view the lifecycle graph in Administrator or view the logs on your AWS account.

View dependencies.

View dependencies to find the tasks or connections that are using the elastic runtime environment or to find the objects that are used by the elastic runtime environment. To view dependencies, expand the **Actions** menu and select **Show Dependencies**.

View images.

An elastic runtime environment uses images to start up IDMC and Secure Agent services.

Informatica allows you to access its image repository so that you can download the images and scan them to ensure that they meet your enterprise-level security and compliance needs. Informatica notifies you when a new image is available so that you can scan the image before the deployment window for the next release.

To view the images that an elastic runtime environment uses, use one of the following options:

- Drill down on the elastic runtime environment and open the **Images** tab. Each image includes the image name, image tag, and repository path in Informatica's image repository.
- Use the REST API to get a token and pull the images from Informatica's image repository.

Connectors in an elastic runtime environment

You can use connectors in tasks that run in an elastic runtime environment.

An elastic runtime environment can run tasks that use the following connectors:

- ActiveCampaign
- Adobe Analytics
- Amazon Athena
- Amazon Redshift V2
- Amazon S3 V2
- Amplitude
- Ariba V2
- BigMachines
- Box
- Concur V2
- Coupa V2
- Databricks
- Eloqua Bulk API
- Eloqua REST
- File Processor
- FileIo
- Hadoop Files V2
- IBM MQ
- JDBC V2
- JIRA
- JIRA Cloud
- JMS
- Kafka
- Large Language Model
- LDAP
- Magento V1
- Mailchimp
- Marketo V3
- Microsoft Dynamics 365 for Operations
- Microsoft Dynamics 365 for Sales
- Microsoft Excel
- Microsoft SharePoint
- Microsoft SharePoint Online
- Microsoft SQL Server*
- Mixpanel
- MongoDB V2
- MySQL
- NetSuite
- NetSuite RESTlet V2
- OData
- OData Consumer
- OData V2 Protocol Reader
- OData V2 Protocol Writer
- ODBC
- Oracle*
- Oracle Autonomous Database
- Oracle Business Intelligence Publisher V1
- Oracle Cloud Object Storage
- Oracle CRM Cloud V1
- Oracle Financials Cloud V1
- Oracle HCM Cloud V1
- PostgreSQL*
- Quickbooks V2
- REST V2
- Salesforce
- Salesforce Analytics
- Salesforce Data Cloud
- Salesforce Marketing Cloud
- Salesforce Pardot
- SAP ADSO Writer
- SAP BAPI
- SAP BW BEx Query
- SAP HANA
- SAP OData V2
- SAP OData V4
- SAP ODP Extractor
- ServiceNow
- Shopify
- Snowflake Data Cloud
- SuccessFactors ODATA
- SurveyMonkey
- Tableau V2
- Tableau V3
- Veeva Vault
- Web Service Consumer
- Workday V2
- Xero
- Zendesk V2
- Zuora AQuA

*Applicable only if deployed on Cloud instances.

Downloading images using the REST API

Download images from Informatica's image repository using the REST API so that you can scan them for vulnerabilities according to your compliance needs.

To download images, you need the "Elastic runtime environment - manage tokens" privilege in your organization. For more information about feature privileges, see *User Administration*. To run Docker commands, install Docker on the machine.

1. Use the platform REST API version 3 containerimagetoken resource to get a token for the image repository.

   You can create a token or use an existing token for your organization. For more information, see Getting a new token or Getting all tokens for an organization, respectively.

2. Use the platform REST API version 3 ais resource to get image information for the elastic runtime environment.

   For more information, see Getting runtime environment image information.

3. On the machine that you use to scan images, run the following Docker command to connect to the image repository:

   ```
   docker login infacloud-k8s-agent-docker-dev.jfrog.io --username <Informatica
   organization ID> --password <token value>
   ```

4. Run a Docker command to download the image.

   For example, the following command downloads version 64.1:

   ```
   docker pull infacloud-k8sagent-docker-dev-jfrog.io/Data Integration:64.1
   ```

IAM policy reference

The policy reference describes the statements in the cluster installer and worker policies so that you can understand the level of access that the cluster installer and worker roles have in your AWS environment.

Cluster installer policy statements

Use the statement descriptions in the cluster installer policy to understand the level of access that the cluster installer has in your AWS environment. The cluster installer policy includes the permissions that the cluster installer needs to deploy an elastic runtime environment.

Auto Scaling

The following statement allows the role to manage Auto Scaling groups:

```
{
    "Sid": "VisualEditor0",
    "Effect": "Allow",
    "Action": [
        "autoscaling:CreateAutoScalingGroup",
        "autoscaling:DeleteAutoScalingGroup",
        "autoscaling:TerminateInstanceInAutoScalingGroup",
        "autoscaling:DescribeAutoScalingGroups",
        "autoscaling:DescribeScalingActivities",
        "autoscaling:DescribeTags",
        "autoscaling:UpdateAutoScalingGroup",
```

```
                "autoscaling:CreateOrUpdateTags",
                "autoscaling:DeleteTags",
                "autoscaling:DescribeAutoScalingInstances",
                "autoscaling:DescribeLaunchConfigurations",
                "autoscaling:DescribeLifecycleHooks",
                "autoscaling:ResumeProcesses",
                "autoscaling:SetDesiredCapacity",
                "autoscaling:SuspendProcesses"
            ],
            "Resource": "*"
        },
```

Describing EC2 resources and managing security groups

The following statement allows the role to describe and update attributes of EC2 resources and authorize security group ingress:

```
        {
            "Sid": "VisualEditor1",
            "Effect": "Allow",
            "Action": [
                "ec2:DescribeAvailabilityZones",
                "ec2:DescribeImages",
                "ec2:DescribeInstanceAttribute",
                "ec2:DescribeInstances",
                "ec2:DescribeInstanceTypes",
                "ec2:DescribeSecurityGroups",
                "ec2:DescribeLaunchTemplateVersions",
                "ec2:DescribeLaunchTemplates",
                "ec2:DescribeTags",
                "ec2:AuthorizeSecurityGroupIngress",
                "ec2:DeleteLaunchTemplate",
                "ec2:ModifyInstanceAttribute",
                "ec2:DescribeVpcs"
            ],
            "Resource": "*"
        },
```

Running EC2 instances

The following statement allows the role to run EC2 instances:

```
        {
            "Sid": "VisualEditor2",
            "Effect": "Allow",
            "Action": "ec2:RunInstances",
            "Resource": "*"
        },
```

Creating keys, tags, and launch templates

The following statement allows the role to create keys, tags, and launch templates:

```
        {
            "Sid": "VisualEditor3",
            "Effect": "Allow",
            "Action": [
                "ec2:CreateKeyPair",
                "ec2:CreateTags",
                "ec2:CreateLaunchTemplate"
            ],
            "Resource": [
                "arn:aws:ec2:*:<AWS account ID>:security-group/*",
                "arn:aws:ec2:*:<AWS account ID>:network-interface/*",
                "arn:aws:ec2:*:<AWS account ID>:launch-template/*",
                "arn:aws:ec2:*:<AWS account ID>:instance/*",
                "arn:aws:ec2:*:<AWS account ID>:subnet/*",
                "arn:aws:ec2:*:<AWS account ID>:volume/*",
                "arn:aws:ec2::*::image/*",
                "arn:aws:ec2:*:<AWS account ID>:key-pair/*"
```

```
        ]
    },
```

Tagging EC2 resources

The following statement provides conditional permissions to create tags on EC2 resources:

```
    {
        "Sid": "VisualEditor5",
        "Effect": "Allow",
        "Action": "ec2:CreateTags",
        "Resource": [
            "arn:aws:ec2:*:<AWS account ID>:network-interface/*",
            "arn:aws:ec2:*:<AWS account ID>:key-pair/*",
            "arn:aws:ec2:*:<AWS account ID>:launch-template/*",
            "arn:aws:ec2:*:<AWS account ID>:instance/*",
            "arn:aws:ec2:*:<AWS account ID>:volume/*",
            "arn:aws:ec2:*:<AWS account ID>:subnet/*"
        ],
        "Condition": {
            "StringEquals": {
                "ec2:CreateAction": [
                    "RunInstances",
                    "CreateKeyPair",
                    "CreateLaunchTemplate",
                    "CreateLaunchTemplateVersion",
                    "network-interface",
                    "CreateTags",
                    "CreateAutoScalingGroup"
                ]
            }
        }
    },
```

Passing role information to AWS services

The following statement allows the cluster installer to pass role details to AWS services so they can assume the permissions:

```
    {
        "Sid": "VisualEditor13",
        "Effect": "Allow",
        "Action": "iam:PassRole",
        "Resource": [
            "arn:aws:iam::<AWS account ID>:role/<cluster installer role>",
            "arn:aws:iam::<AWS account ID>:role/<worker role>"
        ]
    },
```

AWS Secrets Manager

The following statement allows the role to manage secrets in AWS Secrets Manager:

```
    {
        "Sid": "VisualEditor21",
        "Effect": "Allow",
        "Action": [
            "secretsmanager:DescribeSecret",
            "secretsmanager:GetSecretValue",
            "secretsmanager:ListSecrets",
            "secretsmanager:CreateSecret",
            "secretsmanager:TagResource",
            "secretsmanager:DeleteSecret",
            "secretsmanager:UpdateSecret"
        ],
        "Resource": "arn:aws:secretsmanager:*:<AWS account ID>:secret:*"
    },
```

Managing EFS file systems

The following statement allows the role to manage EFS file systems:

```
{
    "Sid": "VisualEditor10",
    "Effect": "Allow",
    "Action": [
        "elasticfilesystem:CreateMountTarget",
        "elasticfilesystem:DescribeAccessPoints",
        "elasticfilesystem:DescribeFileSystems",
        "elasticfilesystem:DescribeMountTargets"
    ],
    "Resource": "*"
},
```

Describing subnets

The following statement provides read-only access to describe subnets:

```
{
    "Sid": "VisualEditor11",
    "Effect": "Allow",
    "Action": [
        "ec2:DescribeSubnets"
    ],
    "Resource": "*"
},
```

EC2 management

The following statement provides EC2 management permissions:

```
{
    "Sid": "EC2Management",
    "Effect": "Allow",
    "Action": [
        "ec2:AssociateIamInstanceProfile",
        "ec2:CreateKeyPair",
        "ec2:CreateLaunchTemplate",
        "ec2:CreateLaunchTemplateVersion",
        "ec2:CreateTags",
        "ec2:DeleteLaunchTemplate",
        "ec2:DeleteTags",
        "ec2:ModifyLaunchTemplate",
        "ec2:RunInstances",
        "ec2:TerminateInstances",
        "iam:PassRole"
    ],
    "Resource": "*"
}
```

Worker policy statements

Use the statement descriptions in the worker policy to understand the level of access that the worker nodes have in your AWS environment. The worker policy includes the permissions that worker nodes need to process data in an elastic runtime environment.

Describing resources

The following statement provides read-only resource descriptions:

```
{
    "Sid": "DescribeActions",
    "Effect": "Allow",
    "Action": [
        "elasticfilesystem:DescribeMountTargets",
        "autoscaling:DescribeAutoScalingGroups",
        "autoscaling:DescribeAutoScalingInstances",
```

```
                "autoscaling:DescribeLaunchConfigurations",
                "autoscaling:DescribeTags",
                "ec2:DescribeInstances",
                "ec2:DescribeLaunchTemplateVersions",
                "ec2:DescribeTags",
                "autoscaling:DescribeLifecycleHooks",
                "autoscaling:DescribeScalingActivities",
                "ec2:DescribeLaunchTemplates",
                "ec2:DescribeInstanceTypes",
                "ec2:DescribeVpcs",
                "secretsmanager:DescribeSecret",
                "secretsmanager:ListSecrets"
            ],
            "Resource": "*"
        },
```

Auto Scaling

The following statement allows the role to manage Auto Scaling groups:

```
        {
            "Sid": "AutoscalingCapacity",
            "Effect": "Allow",
            "Action": [
                "autoscaling:SetDesiredCapacity",
                "autoscaling:TerminateInstanceInAutoScalingGroup",
                "autoscaling:CreateAutoScalingGroup",
                "autoscaling:CreateOrUpdateTags",
                "autoscaling:DeleteAutoScalingGroup",
                "autoscaling:DeleteTags",
                "autoscaling:SuspendProcesses",
                "autoscaling:UpdateAutoScalingGroup"
            ],
            "Resource": "*"
        },
```

Amazon EC2

The following statement allows the role to manage EC2 resources:

```
        {
            "Sid": "EC2Management",
            "Effect": "Allow",
            "Action": [
                "ec2:AssociateIamInstanceProfile",
                "ec2:CreateKeyPair",
                "ec2:CreateLaunchTemplate",
                "ec2:CreateLaunchTemplateVersion",
                "ec2:CreateTags",
                "ec2:DeleteLaunchTemplate",
                "ec2:DeleteTags",
                "ec2:ModifyLaunchTemplate",
                "ec2:RunInstances",
                "ec2:TerminateInstances",
                "iam:PassRole"
            ],
            "Resource": "*"
        },
```

Amazon EFS

The following statement allows the role to create and delete access points on EFS file systems:

```
        {
            "Sid": "EFSManagement",
            "Effect": "Allow",
            "Action": [
                "elasticfilesystem:CreateAccessPoint",
                "elasticfilesystem:DeleteAccessPoint"
            ],
```

```
        "Resource": "*"
    },
```

AWS Secrets Manager

The following statement allows the role to access and update secrets in AWS Secrets Manager:

```
    {
        "Sid": "SecretsManagerAccess",
        "Effect": "Allow",
        "Action": [
            "secretsmanager:GetSecretValue",
            "secretsmanager:UpdateSecret"
        ],
        "Resource": "*"
    }
```

CHAPTER 5

Serverless runtime environments

A serverless runtime environment is an advanced serverless deployment solution that doesn't require downloading, installing, configuring, or maintaining a Secure Agent or Secure Agent group. You can use a serverless runtime environment in the same way that you use a runtime environment when you configure a connection or tasks.

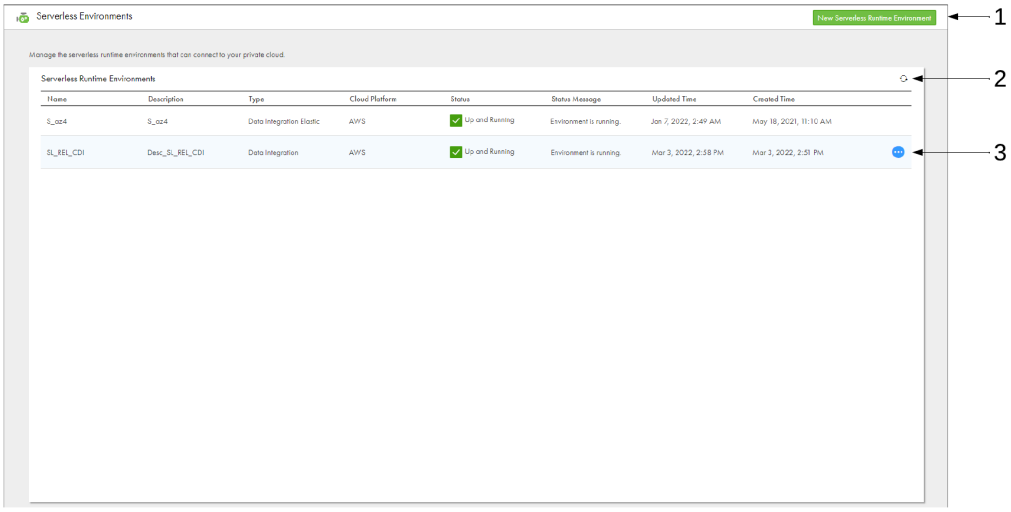
A serverless runtime environment is an advanced serverless deployment solution that uses an isolated, single-tenant model, unlike the multi-tenant model on the Hosted Agent. The single-tenant model provides a dedicated server with virtual machine resources to run tasks for your organization. The serverless runtime environment auto-scales with the size of the workload while your data remains in your cloud environment.

When you use a serverless runtime environment to run mappings in advanced mode, the advanced serverless deployment fulfills the prerequisites to create an advanced cluster and to run jobs on the cluster. The serverless runtime environment manages the advanced cluster while the cluster adapts to workload changes by provisioning and de-provisioning resources. Worker nodes in the advanced cluster are highly available. High availability mitigates job failures and maintains job performance when a worker node crashes.

Serverless runtime environments are available in AWS and Microsoft Azure environments. A serverless runtime environment supports local regions in each geo-location. For example, an AWS cloud platform in the United States (US) supports all US regions and an AWS cloud platform in Asia-Pacific (APAC) supports all APAC regions.

You can view serverless runtime environments on the **Serverless Environments** page. To view environments, you must have at least read permissions for Secure Agent groups. For more information, see "Working with Secure Agent groups" on page 16.

The following image shows the **Serverless Environments** page:



1. Option to create a serverless runtime environment
2. Refresh icon
3. Actions menu

You can create a maximum of 10 serverless runtime environments in your organization. If you have a trial license, you can create a maximum of two environments.

Serverless setup using AWS PrivateLink for service network endpoints

A serverless runtime environment using AWS PrivateLink for service network endpoints uses a resource share to connect to data sources through your AWS account. Then, it runs tasks using the infrastructure in Informatica's AWS account.

Starting in the October 2025 release, serverless runtime environments in new IDMC organizations use AWS PrivateLink. Existing organizations can continue to create serverless runtime environments using a cross-account elastic network interface. AWS continues to support cross-account elastic network interfaces but they're no longer on AWS's roadmap, so Informatica recommends all organizations transition to AWS PrivateLink going forward.

To create a serverless runtime environment using AWS PrivateLink for service network endpoints, complete the following tasks:

1. Set up your AWS environment to create and configure the required AWS resources
2. In Administrator, create a serverless runtime environment to deploy it to AWS.

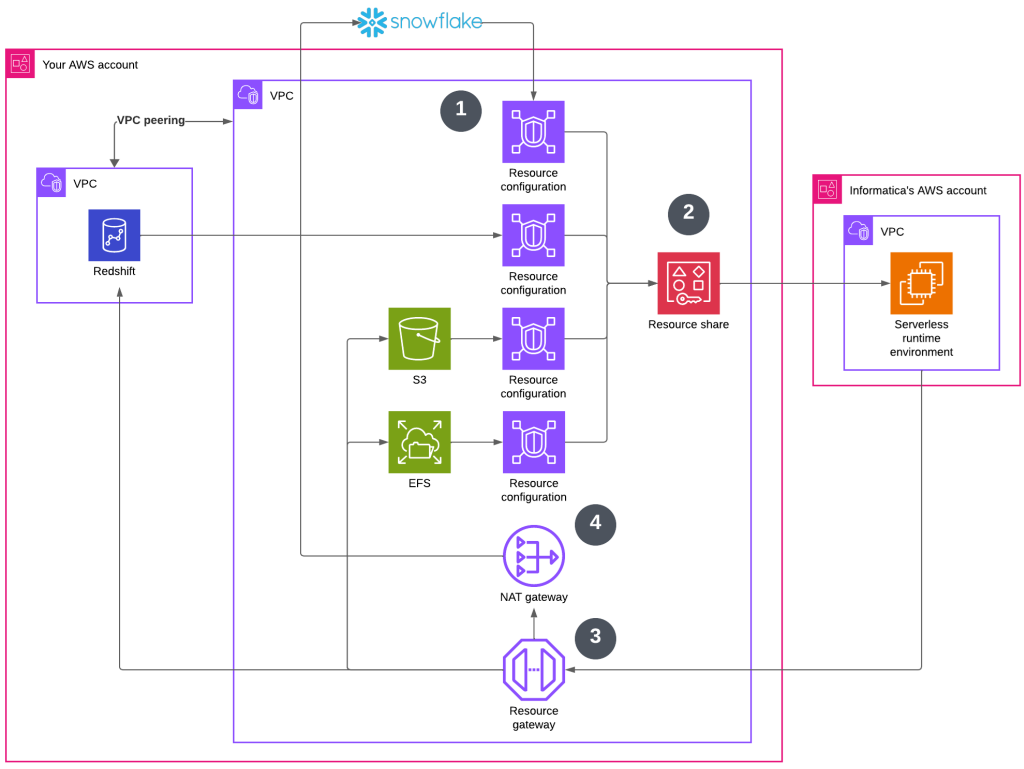
Set up the AWS environment

Set up your AWS environment to create and configure the required AWS resources before creating a serverless runtime environment.

To set up the AWS environment, complete the following tasks:

1. Create and configure AWS resources to connect to the serverless runtime environment.
2. Verify that your VPC can connect to all data sources in the resource share, including private resources such as Amazon S3 and public resources such as Snowflake.
3. Create an IAM role that the serverless runtime environment can use to access the supplementary file location.
4. Create a system disk to improve mapping performance in Data Integration.
5. Create a data disk if you have files on Amazon EFS or an NFS file server that you want to use in tasks.
6. If tasks use JAR files and external libraries, dedicate a location on Amazon S3 to store the files and create folders for each file type.
7. If your organization allows only trusted IP addresses, add the IP addresses for the Informatica NAT gateway to the trusted list.

The following image shows how you share data sources and the serverless runtime environment connects to them:



1. You add each data source to a resource configuration.
2. You create a resource share to share all resource configurations.
3. Informatica uses a resource gateway to connect to data sources like S3, EFS, and Redshift.
4. Informatica uses a NAT gateway to connect to data sources like Snowflake.

Step 1. Create AWS resources

Create and configure AWS resources to connect to the serverless runtime environment.

Create and configure the following AWS resources:

VPC

A VPC is a virtual network that contains other AWS resources. You can use an existing VPC or create a VPC using the following guidelines:

- Use IPv4 CIDR manual input and enter the CIDR block for the VPC to use.
- Make sure that enableDnsSupport and enableDnsHostnames are set to true.

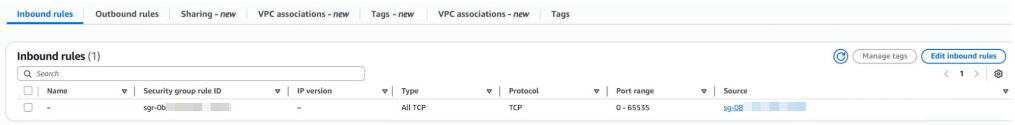
Subnet

A subnet in the VPC connects to the data sources. You can use an existing public or private subnet, or you can create a subnet using the following guidelines:

- Use any availability zone in the region where you created the VPC.
- To set the IPv4 VPC CIDR block, use the same IPv4 CIDR block that you specified when you created the VPC.
- To set the IPv4 subnet CIDR block, use a CIDR range that's within the VPC CIDR range. For example, if you use 10.1.0.0/16 for the VPC, you might use 10.1.240.0/20 for the subnet.

Security group for the resource gateway

The resource gateway uses a security group to allow all inbound traffic from the same security group. The following image shows the inbound rule in the security group:



Note: You use this security group to create the serverless runtime environment in Administrator.

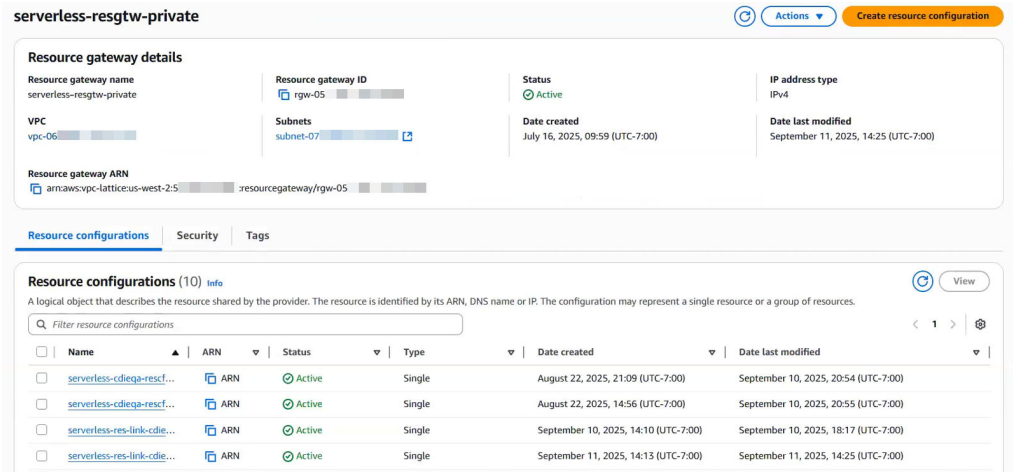
Resource gateway

A resource gateway in Amazon VPC Lattice is an entry point into the VPC that allows access to data sources. It routes traffic to the data sources that you want the serverless runtime environment to access.

To create the resource gateway, use the following guidelines:

- Use your existing VPC or the VPC that you created.
- Select an existing subnet or the subnet that you created.
- Use the same availability zone that the subnet uses.
- Use the security group that you created for the resource gateway.

The following image shows an example of a resource gateway:



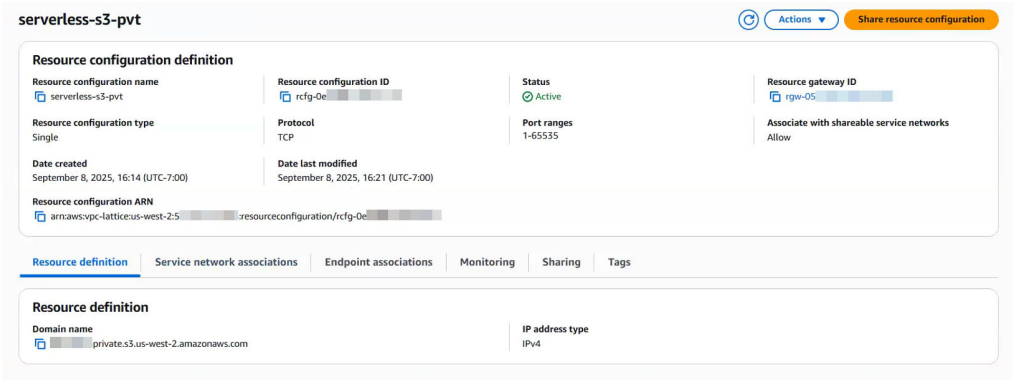
Resource configuration

A resource configuration in Amazon VPC Lattice is a logical representation of a data source like an Amazon S3 bucket or Snowflake database. The serverless runtime environment first connects to the resource configuration to connect to the data source.

Create a resource configuration for each data source that you want the serverless runtime environment to access. To create a resource configuration, use the following guidelines:

- Use the **Resource** configuration type and set it to **Single** to create the resource configuration for a single resource.
- Use the resource gateway that you created.
- For data sources that aren't Amazon EFS, use **DNS resource** as the resource definition and specify the public DNS name, such as `serverless-private.s3.us-west-2.amazonaws.com` or `serverless.snowflakecomputing.com`.
- For an Amazon EFS data source, use **IP resource** as the resource definition and use **IPv4** as the IP address type. Then, specify the private IP address.

The following image shows an example of a resource configuration:



Resource share

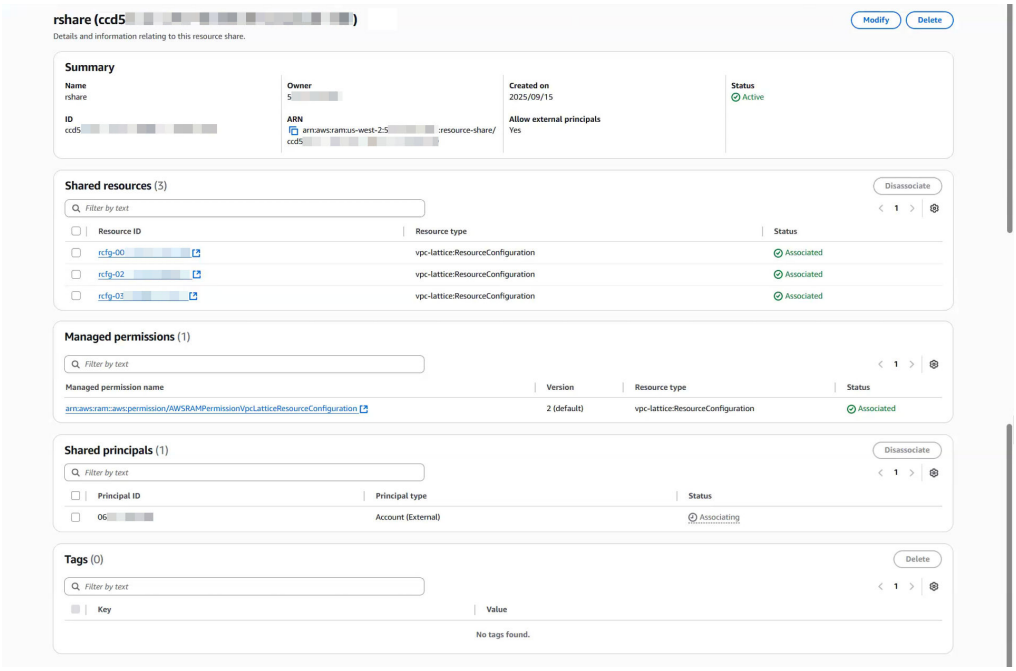
A resource share in AWS Resource Access Manager is a logical group that allows you to share resource configurations with Informatica.

To create the resource share, use the following guidelines:

- To specify the resource share details, complete the following tasks:
  - Use **VPC Lattice Resource Configurations** as the resource type.
  - Select all the resource configurations that you want to add to the resource share.

    Note: Make sure you add all of the resource configurations as part of this step. After you create a serverless runtime environment in Administrator, you can update the resource share only by creating a new serverless runtime environment.
- To grant access to Informatica as a principal, complete the following tasks:
  - Allow sharing with anyone.
  - Use the **AWS account** principal type and add Informatica's AWS account number as a principal. To find Informatica's AWS account number, create a serverless runtime environment in Administrator and copy the number from the **Informatica Account Number** property.

The following image shows an example of a resource share:



After you create the resource share, AWS sends a resource share invite to Informatica and the status of the resource share is **Associating**. When you create the serverless runtime environment, the environment accepts the resource share and the status changes to **Associated**.

If you don't create the serverless runtime environment within a certain amount of time, the invite expires. You'll need to edit the resource share and re-add Informatica as a principal so that AWS sends a new resource share invite.

Step 2. Verify resource connectivity in your AWS account

Verify that your VPC can connect to all data sources in the resource share, including private resources such as Amazon S3 and public resources such as Snowflake.

For example, you might need to create an S3 gateway endpoint to connect to a private S3 resource, configure VPC peering to connect to a Redshift cluster, or create a NAT gateway to connect to a Snowflake database. You might need to update the inbound rules for your security groups or update route tables.

Step 3. Create an IAM role for the supplementary file location

Create an IAM role that the serverless runtime environment can use to access the supplementary file location.

Note: If you don't use a supplementary file location, skip this step.

1. In AWS, create a policy that includes the following permissions:

```
"Version": "2012-10-17",
"Statement": [
  {
    "Effect": "Allow",
    "Action": [
      "s3:PutObject",
      "s3:GetObject",
      "s3:DeleteObject",
      "s3:ListBucket",
      "s3:ListBucketMultipartUploads"
    ],
    "Resource": [
      "arn:aws:s3:::dev",
      "arn:aws:s3:::dev/*"
    ]
  }
]
```

2. Create a role.
   a. Set the trusted entity to **AWS service** and the use case to **Amazon EC2**.
   b. Attach the policy from step 1 to the role.
   c. Edit the trust relationship of the role and specify the following policy:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Principal": {
        "AWS": [
          "arn:aws:iam::<your account ID>:root"
        ]
      },
      "Action": "sts:AssumeRole",
      "Condition": {
        "StringEquals": {
          "sts:ExternalId":"<external ID>"
        }
      }
    }
  ]
}
```

Note: You enter the same role name and external ID when you create the serverless runtime environment in Administrator.

Step 4. Create a system disk

Create a system disk to improve mapping performance in Data Integration.

You can create a system disk using Amazon EFS or an NFS. File system connections in Amazon EFS use TLS by default. File system connections in NFS use NFSv4.

When you use a system disk, the serverless runtime environment creates the following directory on the system disk to store job metadata and logs:

```
<organization ID>/<serverless runtime environment ID>
```

Creating a system disk on Amazon EFS

To create a system disk on Amazon EFS, use the following guidelines:

- Create any folder required by an access point before creating the access point itself. For example, if the access point refers to the folder `/my-company/dev`, then define this folder first before you set up the access point.
- Configure connectivity between the subnet in the serverless runtime environment and the EFS file system.
- Configure the EFS security group to allow inbound access from the security group of the serverless runtime environment.
- Configure an Amazon EFS file system policy to grant access to the serverless role. Use the same IAM role that you created in "Step 3. Create an IAM role for the supplementary file location" on page 71.
  For example, the following file system policy allows root access to the IAM role for file system fs-12345 and allows SecureTransport only:

```
"Version": "2012-10-17",
    "Id": "efs-policy-wizard-<EFS policy wizard ID>",
    "Statement": [
        {
            "Sid": "efs-statement-<EFS statement ID>",
            "Effect": "Allow",
            "Principal": {
                "AWS": "arn:aws:iam::<your account ID>:role/<IAM role for
supplementary file location>"
            },
            "Action": [
                "elasticfilesystem:ClientWrite",
                "elasticfilesystem:ClientMount",
                "elasticfilesystem:ClientRootAccess"
            ],
            "Resource": "arn:aws:elasticfilesystem:us-west-2:<your account ID>:file-
system/fs-12345",
            "Condition": {
                "Bool": {
                    "elasticfilesystem:AccessedViaMountTarget": "true"
                }
            }
        },
        {
            "Sid": "efs-statement-<EFS statement ID>",
            "Effect": "Deny",
            "Principal": {
                "AWS": "*"
            },
            "Action": "*",
            "Resource": "arn:aws:elasticfilesystem:us-west-2:123456789:file-system/
fs-12345",
            "Condition": {
                "Bool": {
                    "aws:SecureTransport": "false"
                }
            }
        }
    ]
}
```

The following table describes the actions in the policy:

Action	Description
elasticfilesystem:ClientMount	Provides read-only access to a file system.
elasticfilesystem:ClientWrite	Provides write permissions on a file system.
elasticfilesystem:ClientRootAccess	Provides use of the root user when accessing a file system.

- Configure the IAM role to restrict access to specific access points on the file system. For more information, see the AWS documentation.

Creating a system disk using an NFS

To create a system disk using an NFS, use the following guidelines:

- Set the file system to the DNS of the NFS server.
- Configure connectivity between the subnet in the serverless runtime environment and the NFS file server.
- Configure the NFS security group to allow inbound access from the security group of the serverless runtime environment.

Step 5. Create a data disk

Create a data disk if you have files on Amazon EFS or an NFS file server that you want to use in tasks.

You can store the following types of files on a data disk:

- Flat files for a flat file connection in tasks
- Parameter files for mapping tasks

1. Mount the Amazon EFS or NFS directory on an Amazon EC2 instance.
2. Log in to the EC2 instance.
3. Search for a user with ID 501. If the user doesn't exist, create a user with this ID.

   User ID 501 corresponds to the user cldagnt. The serverless runtime environment uses this user to access mounted EFS or NFS directories.
4. Assign read and write permissions for the mounted directories to the user with ID 501.

For more information about creating a data disk using NFS, see Network File System (NFS) level users, groups, and permissions in the AWS documentation.

Step 6. Create a supplementary file location

If tasks use JAR files and external libraries, dedicate a location on Amazon S3 to store the files and create folders for each file type.

To create the supplementary file location, use the following guidelines:

1. Create the following file structure on Amazon S3:

```
<Supplementary file location>
├── ext
├── odbc
│   └── lib
└── serverless_agent_config
    ├── jars
```

```
├── SSL
├── j_depends
└── py_depends
```

2. Create a serverlessUserAgentConfig.yml file. For more information, see "Configure the serverlessUserAgentConfig.yml file" on page 100.
3. Add the serverlessUserAgentConfig.yml file directly under the `serverless_agent_config` directory.

The following table lists the file types that you can store in each location:

Location	Files
<Supplementary file location>/ext	JDBC JAR files
<Supplementary file location>/odbc	The following files: - odbc.ini - odbcinst.ini - exports.ini
<Supplementary file location>/odbc/lib	ODBC shared libraries for a Linux operating system
<Supplementary file location>/serverless_agent_config	The following files: - serverlessUserAgentConfig.yml - JDBC V2 Connector JAR files - REST V3 Connector truststore and keystore certificates - JAR files for the Java transformation - Installation and resource files for the Python transformation You can customize the directory structure under the serverless_agent_config folder and specify the relative path to each file in the serverlessUserAgentConfig.yml file.

Step 7. Add trusted Informatica IP addresses

If your organization allows only trusted IP addresses, add the IP addresses for the Informatica NAT gateway to the trusted list.

For information about adding trusted IP addresses, see *Organization Administration*.

The following table lists the IP addresses for the Informatica NAT gateway to allow for each POD:

POD	Trusted IP addresses
NA West 1, NA East 2, US West 3, US East 4, US West 5, US East 6	44.242.20.143 44.239.8.148 54.221.247.69 54.160.9.90 3.131.176.232 18.142.14.24 3.1.147.184 18.220.76.98 13.56.74.27 52.52.220.198
Canada Central 1	3.97.103.68 3.96.182.201

POD	Trusted IP addresses
EM West 1 (Ireland)	3.64.66.226 3.125.185.124 54.76.54.130 54.78.183.88 35.176.60.118 18.135.50.152 35.152.49.63 35.152.45.151 15.237.157.126 15.237.97.211 13.49.61.89 13.53.141.231
UK	34.250.251.16 18.170.170.192 13.37.37.71 18.157.124.91 13.53.147.238 15.161.184.93 15.160.41.209
AP Southeast 1 (Australia)	54.253.179.190 3.24.111.61 35.72.149.44 13.112.143.134 3.34.56.248 52.79.244.47 52.76.184.230 18.140.193.120 65.1.80.5 13.234.141.216 18.163.244.73 18.167.71.151
AP Northeast 2 (Japan)	18.181.99.0 13.208.101.21

Create a serverless runtime environment

In Administrator, create a serverless runtime environment to deploy it to AWS.

1. In Administrator, open the **Serverless Environments** page.
2. Select **New Serverless Runtime Environment**.
3. Select **AWS PrivateLink for service network endpoints**.
4. Configure the serverless runtime environment properties.

   For information about how to configure the properties, see "Serverless runtime environment properties" on page 76.
5. To save and deploy the serverless runtime environment, select **Save**.

It takes at least five minutes for the serverless runtime environment to become available. Use the **Serverless Environments** page to track the status of the environment and review any status messages.

Serverless runtime environment properties

Configure the serverless runtime environment properties so that IDMC can accept the resource share and connect to the data sources in your AWS account.

Basic configuration

The following table describes the basic properties:

Property	Description
Name	Name of the serverless runtime environment.
Description	Description of the serverless runtime environment.
Task Type	Type of tasks that run in the serverless runtime environment. - Select **Data Integration** to run mappings outside of advanced mode. - Select **Advanced Data Integration** to run mappings in advanced mode.
Cloud Platform	Cloud platform to host the serverless runtime environment. Use Amazon Web Services (AWS).
Max Compute Units Per Task	Maximum number of serverless compute units corresponding to CPU and memory that a task can use. The property value configured in the serverless runtime environment specifies the maximum number of serverless compute units that each task can request from the environment. When you create a mapping task, you can override the maximum number of compute units that the task can request. In Monitor, you can view the number of compute units that the task requested and consumed. For information about metering, see *Organization Administration*.
Task Timeout	Amount of time in minutes to wait for a task to complete before it is terminated. The timeout ensures that serverless compute units are not unproductive when a task hangs. By default, the timeout is 2880 minutes (48 hours). You can set the timeout to a value that is less than 2880 minutes.
Informatica Account Number	Informatica's account number on the cloud platform where the serverless runtime environment will be created. The account number is populated automatically.

Platform configuration

The following table describes the platform configuration properties:

Property	Description
Configuration Name	Name of the serverless configuration.
Configuration Description	Description of the serverless configuration. The description can be up to 256 characters and can contain alphanumeric characters and the following special characters: `._-:/()#,@[]+=&;{}!$"*`
Account Number	Your account number on the cloud platform.
Resource Share ID	ID of the resource share.

Property	Description
Region	Region of your cloud environment. The sources and targets that you use in mappings must either reside in or be accessible from this region.
AZ ID	ID of the availability zone. The sources and targets that you use in mappings must either reside or be accessible from the availability zone.
VPC ID	ID of the VPC. The VPC must be configured with an endpoint to access the sources and targets that you use in mappings. For example, `vpc-2f09a348`
Subnet ID	ID of the subnet within the VPC. The subnet must be have an entry point to access the sources and targets that you use in mappings. For example, `subnet-b46032ec`
Security Group ID	ID of the security group for the resource gateway. For example, `sg-e1fb8c9a`
Supplementary File Location	Location on Amazon S3 to store supplementary files, such as JAR files and external libraries for certain transformations and connectors. Use the format: `s3://<bucket name>/<folder name>` Place script files in a folder named `command_scripts`. The folder can have subfolders. If you update files in the `command_scripts` folder, IDMC automatically uploads them to the Secure Agent.
Role Name	Name of the role that the serverless runtime environment can use to access the supplementary file location.
External ID	External ID to associate with the role for the supplementary file location. You can use the generated external ID or specify your own external ID.
Advanced Custom Properties	Domain map of all resource configurations in the resource share. Use the format `<resource configuration ID>:<public DNS name>` and separate each resource configuration using a comma. For example: `ResourceConfigDomainNameMap=rcfg-0123456abcdef:informatica-private.s3.us-west-2.amazonaws.com,rcfg-6543210fedcba:serverless-redshift-cluster-1.abcdef0123456.us-west-2.redshift.amazonaws.com,rcfg-fedcba6543210:serverless-postgresql-db.cluster-0a1b2c3d4e5f6.us-west-2.rds.amazonaws.com,rcfg-6f5e4d3c2b1a0:serverless-private.s3-us-west-2.amazonaws.com` **Note:** If the resource share includes a resource configuration for an Amazon EFS data source, use the DNS name to configure this domain map even though the resource configuration on AWS uses the IP address.

System disk

The following table describes the system disk properties:

Property	Description
Type	System disk type, either EFS or NFS.
File System	For EFS disks, the file system is the file system ID of the EFS disk. For NFS disks, the file system is the DNS of the file system.

Property	Description
Source Mount	File system path to be mounted in the serverless runtime environment.
Access Point	ID of the Amazon EFS file system access point. The access point ensures isolation for tenants in a multi-tenant EFS file system. When an access point is set up, you can configure the file system policy to allow access only to the access point.

Data disk

The following table describes the data disk properties:

Property	Description
Type	Data disk type, either EFS or NFS.
File System	For EFS disks, the file system is the file system ID of the EFS disk. For NFS disks, the file system is the DNS of the file system.
Source Mount	File system path to be mounted in the serverless runtime environment.
Target Mount	File system to be mounted on the Secure Agent.
Access Point	ID of the Amazon EFS file system access point. The access point ensures isolation for tenants in a multi-tenant EFS file system. When an access point is set up, you can configure the file system policy to allow access only to the access point.

Serverless setup using AWS cross-account elastic network interfaces

A serverless runtime environment using AWS cross-account elastic network interfaces (ENIs) uses an ENI to connect to the AWS resources in your AWS account. Then, it runs tasks using the infrastructure in Informatica's AWS account.

To create the serverless runtime environment, complete the following tasks:

1. Set up your AWS environment to create and configure the required AWS resources.
2. In Administrator, create a serverless runtime environment to deploy it to AWS.
3. If your organization uses an outgoing proxy server to connect to the internet, you can configure the serverless runtime environment to connect to IDMC through the proxy server.
4. Prepare for disaster recovery by setting up a temporary serverless runtime environment where you can redirect jobs as part of your organization's disaster recovery plan.

Note: Your VPC must have default tenancy. A serverless runtime environment can't connect to a VPC with dedicated instance tenancy.

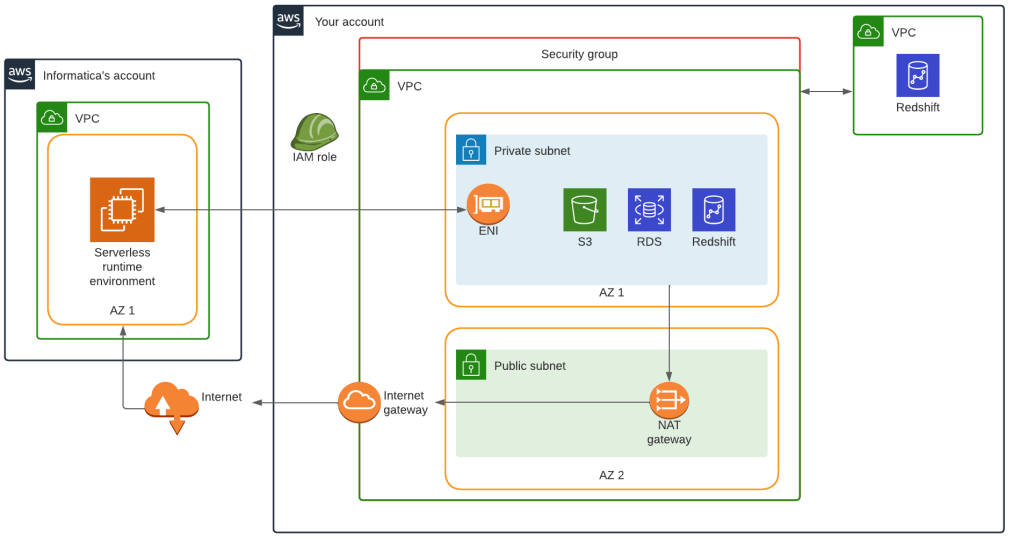
Set up the AWS environment

Set up your AWS environment to create and configure the required AWS resources before creating a serverless runtime environment.

To set up the AWS environment, complete the following tasks:

1. Create and configure AWS resources to connect to the serverless runtime environment.
2. Configure VPC to VPC connectivity to access data across VPCs.
3. Create an IAM role that the serverless runtime environment and advanced cluster worker nodes can use to create, attach, detach, and delete the ENI that's associated with the private subnet in your VPC.
4. Create a system disk to improve mapping performance in Data Integration.
5. Create a data disk if you have files on Amazon EFS or an NFS file server that you want to use in tasks.
6. If tasks use JAR files and external libraries, dedicate a location on Amazon S3 to store the files and create folders for each file type.
7. If your organization allows only trusted IP addresses, add the IP addresses for the Informatica NAT gateway to the trusted list.

The following image shows how the serverless runtime environment connects to your AWS resources:



Step 1. Create AWS resources

Create and configure AWS resources in your VPC to connect to the serverless runtime environment in Informatica's VPC.

Create and configure the following AWS resources:

VPC

A VPC is a virtual network that contains other AWS resources. You can use an existing VPC or create a VPC.

Enable DNS hostnames and DNS resolution for the VPC. Also, ensure that at least one of the following scenarios applies to you:

- Your VPC's DHCP option is set with AmazonProvidedDNS.

- If you have custom DNS servers in your DHCP option set, ensure that AmazonProvidedDNs is part of the option set or that the DNS servers can resolve EC2 internal hostnames. To ensure that the DNS servers can resolve EC2 internal hostnames, internally redirect the DNS query to AmazonProvidedDNS.

Public subnet for internet access

A public subnet provides internet access through a NAT gateway. To create the public subnet, use the following guidelines:

- Use any availability zone in the region where you created the VPC.
- The CIDR range must be within the VPC CIDR range. Choose a range based on the number of IP addresses that you want to have within the subnet.

Private subnet to host the ENI

A private subnet hosts the ENI that the serverless runtime environment uses to connect to your VPC.

Create a private subnet and configure a CIDR range to determine the maximum number of IP addresses and therefore, the scalability, of the serverless runtime environment. Configure the CIDR range to have at least 25 IP addresses per serverless runtime environment so that the serverless runtime environment can scale effectively when developers run concurrent workloads.

After your organization administrator creates a serverless runtime environment in Administrator, the serverless runtime environment creates an ENI in your private subnet.

Security group

A security group controls the traffic flow from the serverless runtime environment. Create a security group in the VPC. The security group is associated with all ENIs that the serverless runtime environment creates.

To create the security group, use the following guidelines:

- Leave the inbound rules empty to restrict all incoming traffic.
- The outbound rules can either allow all traffic or limit traffic to all Amazon S3 resources and all source and target systems that the serverless runtime environment accesses.

You specify this security group in the serverless runtime environment properties in Administrator.

NAT gateway for internet access from the private subnet

A NAT gateway allows outbound traffic to the internet from private instances. All compute instances in the serverless runtime environment that are associated with the ENI are private.

Create a NAT gateway to route outbound traffic from the private subnet to the internet. AWS provides several ways to configure subnet routing rules, such as route tables and NACL. For more information, see the AWS documentation.

Step 2. Configure VPC to VPC connectivity (optional)

Configure VPC to VPC connectivity to access data across VPCs.

For example, configure VPC to VPC connectivity if a mapping reads data from an Amazon Redshift cluster in one VPC and writes data to an Amazon Redshift cluster in another VPC.

AWS provides several ways to configure VPC to VPC connectivity, such as VPC peering or AWS Transit Gateway. Use AWS PrivateLink if applicable. For more information, see the AWS documentation.

Step 3. Create an IAM role

Create an IAM role that the serverless runtime environment and advanced cluster worker nodes can use to create, attach, detach, and delete the ENI that's associated with the private subnet in your VPC.

The IAM role must be able to access the S3 location for supplementary files as well as the sources and targets that you use in mappings and tasks. You can use the following template:

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Sid": "VisualEditor0",
            "Effect": "Allow",
            "Action": [
                "ec2:DetachNetworkInterface",
                "ec2:DeleteTags",
                "ec2:DescribeTags",
                "ec2:CreateTags",
                "ec2:DeleteNetworkInterface",
                "ec2:DescribeSecurityGroups",
                "ec2:CreateNetworkInterface",
                "ec2:DeleteNetworkInterfacePermission",
                "ec2:DescribeNetworkInterfaces",
                "ec2:DescribeAvailabilityZones",
                "ec2:CreateNetworkInterfacePermission",
                "ec2:AttachNetworkInterface",
                "ec2:DescribeNetworkInterfacePermissions",
                "ec2:DescribeSubnets",
                "ec2:DescribeNetworkAcls"
                "ec2:DescribeVpcs"
                "ec2:DescribeRegions"
            ],
            "Resource": "*"
        },
        {
            "Sid": "VisualEditor1",
            "Effect": "Allow",
            "Action": [
                "s3:ListBucket",
                "s3:GetBucketAcl"
            ],
            "Resource": "arn:aws:s3:::<S3 bucket name>"
        },
        {
            "Sid": "VisualEditor2",
            "Effect": "Allow",
            "Action": [
                "s3:PutObject",
                "s3:GetObject",
                "s3:DeleteObject"
            ],
            "Resource": [
                "arn:aws:s3:::<Supplementary file location>/*"
            ]
        }
    ]
}
```

In the trust relationship, specify the Informatica account number as a trusted entity and create an external ID. To find the Informatica account number, create a serverless runtime environment in Administrator and check the environment properties. You can use the following template:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Principal": {
        "AWS": "arn:aws:iam::<Informatica account>:root"
```

```
        },
        "Action": "sts:AssumeRole",
        "Condition": {
          "StringEquals": {
            "sts:ExternalId": "<External ID>"
          }
        }
      }
    ]
}
```

Step 4. Create a system disk

Create a system disk to improve mapping performance in Data Integration.

You can create a system disk using Amazon EFS or an NFS. File system connections in Amazon EFS use TLS by default. File system connections in NFS use NFSv4.

When you use a system disk, the serverless runtime environment creates the following directory on the system disk to store job metadata and logs:

```
<organization ID>/<serverless runtime environment ID>
```

Creating a system disk on Amazon EFS

To create a system disk on Amazon EFS, use the following guidelines:

- Create any folder required by an access point before creating the access point itself. For example, if the access point refers to the folder `/my-company/dev`, then define this folder first before you set up the access point.
- Configure connectivity between the subnet in the serverless runtime environment and the EFS file system.
- Configure the EFS security group to allow inbound access from the security group of the serverless runtime environment.
- Configure an Amazon EFS file system policy to grant access to the serverless role. Use the same IAM role that you created in "Step 3. Create an IAM role for the supplementary file location" on page 71.
  For example, the following file system policy allows root access to the IAM role for file system fs-12345 and allows SecureTransport only:

```
"Version": "2012-10-17",
    "Id": "efs-policy-wizard-<EFS policy wizard ID>",
    "Statement": [
        {
            "Sid": "efs-statement-<EFS statement ID>",
            "Effect": "Allow",
            "Principal": {
                "AWS": "arn:aws:iam::<your account ID>:role/<IAM role for
supplementary file location>"
            },
            "Action": [
                "elasticfilesystem:ClientWrite",
                "elasticfilesystem:ClientMount",
                "elasticfilesystem:ClientRootAccess"
            ],
            "Resource": "arn:aws:elasticfilesystem:us-west-2:<your account ID>:file-
system/fs-12345",
            "Condition": {
                "Bool": {
                    "elasticfilesystem:AccessedViaMountTarget": "true"
                }
            }
        },
        {
            "Sid": "efs-statement-<EFS statement ID>",
```

```
            "Effect": "Deny",
            "Principal": {
                "AWS": "*"
            },
            "Action": "*",
            "Resource": "arn:aws:elasticfilesystem:us-west-2:123456789:file-system/
fs-12345",
            "Condition": {
                "Bool": {
                    "aws:SecureTransport": "false"
                }
            }
        }
    ]
}
```

The following table describes the actions in the policy:

Action	Description
elasticfilesystem:ClientMount	Provides read-only access to a file system.
elasticfilesystem:ClientWrite	Provides write permissions on a file system.
elasticfilesystem:ClientRootAccess	Provides use of the root user when accessing a file system.

- Configure the IAM role to restrict access to specific access points on the file system. For more information, see the AWS documentation.

Creating a system disk using an NFS

To create a system disk using an NFS, use the following guidelines:

- Set the file system to the DNS of the NFS server.
- Configure connectivity between the subnet in the serverless runtime environment and the NFS file server.
- Configure the NFS security group to allow inbound access from the security group of the serverless runtime environment.

Step 5. Create a data disk

Create a data disk if you have files on Amazon EFS or an NFS file server that you want to use in tasks.

You can store the following types of files on a data disk:

- Flat files for a flat file connection in tasks
- Parameter files for mapping tasks

1. Mount the Amazon EFS or NFS directory on an Amazon EC2 instance.
2. Log in to the EC2 instance.
3. Search for a user with ID 501. If the user doesn't exist, create a user with this ID.

   User ID 501 corresponds to the user cldagnt. The serverless runtime environment uses this user to access mounted EFS or NFS directories.
4. Assign read and write permissions for the mounted directories to the user with ID 501.

For more information about creating a data disk using NFS, see Network File System (NFS) level users, groups, and permissions in the AWS documentation.

Step 6. Create a supplementary file location

If tasks use JAR files and external libraries, dedicate a location on Amazon S3 to store the files and create folders for each file type.

To create the supplementary file location, use the following guidelines:

1. Create the following file structure on Amazon S3:

```
<Supplementary file location>
├── ext
├── odbc
│   └── lib
└── serverless_agent_config
    ├── jars
    ├── SSL
    ├── j_depends
    └── py_depends
```

2. Create a serverlessUserAgentConfig.yml file. For more information, see "Configure the serverlessUserAgentConfig.yml file" on page 100.
3. Add the serverlessUserAgentConfig.yml file directly under the `serverless_agent_config` directory.

The following table lists the file types that you can store in each location:

Location	Files
<Supplementary file location>/ext	JDBC JAR files
<Supplementary file location>/odbc	The following files: - odbc.ini - odbcinst.ini - exports.ini
<Supplementary file location>/odbc/lib	ODBC shared libraries for a Linux operating system
<Supplementary file location>/serverless_agent_config	The following files: - serverlessUserAgentConfig.yml - JDBC V2 Connector JAR files - REST V3 Connector truststore and keystore certificates - JAR files for the Java transformation - Installation and resource files for the Python transformation You can customize the directory structure under the serverless_agent_config folder and specify the relative path to each file in the serverlessUserAgentConfig.yml file.

Step 7. Add trusted Informatica IP addresses

If your organization allows only trusted IP addresses, add the IP addresses for the Informatica NAT gateway to the trusted list.

For information about adding trusted IP addresses, see *Organization Administration*.

The following table lists the IP addresses for the Informatica NAT gateway to allow for each POD:

POD	Trusted IP addresses
NA West 1, NA East 2, US West 3, US East 4, US West 5, US East 6	44.242.20.143 44.239.8.148 54.221.247.69 54.160.9.90 3.131.176.232 18.142.14.24 3.1.147.184 18.220.76.98 13.56.74.27 52.52.220.198
Canada Central 1	3.97.103.68 3.96.182.201
EM West 1 (Ireland)	3.64.66.226 3.125.185.124 54.76.54.130 54.78.183.88 35.176.60.118 18.135.50.152 35.152.49.63 35.152.45.151 15.237.157.126 15.237.97.211 13.49.61.89 13.53.141.231
UK	34.250.251.16 18.170.170.192 13.37.37.71 18.157.124.91 13.53.147.238 15.161.184.93 15.160.41.209
AP Southeast 1 (Australia)	54.253.179.190 3.24.111.61 35.72.149.44 13.112.143.134 3.34.56.248 52.79.244.47 52.76.184.230 18.140.193.120 65.1.80.5 13.234.141.216 18.163.244.73 18.167.71.151
AP Northeast 2 (Japan)	18.181.99.0 13.208.101.21

Create a serverless runtime environment

In Administrator, create a serverless runtime environment to deploy it to AWS.

1. In Administrator, open the **Serverless Environments** page.
2. Select **New Serverless Runtime Environment**.
3. Select **AWS Cross-account elastic network interface**.

4. Configure the serverless runtime environment properties.

   For information about how to configure the properties, see "Serverless runtime environment properties" on page 86.

5. To save and deploy the serverless runtime environment, select **Save**.

It takes at least five minutes for the serverless runtime environment to become available. Use the **Serverless Environments** page to track the status of the environment and review any status messages.

Serverless runtime environment properties

Configure the serverless runtime environment properties so that IDMC can connect to your AWS account.

Basic configuration

The following table describes the basic properties:

Property	Description
Name	Name of the serverless runtime environment.
Description	Description of the serverless runtime environment.
Task Type	Type of tasks that run in the serverless runtime environment. - Select **Data Integration** to run mappings outside of advanced mode. - Select **Advanced Data Integration** to run mappings in advanced mode.
Cloud Platform	Cloud platform to host the serverless runtime environment. Use Amazon Web Services (AWS).
Max Compute Units Per Task	Maximum number of serverless compute units corresponding to CPU and memory that a task can use. The property value configured in the serverless runtime environment specifies the maximum number of serverless compute units that each task can request from the environment. When you create a mapping task, you can override the maximum number of compute units that the task can request. In Monitor, you can view the number of compute units that the task requested and consumed. For information about metering, see *Organization Administration*.
Task Timeout	Amount of time in minutes to wait for a task to complete before it is terminated. The timeout ensures that serverless compute units are not unproductive when a task hangs. By default, the timeout is 2880 minutes (48 hours). You can set the timeout to a value that is less than 2880 minutes.
Informatica Account Number	Informatica's account number on the cloud platform where the serverless runtime environment will be created. The account number is populated automatically.
External ID	External ID to associate with the role that you create for the serverless runtime environment. You can use the generated external ID or specify your own external ID.

Platform configuration

The following table describes the platform configuration properties:

Property	Description
Configuration Name	Name of the serverless configuration.
Configuration Description	Description of the serverless configuration. The description can be up to 256 characters and can contain alphanumeric characters and the following special characters: `._-:/()#,@[]+=&;{}!$"*`
Account Number	Your account number on the cloud platform.
Region	Region of your cloud environment. The sources and targets that you use in mappings must either reside in or be accessible from this region.
AZ ID	ID of the availability zone. The sources and targets that you use in mappings must either reside or be accessible from the availability zone.
VPC ID	ID of the VPC. The VPC must be configured with an endpoint to access the sources and targets that you use in mappings. For example, `vpc-2f09a348`
Subnet ID	ID of the subnet within the VPC. The subnet must be have an entry point to access the sources and targets that you use in mappings. For example, `subnet-b46032ec`
Security Group ID	ID of the security group that the serverless runtime environment will attach to the ENI. The security group allows access to the sources and targets that you use in tasks. For example, `sg-e1fb8c9a`.
Role Name	Name of the IAM role that the serverless runtime environment can assume on your AWS account. The role must have permissions to create, read, delete, list, detach, and attach an ENI. It also requires read and write permissions on supplementary file location. Use the Informatica account number and the external ID when you create a policy for the role.
AWS Tags	AWS tags to label the ENI that's created in your AWS account. Each tag must be a key-value pair in the format: `Key=string,Value=string` where `Key` and `Value` are case-sensitive. Use a space to separate tags. Follow the rules and guidelines for tagging that AWS specifies. For more information, refer to the AWS documentation.
Supplementary File Location	Location on Amazon S3 to store supplementary files, such as JAR files and external libraries for certain transformations and connectors. Use the format: `s3://<bucket name>/<folder name>` Place script files in a folder named `command_scripts`. The folder can have subfolders. If you update files in the `command_scripts` folder, IDMC automatically uploads them to the Secure Agent.

System disk

The following table describes the system disk properties:

Property	Description
Type	System disk type, either EFS or NFS.
File System	For EFS disks, the file system is the file system ID of the EFS disk. For NFS disks, the file system is the DNS of the file system.
Source Mount	File system path to be mounted in the serverless runtime environment.
Access Point	ID of the Amazon EFS file system access point. The access point ensures isolation for tenants in a multi-tenant EFS file system. When an access point is set up, you can configure the file system policy to allow access only to the access point.

Data disk

The following table describes the data disk properties:

Property	Description
Type	Data disk type, either EFS or NFS.
File System	For EFS disks, the file system is the file system ID of the EFS disk. For NFS disks, the file system is the DNS of the file system.
Source Mount	File system path to be mounted in the serverless runtime environment.
Target Mount	File system to be mounted on the Secure Agent.
Access Point	ID of the Amazon EFS file system access point. The access point ensures isolation for tenants in a multi-tenant EFS file system. When an access point is set up, you can configure the file system policy to allow access only to the access point.

Set up a proxy server

If your organization uses an outgoing proxy server to connect to the internet, you can configure the serverless runtime environment to connect to IDMC through the proxy server.

When you configure a proxy server for the serverless runtime environment, you define the required proxy server settings in the serverlessUserAgentConfig.yml file before you can import metadata or design your mappings. Data Integration copies the proxy entries in the file to the serverless runtime environment.

To apply the proxy when you run mappings, set the proxy configurations on the **Serverless Environments** page in Administrator.

You can configure proxy settings for the serverless runtime environment in certain connectors. To see if the proxy applies in a connector, see the help for the appropriate connector.

Configuring the proxy in the serverlessUserAgentConfig.yml file

To apply proxy server settings when you design mappings and import metadata, add the proxy server details to the `serverlessUserAgentConfig.yml` file.

Use the following code snippet as a template to provide the values for the proxy server in the `serverlessUserAgentConfig.yml` file:

```
agent:
  agentAutoDeploy:
    general:
        proxy:
          proxyHost: <Host_name of proxy server>
          proxyPort: <Port number of the proxy server>
          proxyUser: <User name of the proxy server>
          proxyPassword: <Password to access the proxy server>
          nonProxyHost: <Non-proxy host>
```

Configuring the proxy in the JVM options

To apply proxy server settings when you run mappings or tasks, configure JVM options in Administrator.

1. On the **Serverless Environments** page, click the name of the serverless runtime environment.
2. Click **Edit**.
3. In the **Runtime Configuration Properties** section, select the **Service** as **Data Integration Server** and the **Type** as **DTM**.
4. Edit any of the JVMOption fields and specify appropriate values for each parameter based on whether you use an HTTPS or HTTP proxy server.

   The following table describes the parameters:

Parameter	Description
-Dhttp.proxySet=	Determines if the serverless runtime environment must use the proxy settings when the outgoing proxy server is HTTP. Select -Dhttp.proxySet=True to use the proxy.
-Dhttps.proxySet=	Determines if the serverless runtime environment must use the proxy settings when the outgoing proxy server is HTTPS. Select -Dhttps.proxySet=True to use the proxy.
-Dhttp.proxyHost=	Host name of the outgoing HTTP proxy server.
-Dhttp.proxyPort=	Port number of the outgoing HTTP proxy server.
-Dhttp.proxyUser=	Authenticated user name for the HTTP proxy server.
-Dhttp.proxyPassword=	Password for the authenticated user.
-Dhttps.proxyHost=	Host name of the outgoing HTTPS proxy server.
-Dhttps.proxyPort=	Port number of the outgoing HTTPS proxy server.
-Dhttps.proxyUser=	Authenticated user name for the HTTPS proxy server.
-Dhttps.proxyPassword=	Password for the authenticated user.

5. Click **Save**.

Allowing domains in the proxy server

To run a mapping successfully, the proxy server must allow traffic from the AWS endpoints that are required to process the data in the mapping.

Allow traffic from the following domains:

```
s3.<region>.amazonaws.com
s3.amazonaws.com
ec2.<region>.amazonaws.com
sts.<region>.amazonaws.com
efs.<region>.amazonaws.com
elasticfilesystem.<region>.amazonaws.com
```

Specify the region that contains the VPC that connects to the serverless runtime environment.

Prepare for disaster recovery

Prepare for disaster recovery by setting up a temporary serverless runtime environment where you can redirect jobs as part of your organization's disaster recovery plan.

Disaster recovery procedure

During a disaster, all virtual machines in the serverless runtime environment shut down and jobs can no longer run in the environment.

To minimize data loss and downtime, complete the following tasks:

1. Create a temporary serverless runtime environment in a stable region or availability zone.
2. Make sure that the connections used in jobs are available in the stable region or availability zone.
3. Clean up data related to incomplete job runs. If data was partially loaded to a target, manually delete the data or update the mapping to truncate the target before writing new rows.
4. Redirect jobs to the temporary environment.

Restoring the primary environment

When the region or availability zone that hosts the primary serverless runtime environment has recovered, you can restore the primary environment.

To restore the primary environment, complete the following tasks:

1. Clean up the ENIs that were created in your AWS account for the primary environment.
2. Redeploy the primary environment.
3. Redirect jobs to the primary environment.
4. Delete the temporary environment.

Serverless runtime environment validation

Informatica validates the serverless configuration properties and several network settings in the serverless runtime environment when you create, clone, or redeploy it; or when you edit and save a failed environment.

To validate the serverless runtime environment, Informatica connects to your AWS account using the IAM role that you created. Informatica uses the role to verify AWS resource properties, such as the role name, subnet ID, and availability zone ID. To access these resources, Informatica uses the following role permissions:

```
ec2:DescribeRegions
ec2:DescribeAvailabilityZones
ec2:DescribeVpcs
```

```
ec2:DescribeSubnets
ec2:DescribeSecurityGroups
```

Note: Informatica doesn't validate the VPC ID if the subnet ID doesn't exist.

Informatica also checks the number of IP addresses in the subnet.

If validation fails for an AWS resource or there aren't enough IP addresses available in the subnet, the serverless runtime environment fails to start. You can download the validation log on the **Serverless Environments** page.

Serverless setup on Microsoft Azure

You can host a serverless runtime environment on an Azure Virtual Network (VNet). You configure the serverless runtime environment on Azure Native ISV Services.

To configure a serverless runtime environment on Azure, complete the following steps:

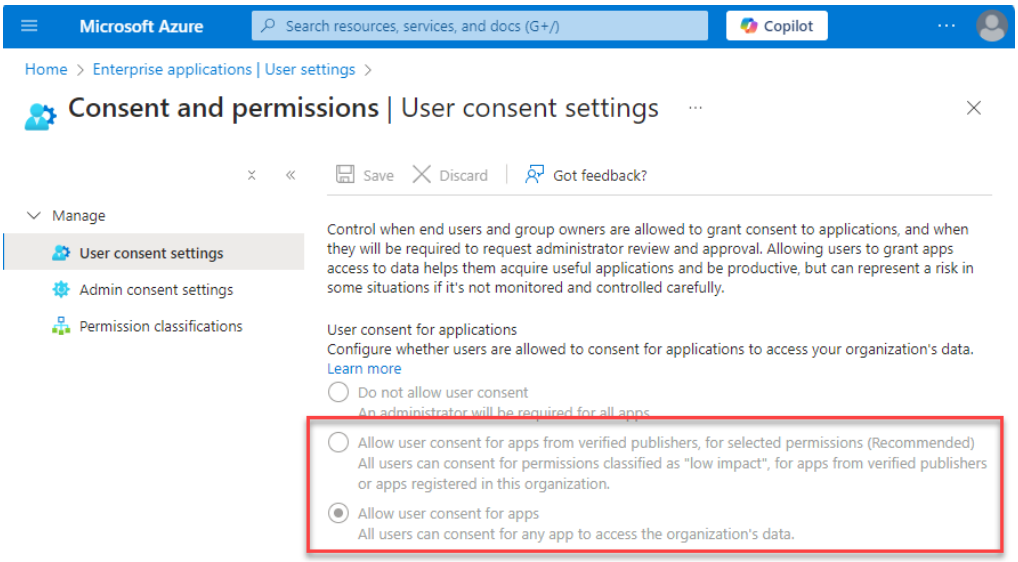
1. Verify your user consent settings.
2. Configure the subnet.
3. Configure the serverlessUserAgentConfig.yml file.
4. Create an organization on Azure Native ISV Services.
5. Create a serverless runtime environment.

Step 1. Verify your user consent settings

Your Azure administrator might need to approve the Informatica app, depending on your user consent settings.

1. From the Azure Portal home page, select **Enterprise applications**.
2. Expand **Security** and then click **Consent and permissions**.

   The **User consent settings** page is displayed:



3. Review the user consent setting assigned to you:
   - If your **User consent settings** page is set to **Allow user consent for apps from verified publishers, for selected permissions**, you must ask your Azure administrator to approve the Informatica application before you can create serverless runtime environments.
     For more information on approving an appliccation, see Review admin consent requests in the Microsoft documentation.
   - If your consent settings show **Allow user consent for apps**, then no approval is required from the Azure administrator. You can proceed to create your serverless runtime environment.

Step 2. Configure the subnet

Assign a specific subnet for use with the Azure serverless runtime environment.

Ensure that the subnet has internet connectivity. There are several ways to achieve this connectivity, including configuring a NAT gateway with a public IP address.

Ensure that the service endpoint `Microsoft.AzureActiveDirectory` is enabled on the delegated subnet. This is not required if you have configured a NAT gateway with a public IP address.

There is no limit to the number of serverless runtime environments on the same subnet, provided there are enough IP addresses available.

The following table lists the regions for VNET/subnets linked via subnet delegation:

POD	Regions for VNET/subnets linked via subnet delegation
US West 1 Azure (USW1-1)	West US East US East US2 US South Central US Central
US West 3 Azure (USW3-1)	West US East US2 US South Central US Central
Canada Central 1 Azure (CAC2)	Canada Central Canada East
EM Central 1 Azure (EMC1)	West Europe Germany West Central North Europe
EM SouthEast 1 Azure (EMSE1)	(Middle East) UAE North

POD	Regions for VNET/subnets linked via subnet delegation
AP East 2 Azure (APSE2)	(Asia Pacific) Southeast Asia
Australia Azure (APAUC1)	Australia Central Australia Southeast Australia East

1. Use the Azure Cloud Shell to run the following command:

   ```
   az provider register --namespace 'Informatica.DataManagement'
   ```

   This command registers the "Informatica.DataManagement" resource provider in your subscription where your subnet is located.
2. Create a new subnet and delegate this subnet to the following service: `Informatica.DataManagement/organizations`.

   Note: Delegating the subnet to the service `Informatica.DataManagement/organizations` must be done after all subnet-related configuration is complete, otherwise issues might occur during subnet configuration.

Step 3. Configure the serverlessUserAgentConfig.yml file

A serverlessUserAgentConfig.yml file is required when you create a supplementary file location later in the configuration process.

For information on the complete procedure, see "Configure the serverlessUserAgentConfig.yml file" on page 100.

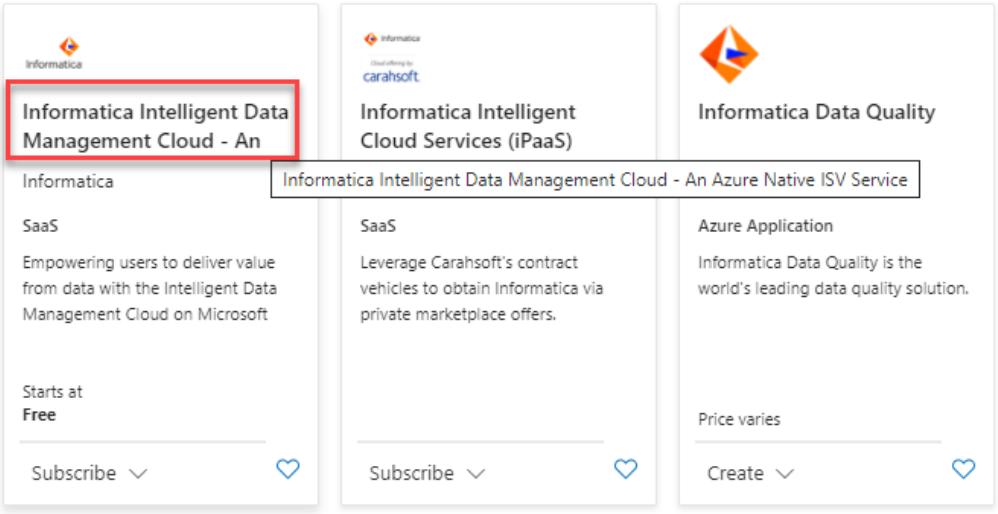
Step 4. Create an Informatica organization

After you've subscribed to the Informatica single sign-on app and configured your subnet, you can create a new organization or link to an existing organization.

1. From the Azure Portal home page, select **Marketplace**.
2. Search for **Informatica**.

   You might see several plans with "Informatica" in their names.
3. Select the plan named **Informatica Intelligent Data Management Cloud - An Azure Native ISV Service**.

If you are in tile or list view, the name is truncated. To confirm that you've selected the correct plan, hover over the name. The following image shows the correct plan:



4. Select the plan specified in your contract, click **Subscribe** > **Yes, continue**.

   The **Create an Informatica organization** page appears.

5. Enter the following information on the **Basics** tab:

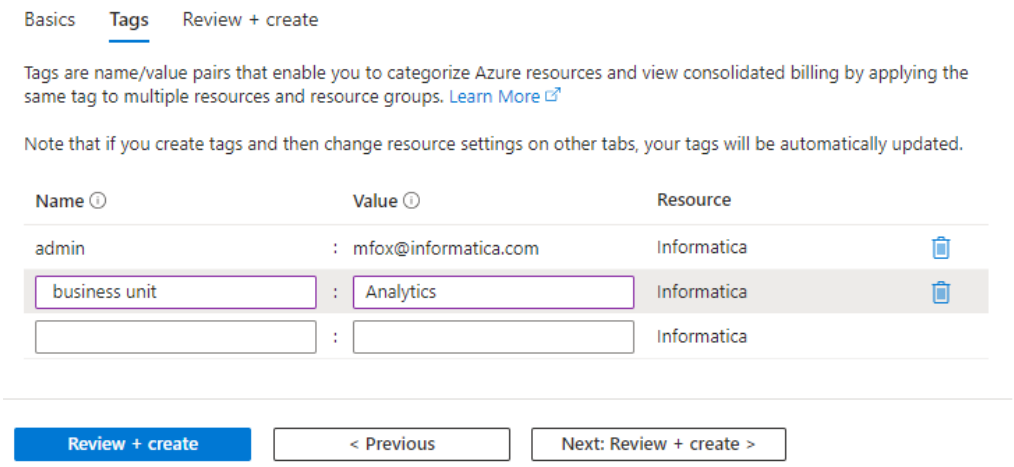
Field	Value
Subscription	Select the Azure subscription you want to use to create the Informatica organization. You must have an owner or contributor role for this subscription.
Resource group	Select a resource group or create a new one. You must have a contributor role to this resource group. A resource group is a container that holds related resources for an Azure solution.
Resource Name	Enter a name for the Azure resource.
Region	Select the region where the resource of type "Informatica organization" will be provisioned.
Informatica Region	Select the region that is closest to you. This determines the POD that you'll connect to.
Organization	Choose whether to create a new organization or link to an existing organization. If you link to an existing organization, you can choose either to continue with an existing Informatica billing or use the Azure Marketplace billing. **Note:** You can only link to a parent organization that has a Production or Sandbox license type. You can't link to a sub-organization or if the license type is Trial. For more information, see *Organization Administration*.

Field	Value
Organization name	If you are linking to an existing organization, click **Link to an existing Informatica organization** and provide your login credentials. The Informatica organization created on the Azure portal automatically links to your existing organization.
Plan	The plan you selected earlier is displayed. If you decide to change it, click **Change plan**.

6. Optionally, click **Next: Tags** to create tags to help you categorize Azure resources.

   The following are examples of tags:

   - Name: <name of admin>, Value: <email address>
   - Name: <business unit>, Value: <user's business unit>



7. Click **Next: Review + create**.

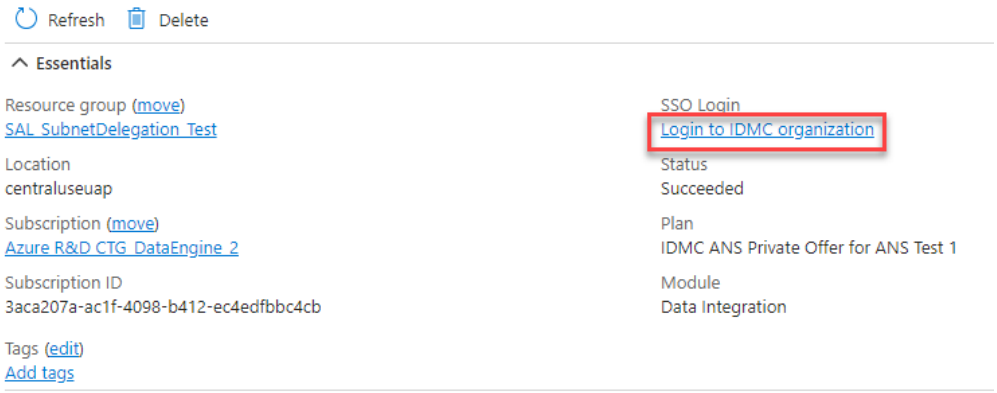
   Review the information for accuracy and ensure that **Validation passed** displays at the top of the page. During service creation, the process validates your Active Directory properties according to the information in the following table:

Field	Validation
First Name	The first name can contain only letters, hyphens, and single quotation marks (').
Last Name	The last name can contain only letters, hyphens, and single quotation marks (').
Company	Company name must be 255 characters or less.
Country	Company name must be 255 characters or less.
Email	The email must be in a valid email format.
Phone Number	A phone number must contain 10-25 characters. It can contain only numbers, spaces, parentheses, hyphens, periods, and a plus sign as the first character.

   If any field fails validation, then the service creation fails.

8. To begin the deployment process, click **Create**.

   The deployment is complete when you see **Deployment succeeded**.
9. To see the resource dashboard for your organization, click **Go to resource**.
10. Click **Login to IDMC organization** on the resource dashboard.



   Note: It is important to log in to IDMC before you create a serverless runtime environment.
11. Verify that your personal information is accurate and confirm that you agree to the subscription agreement.
12. Click **Confirm** to log in to IDMC.

    This step is necessary to add you as a user to the system.

Step 5. Create a serverless runtime environment

When you use Azure Native ISV Services, you create and maintain your serverless runtime environments on the Azure portal instead of in IDMC.

Note: The System Disk and Proxy Servers features are currently not supported on Azure serverless runtime environments.

1. On your organization's resource page on the Azure portal, click **Serverless Runtime Environment**.

   Right now, the list is empty since you have no serverless runtime environments.
2. Click **Create New Serverless Environment**.

   The configuration is divided across four tabs: **Basics**, **Platform details**, **Runtime configuration**, and **Tags**.
3. On the **Basics** tab, enter the following information:

Field	Description
Name	Name of the serverless runtime environment.
Description	Description of the serverless runtime environment.
Task type	Type of tasks that run in the serverless runtime environment: - Select **Data Integration and Data Ingestion/Replication** to run mappings outside of advanced mode. - Select **Advanced Data Integration** to run mappings in advanced mode.

Field	Description
Maximum computer units per task	Maximum number of serverless compute units corresponding to CPU and memory that a task can use. The property value configured in the serverless runtime environment specifies the maximum number of serverless compute units that each task can request from the environment. When you create a mapping task, you can override the maximum number of compute units that the task can request. In Monitor, you can view the number of compute units that the task requested and consumed. For information about metering, see *Organization Administration*. This property does not apply to Data Ingestion and Replication.
Task time out (Minutes)	By default, the timeout is 2880 minutes (48 hours). You can set the timeout to a value that is less than 2880 minutes. This property does not apply to Data Ingestion and Replication.

4. On the **Platform Detail** tab, enter the following information:

Field	Description
Region	Select the region where the serverless runtime environment is hosted.
Virtual network	Select a virtual network to use.
Subnet	Select a subnet within the virtual network to use. **Note:** Be sure you've completed "Step 2. Configure the subnet" on page 92 .

Supplementary file location	ADLS Gen2 location of any supplementary files. Use the following format:
	abfs://<file_system>@<account_name>.dfs.core.windows.net/<path>
	For example, to use a JDBC connection, you place the JDBC JAR files in the supplementary file location and then enter this location:
	`abfs://discaleqa@serverlessadlsgen2acct.dfs.core.windows.net/serverless`
	If you intend to run command tasks within the Azure serverless runtime environment, create a folder named `command_scripts` within supplementary file location.
	For example, you would create the folder here:
	`abfs://discaleqa@serverlessadlsgen2acct.dfs.core.windows.net/serverless/command_scripts`
	For more information about command tasks, see "Command task steps" in *Taskflows*.
	Note: Ensure that there are no extra spaces or special characters in the supplementary file location, because this might cause the deployment to fail.
	When you specify a supplementary file location, select a managed identity or specify service principal credentials in the custom properties to grant the serverless runtime environment access to the file location.

Field	Description
Data disks	Access files in an NFS file system. Configuring a data disk in your serverless runtime environment allows you to access files in NFS directories. For more information about configuring data disks, see "Working with data disks" on page 98. Configure the following properties to configure a data disk: - Type. Only NFS directories are currently supported. - Server host/IP. The hostname or IP address of the NFS server. - Source mount. File system path to be mounted in the serverless runtime environment. - Target mount. File system to be mounted on the Secure Agent. - Mount options. If mount options are required, enter them in a comma-separated list in the following format: `<key1>=<value1>,<key2>=<value2>`
Managed identity	User-assigned managed identity that grants the serverless runtime environment access to the supplementary file location. Required if you don't provide access through service principal credentials in the custom properties.
Custom properties	Additional properties for the serverless runtime environment. To grant access to the supplementary file location through service principal credentials, add the following custom properties and their appropriate values: - ServicePrincipalId - ServicePrincipalSecret. Check for accuracy, as the secret is masked once the configuration is saved. Service principal credentials are required if you don't configure a managed identity. If you configure both a managed identity and service principal credentials, the serverless runtime environment uses the managed identity to access the file location. Informatica Global Customer Support might direct you to add other custom properties. If you clone an Azure serverless runtime environment, you need to re-enter the *ServicePrincipalSecret* because the original value is replaced by a masked string.

Note: You can't change the supplementary file location, managed identity, or custom properties once the configuration is saved. If you need to make corrections, clone the configuration. For more information about cloning, see "Working with serverless runtime environments" on page 105.

5. On the **Runtime Configuration** tab, enter properties that determine how the serverless runtime environment behaves.

   Don't make any changes to this tab unless directed by Informatica Global Customer Support.

6. On the **Tags** page, create tags that help you categorize Azure resources and view consolidated billing.

   Enter as many tags as necessary as name-value pairs.

7. On the **Review+Create** page, verify that all entries are correct and then click **Create**.

   If you need to go back to make corrections, click **Previous** to return to the appropriate tab.
   For more information on using and managing serverless runtime environments, see Serverless Runtime Environments on the Informatica Documentation Portal.

Working with data disks

Create a data disk in your serverless runtime environment if you have files in NFS directories that you want to use in the serverless runtime environment without needing to update all your mappings. This feature works only with the NFS version 4.x protocol.

Once you mount your NFS location in a data disk, you have access to the following capabilities:

- Flat file support. You can use flat files from the mounted NFS location in your mappings.

- Parameter file support. You can use parameter files stored in the mounted NFS location. This simplifies migrating jobs from a Secure Agent group to a serverless runtime environment, since you don't need to modify your mappings.

When using existing NFS directories as data disks in an Azure serverless runtime environment, ensure that the user with ID=501 has permissions to read and write from the mount directory of the NFS storage. User ID 501 is the `cldagnt` user, which the Azure serverless runtime environment uses to access mounted NFS directories.

If user 501 doesn't exist, create that user and assign the appropriate permissions.

Note: If you have a network firewall that prevents access to the NFS Server, you must allow access to the NFS Server from the delegated subnet used to create the Azure serverless runtime environment.

Configure VNet for databases or endpoints outside of Azure

You need to perform additional configuration on the VNet if you use Oracle or any database or endpoint hosted on-premises or outside of the Azure ecosystem.

Perform the following tasks:

- Add trusted IP addresses. If your organization filters based on IP addresses, add the safe Informatica addresses so that they won't get blocked by the firewall. For more information, see "Step 7. Add trusted Informatica IP addresses" on page 74.
- Create a supplementary file location. If your mappings use JAR files and external libraries, set up a location on an Azure storage account with either Gen2 or blob. Use the following format:
  abfs://<file_system>@<account_name>.dfs.core.windows.net/<path>

  For example, to use a JDBC connection, you place the JDBC JAR files in the supplementary file location and then enter this location:

  `abfs://discaleqa@serverlessadlsgen2acct.dfs.core.windows.net/serverless`
- Configure TLS to authenticate REST APIs. If you use a REST V3 Connector, you can configure TLS to authenticate REST APIs. For more information, see "Configure TLS to authenticate REST APIs" on page 104.

Run command tasks

To run command tasks in an Azure serverless runtime environment, you must place the shell scripts in a specific folder.

Ensure that there is a folder named `/command_scripts` within the "Step 5. Create a serverless runtime environment" on page 96 of the Azure serverless runtime environment.

1. Place the script files that you want to run in the command task in the `/command_scripts` folder. The `/command_scripts` folder can have its own subfolders.

   The files placed in the `/command_scripts` folder are synchronized to the Secure Agent machine, to following folder in the Secure Agent installation directory:

   `apps/Common_Integration_Components/data/command/serverless/command_scripts`
2. Add a command task step to your taskflow. The script specified in the command task is executed in the Secure Agent Docker container.

   The files under `apps/Common_Integration_Components/data/command/serverless/command_scripts` are mounted under the `/command_scripts` folder inside docker container. You can reference any other files by using the relative path. The script working directory is set to `/command_scripts`.

   For more information about command tasks, see "Command task step" in *Taskflows*.

Troubleshooting a serverless runtime environment in Azure

Why does my serverless runtime environment fail to start after changing a region in Azure?

Your serverless runtime environment might have failed to start because the region entered was incorrect. Once a region is selected, you can't change it. You need to delete the serverless runtime environment and then recreate it using the correct region.

A network security group is automatically created and attached to the region. When you change regions, the network security group no longer matches the new region and the environment fails to start. The only way to correct this is by deleting the environment and creating a new one.

Configure the serverlessUserAgentConfig.yml file

The serverlessUserAgentConfig.yml contains the paths to all supplementary files. Create the file and then place it in the supplementary file location.

To create the serverlessUserAgentConfig.yml file, use a template. Then, specify the paths of the supplementary files. You can specify the following types of files:

- Files for the Elastic Server
- JAR files for the JDBC V2 Connector
- JAR files for the Java transformation
- Resource files for the Python transformation

Note: Escape any spaces or special characters that appear in paths entered in the serverlessUserAgentConfig.yml file.

For information about populating connector details in the serverlessUserAgentConfig.yml file, see the help for the appropriate connector.

serverlessUserAgentConfig.yml template

Use the following template to create the serverlessUserAgentConfig.yml file:

```
# The Secure Agent is the root element, and configurations are applied to the agent.
# Under the agent, there are three levels:
#1: apps : Application where you need to apply configurations.
#2: event: Event relating to the life cycle of application.
  #autoDeploy: Configurations that need the agent app to restart. Configurations are
applied and minor versions of the app are upgraded. An upgrade event will detect the
difference between the configuration that was last applied and the current request and
apply only those configuration changes. Note that Administrator does not show
notifications during minor version upgrades.
  #autoApply: Configuration that takes effect immediately, such as copying Swagger files.
#3: section: Contains configurations based on connectors.

# How do I apply the YML file?
# Create a serverlessUserAgentConfig.yml file with these contents in
<supplementary_file_location>/serverless_agent_config.
# The path in the serverlessUserAgentConfig.yml file is relative to
<supplementary_file_location>/serverless_agent_config/.

# fileCopy section : Provide the source location of the file that needs to be copied.

version: 1
agent: # At the agent level, provide general configurations that are not specific to the
application.
  agentAutoApply:
    general: # General section for common configurations across applications and
```

```
connectors.
      sslStore: # Use this to copy SSL files to the instance machine. You can provide a
list of fileCopy.
        - fileCopy:
            sourcePath: SSL/RESTV2_JWTpyn.jks
  # Data Integration Server app
  dataIntegrationServer:
    autoApply: # Apply configurations that don't need to upgrade the minor version or a
restart of the app. For example, you can copy files.
      restv2: # Connector section
        swaggers: # List of Swaggers files to copy to the instance machine.
          - fileCopy:
              sourcePath: restv2/<swagger_file_name>.json
        keystores: # List of keystore files to copy to the instance machine.
          - fileCopy:
              sourcePath: restv2/key
        truststores: # List of truststore files to copy to the instance machine.
          - fileCopy:
              sourcePath: restv2/key.ext
      wsconsumer:
        wsdls:
          - fileCopy:
              sourcePath: s3/
      jdbc:
        drivers:
          - fileCopy:
              sourcePath: s3/file
    autoDeploy:
    # A change in this event will trigger a minor version upgrade with the new
configurations.
    # In this case, the Data Integration Server app will get a minor version upgrade.
      general: # General section for Data Integration Server app autoDeploy event.
        ssls:
          - fileCopy:
              sourcePath: SSL/RESTV2_JWTpyn.jks
            importCerts:
              certName: cname
              alias: IICS
      sap:
        jcos: # List of jco related files to copy.
          - fileCopy:
              sourcePath: sap/jco/libsapjco3.so
          - fileCopy:
              sourcePath: sap/jco/sapjco3.jar
        nwrfcs: # List of nwrfc related files to copy.
          - fileCopy:
              sourcePath: sap/nwrfc/libicudata.so.50
          - fileCopy:
              sourcePath: sap/nwrfc/libicudecnumber.so
        hanas: # List of hana related files to copy.
          - fileCopy:
              sourcePath: sap/hana/libicudata.so.50
      odbc:
        # Specify ODBC configurations.
        # This section can be used to configure multiple drivers.
        drivers: # Specify drivers to copy.
          - fileCopy:
              sourcePath: ODBC/DWdb227.so
          - fileCopy:
              sourcePath: ODBC/DWdb227.so
        dns:
          # Specify DNS entries. These entries will be updated in odbc.ini file.
          # If the file is not present, a new odbc.ini file will be created.
          # Make sure to give a name as a unique entry for the ini file configuration.
The file will be read and updated using the name.
          - name: "SQL server" # Section name in ini file unique key.
            entries:
              - key: Driver # Only provide the driver file name without the path.
                value: DWsqls227.so # Because the file is copied, the path to attach
during odbc entry is already known.
              - key: Description
```

```
              value: "SQL Server 2014 Connection for ODL"
            - key: HostName
              value: INVW16SQL19
            - key: PortNumber
              value: 1433
            - key: Database
              value: adapter_semantic
            - key: QuotedId
              value: No
            - key: AnsiNPW
              value: Yes
# Database Ingestion and Replication app
databaseIngestion:
  autoDeploy:
    jdbc:
      drivers:
         - fileCopy:
            sourcePath: google/<file_name>.jar
         - fileCopy:
            sourcePath: oracle/jdbc/<file_name>.jar
    hanami:
      jdbc:
        drivers:
          - fileCopy:
              sourcePath: hanami/<file_name>.jar
    mysql:
      jdbc:
        drivers:
          - fileCopy:
              sourcePath: mysql/jdbc/<file_name>.jar
    netsuiteami:
      jdbc:
        drivers:
          - fileCopy:
              sourcePath: netsuite/<file_name>.jar
    odp:
      jdbc:
        drivers:
          - fileCopy:
              sourcePath: sap-odp/<file_name>.jar
    sapami:
      jdbc:
        drivers:
          - fileCopy:
              sourcePath: sap-ami/<file_name>.jar
    oracle:
      oci:
        drivers:
        - fileCopy:
            sourcePath: oci/<file_name>.zip
```

Copying files for the Elastic Server

In the `serverlessUserAgentConfig.yml` file, you can specify files to copy from the supplementary file location to the serverless runtime environment. When you run mappings in advanced mode in the serverless runtime environment, the Elastic Server and the advanced cluster can use the files to access and process data.

You can copy the following file types for the Elastic Server:

- JDBC V2 Connector JAR files
- JAR files for the Java transformation
- Installation and resource files for the Python transformation

You can customize file paths by specifying the relative path to the file in the supplementary file location. For example, you might store JDBC V2 Connector JAR files in the following locations:

```
<Supplementary file location>/serverless_agent_config/jdbc_v2_jars/common/

<Supplementary file location>/serverless_agent_config/jdbc_v2_jars/spark/
```

You can specify the following relative paths in the `serverlessUserAgentConfig.yml` file:

```
agent:
  elasticServer:
    autoApply:
      jdbcv2:
        common:
          - fileCopy:
              sourcePath: jdbc_v2_jars/common/driver.jar
        spark:
          - fileCopy:
              sourcePath: jdbc_v2_jars/spark/driver.jar
```

Copying JDBC V2 Connector JAR files

To copy JAR files for JDBC V2 Connector, use the following code snippet as a template:

```
agent:
  elasticServer:
    autoApply:.
      jdbcv2:
        common:
          - fileCopy:
              sourcePath: jars/connectors/thirdparty/informatica.jdbc_v2/common/
driver.jar
          - fileCopy:
              sourcePath: jars/connectors/thirdparty/informatica.jdbc_v2/common/
driver.jar
        spark:
          - fileCopy:
              sourcePath: jars/connectors/thirdparty/informatica.jdbc_v2/spark/driver.jar
          - fileCopy:
              sourcePath: jars/connectors/thirdparty/informatica.jdbc_v2/spark/driver.jar
```

Copying Java transformation JAR files

To copy JAR files, native libraries, and/or native binaries for the Java transformation, use the following code snippet as a template:

```
agent:
  elasticServer:
    autoApply:
      javaTx:
        resources:
          - fileCopy:
              sourcePath: j_depends/sapjco3.jar
          - fileCopy:
              sourcePath: j_depends/chilkat.jar
          - fileCopy:
              sourcePath: j_depends/chilkatLoader.jar
          - fileCopy:
              sourcePath: j_depends/NativeBinExecutor.jar
       nativeLib:
        resources:
          - fileCopy:
              sourcePath: native-lib/libchilkat.so
          - fileCopy:
              sourcePath: native-lib/libsapjco3.so
      nativeBin:
        resources:
          - fileCopy:
              sourcePath: native-bin/printEnvVariables.sh
          - fileCopy:
              sourcePath: native-bin/printProcess.sh
```

Copying Python transformation resource files

To copy resource files for the Python transformation, use the following code snippet as a template:

```
agent:
  elasticServer:
    autoApply:
      pythonTx:
        resources:
        - fileCopy:
            sourcePath: py_depends/res1.txt
        - fileCopy:
            sourcePath: py_depends/res2.txt
```

Adding files while the environment is running

You can add files for the Elastic Server to the supplementary file location while the serverless runtime environment is running. Files for the Elastic Server include JDBC V2 Connector JAR files, Java transformation JAR files, and Python transformation resource files.

To add a file while the environment is running, complete the following tasks:

1. Add the file to the appropriate location in `<Supplementary file location>/serverless_agent_config/`.
2. Specify the file in the serverlessUserAgentConfig.yml file.

It can take up to 10 minutes for the file to synchronize to the serverless runtime environment.

Redeploy the serverless runtime environment if you perform any of the following tasks:

- Update an existing file.

  To update an existing file while the serverless runtime environment is running, add the file to the supplementary file location and to the serverlessUserAgentConfig.yml file using a different name.
- Add other file types, such as ODBC shared libraries.
- Add a new folder or directory, such as a Python installation directory for the Python transformation.
- Remove files from the serverless runtime environment.

Configure TLS to authenticate REST APIs

If you use REST V3 Connector with an API collection or Machine Learning transformation that runs in a serverless runtime environment, you can configure TLS to establish one-way or two-way secure communication to authenticate REST APIs.

Contact Informatica Global Customer Support to request the required custom properties. Make sure that truststore and keystore certificates are in JKS format.

1. Navigate to the supplementary file location on Amazon S3.
2. In the `serverless_agent_config` folder, create a subfolder called `SSL`.
3. Add the truststore and keystore certificates to the `SSL` folder.

   For one-way secure communication, add the truststore certificates. For two-way secure communication, add both the truststore and keystore certificates.

4. Copy the following code snippet to a text editor and add the relative path to each certificate in the supplementary file location:

```
version: 1
agent:
  agentAutoApply:
    general:
        sslStore:
          - fileCopy:
                sourcePath: SSL/<REST V3 truststore certificate name>.jks
          - fileCopy:
                sourcePath: SSL/<REST V3 keystore certificate name>.jks
```

5. In the `serverless_agent_config` folder, open the `serverlessUserAgentConfig.yml` file.
6. Add the code snippet to the `serverlessUserAgentConfig.yml` file and save the file.

   The serverless runtime environment will copy the certificates from the supplementary file location to its own reference directory so that it can use the certificates at run time.
7. In the REST V3 connection properties, use the following format to specify each truststore and keystore file path in the serverless runtime environment: `/home/cldagnt/SystemAgent/serverless/configurations/ssl_store/<certificate name>.jks`

Provide the custom properties to your developer. Developers enter the custom properties in mapping tasks that run in the serverless runtime environment.

Working with serverless runtime environments

Work with your serverless runtime environment to edit its configuration, stop and start it, and delete it. Use Administrator to update runtime environments on AWS. Use the Azure portal to update runtime environments on Microsoft Azure.

Edit

Edit a serverless runtime environment to update the environment properties. To edit the platform configuration properties, stop the runtime environment first.

The following table describes the properties that you can edit based on the runtime environment status:

Status	Description
Running	You can edit the maximum number of compute units and the task timeout. The updated values take effect for future tasks.
Stopped	You can edit all properties. After you save the changes, start the runtime environment to run it using the updated values.
Failed	You can edit all properties. After you save the changes, the runtime environment automatically starts using the updated values.

A serverless runtime environment that uses AWS PrivateLink doesn't detect updates to existing resource shares. To update the resource share, complete the following tasks:

1. In AWS, update the resource configurations in the resource share.
2. Generate a new resource share invite.
3. In Administrator, navigate to the serverless runtime environment.

4. Open the **Actions** menu and select **Stop**.
5. In the runtime environment properties, specify the resource share ID and append each resource configuration and DNS name to the domain map.
6. Select **Save** and then select **Start**.

Stop

Stop a serverless runtime environment to update its platform configuration or to reduce resource usage during periods of inactivity. When a runtime environment stops, it becomes unavailable but it's ready to come back up when you need it.

Before you stop a runtime environment, use Monitor to make sure that no jobs are running in the runtime environment.

Start

Start a serverless runtime environment to bring it back up. Before you run tasks, wait until the runtime environment status is **Running**.

Clone

Clone a serverless runtime environment to create another runtime environment with a similar configuration. Cloning is useful when you want to set up a runtime environment that's nearly identical to an existing one. For example, a runtime environment that connects to a different subnet or uses a different security group.

Delete

Delete a serverless runtime environment when you don't need it anymore.

Before you delete a runtime environment, complete the following tasks:

1. In Monitor, make sure that no jobs are running in the runtime environment.
2. In Administrator or on the Azure portal, use the **Show Dependencies** action to view a list of tasks, mappings, and connections that use the runtime environment. If any assets use the runtime environment, navigate to each asset and update the runtime environment that it uses.

Redeploy

Note: Effective in the October 2026 release, redeploying a serverless runtime environment is deprecated. Informatica intends to drop support for redeployment in a future release. Deprecated functionality is supported, but Informatica intends to drop support in a future release. Informatica requests that you transition to different functionality before the functionality is dropped.

To update a serverless runtime environment, stop the runtime environment, edit it, and restart it. If the runtime environment fails or encounters an error, resolve the error and save the runtime environment to automatically restart it.

Configuring performance and storage settings

Keep integrations running efficiently by configuring performance and storage settings for your serverless runtime environment. You can update these settings while the runtime environment is running.

Modify Secure Agent service properties

Modify Secure Agent service properties to optimize performance or to apply settings that Informatica Global Customer Support recommends.

On the **Runtime Configuration** tab of the serverless runtime environment, update service properties or add custom properties. To reset all Secure Agent service properties to their system defaults, select **Reset All**.

For more information about the Secure Agent services and their properties, see *Secure Agent Services*.

Set the number of simultaneous tasks

By default, a serverless runtime environment can run up to 150 tasks at the same time.

To lower the limit, edit the maxDTMProcesses property on the **Runtime Configuration** tab of the serverless runtime environment. Filter the list by setting the service to **Data_Integration_Server** and the type to **Tomcat**. You can set the property to any value between 1 and 150.

Configure where Data Integration reads and writes files

Set default directories to configure where Data Integration reads and writes files, such as source files, target files, and temporary files.

To update a directory, edit the appropriate system variable on the **Runtime Configuration** tab of the serverless runtime environment. Filter the list by setting the service to **Data_Integration_Server** and the type to **PMRDTM_CFG**.

Note: Directory names can't contain the following special characters: `* ? < > " | ,`

The following table describes the system variables:

System Variable	Description
$PMLookupFileDir	Directory for lookup files. Default is `$PMRootDir`
$PMBadFileDir	Directory for reject files. Default is `$PMRootDir/error`
$PMCacheDir	Directory for index and data cache files. Default is `$PMRootDir/cache`
$PMStorageDir	Directory for state of operation files, used for recovery in high availability or recovery-enabled workflows. Default is `$PMRootDir`
$PMTargetFileDir	Directory for target files. Default is `$PMRootDir`
$PMSourceFileDir	Directory for source files. Default is `$PMRootDir`
$PMExtProcDir	Directory for external procedures. Default is `$PMRootDir`
$PMTempDir	Directory for temporary files. Default is `$PMRootDir/temp`

Connectors in a serverless runtime environment

When you create a connection, you can specify a serverless runtime environment configured on AWS or Azure.

The list of connectors applicable to the serverless runtime environment configured on Azure and AWS might vary.

AWS serverless runtime environment using AWS PrivateLink for service network endpoints

You can use the serverless runtime environment configured on AWS for the following connectors:

Amazon Redshift V2 Connector	Oracle Connector
Amazon S3 V2 Connector	PostgreSQL Connector
File Processor	REST V2 Connector
JDBC V2 Connector	Salesforce Connector
Kafka Connector	Salesforce Marketing Cloud Connector
Microsoft SQL Server Connector	ServiceNow Connector
MongoDB V2 Connector	Snowflake Data Cloud Connector
ODBC Connector	Workday V2 Connector

AWS serverless runtime environment using AWS cross-account elastic network interfaces

You can use the serverless runtime environment configured on AWS for the following connectors:

Amazon Athena Connector	Microsoft SQL Server Connector
Amazon Redshift V2 Connector	MongoDB V2 Connector*
Amazon DynamoDB V2 Connector*	MySQL Connector
Amazon S3 V2 Connector	NetSuite RESTlet V2
BigMachines Connector	OData Connector
Box Connector	OData Consumer Connector
Concur V2 Connector	OData V2 Protocol Reader Connector
Coupa V2 Connector	ODBC Connector
Cvent Connector	Oracle Connector
Databricks Delta Connector	PostgreSQL Connector
Dropbox Connector	REST V2 Connector
Eloqua Bulk API Connector	REST V3 Connector
Google Analytics Connector	Salesfore Oauth Connector
Google BigQuery V2 Connector	Salesforce Connector
Google Cloud Storage V2 Connector	Salesforce Marketing Cloud Connector
JDBC (JDBC_IC) Connector	SAP ADSO Writer Connector
JDBC V2 Connector	SAP BAPI Connector
JIRA Connector	SAP BW Bex Query Connector
Kafka Connector*	SAP Table Connector
Marketo REST Connector	SAP ODP Extractor Connector
Marketo V3 Connector	SAP HANA Connector
Microsoft Azure Blob Storage V3 Connector	ServiceNow Connector
Microsoft Azure Cosmos DB SQL API Connector	Snowflake Data Cloud Connector
Microsoft Azure Data Lake Storage Gen1 V3 Connector	SuccessFactors ODATA Connector
Microsoft Azure Data Lake Storage Gen2 Connector	Web Service Consumer Connector
Microsoft Azure Synapse SQL Connector	Workday V2 Connector
Microsoft CDM Folders V2 Connector	Xero Connector
Microsoft Dynamics 365 for Operations Connector	Xactly Connector
Microsoft Dynamics 365 for Sales Connector	Zendesk V2 Connector
MongoDB Connector	

*Serverless runtime environment in AWS applies for the connector in mappings in advanced mode only.

Azure serverless runtime environment

You can use the serverless runtime environment configured on Azure for the following connectors:

- Amazon Athena Connector
- Amazon Redshift V2 Connector
- Amazon S3 V2 Connector
- Azure AI Search Connector*
- BigMachines Connector
- Concur V2 Connector
- Coupa V2 Connector
- Cvent Connector
- Databricks Delta Connector
- Db2 for i Database Ingestion Connector
- Db2 for zOS Database Ingestion Connector
- DB2 Warehouse on Cloud Connector
- Dropbox Connector
- Eloqua Bulk API Connector
- Google Analytics Connector
- Google BigQuery V2 Connector
- Google Cloud Storage V2 Connector
- JDBC (JDBC_IC) Connector
- JDBC V2 Connector
- JIRA Connector
- Kafka Connector*
- Marketo V3 Connector
- Microsoft Azure Blob Storage V3 Connector
- Microsoft Azure Cosmos DB SQL API Connector
- Microsoft Azure Data Lake Storage Gen1 V3 Connector
- Microsoft Azure Data Lake Storage Gen2 Connector
- Microsoft Azure Synapse SQL Connector
- Microsoft Azure Synapse Analytics Database Ingestion Connector
- Microsoft CDM Folders V2 Connector
- Microsoft Dynamics 365 for Operations Connector
- Microsoft Dynamics 365 for Sales Connector
- Microsoft Dynamics 365 Mass Ingestion Connector
- Microsoft Fabric Data Warehouse Connector
- Microsoft Fabric Lakehouse Connector
- Microsoft Fabric OneLake Connector
- Microsoft SQL Server Connector
- MongoDB Mass Ingestion
- MongoDB V2 Connector*
- MySQL Connector
- NetSuite Mass Ingestion Connector
- OData Connector
- OData Consumer Connector
- OData V2 Protocol Reader Connector
- ODBC Connector
- Oracle Connector
- Oracle Database Ingestion Connector
- Oracle Fusion Cloud Mass Ingestion Connector
- PostgreSQL Connector
- REST V2 Connector
- REST V3 Connector
- Salesforce Connector
- Salesforce Marketing Cloud Connector
- Salesforce Mass Ingestion Connector
- SAP ADSO Writer Connector
- SAP BAPI Connector
- SAP ODP Extractor Connector
- SAP HANA Connector
- SAP HANA Database Ingestion Connector
- SAP OData V2 Connector
- SAP Mass Ingestion
- ServiceNow Connector
- ServiceNow Mass Ingestion Connector
- Snowflake Data Cloud Connector
- SuccessFactors ODATA Connector
- Web Service Consumer Connector
- Workday V2 Connector
- Workday Mass Ingestion Connector
- Xero Connector
- Xactly Connector
- Zendesk V2 Connector
- Zendesk Mass Ingestion Connector

*Serverless runtime environment on Azure applies for the connector in mappings in advanced mode only.

CHAPTER 6

Secure Agents

The Informatica Cloud Secure Agent is a lightweight program that runs all tasks and enables secure communication across the firewall between your organization and IDMC. When the Secure Agent runs a task, it connects to the Informatica Cloud hosting facility to access task information. It connects directly and securely to sources and targets, transfers data between them, orchestrates the flow of tasks, runs processes, and performs any additional task requirement.

If the Secure Agent loses connectivity to IDMC, it tries to reestablish connectivity to continue the task. If it cannot reestablish connectivity, the task fails.

The Secure Agent uses pluggable microservices for data processing. For example, the Data Integration Server runs all data integration jobs, and Process Server runs application integration and process orchestration jobs. Each Secure Agent service has a unique set of configuration properties, such as Tomcat and Tomcat JRE settings. For more information about Secure Agent services, see *Secure Agent Services*.

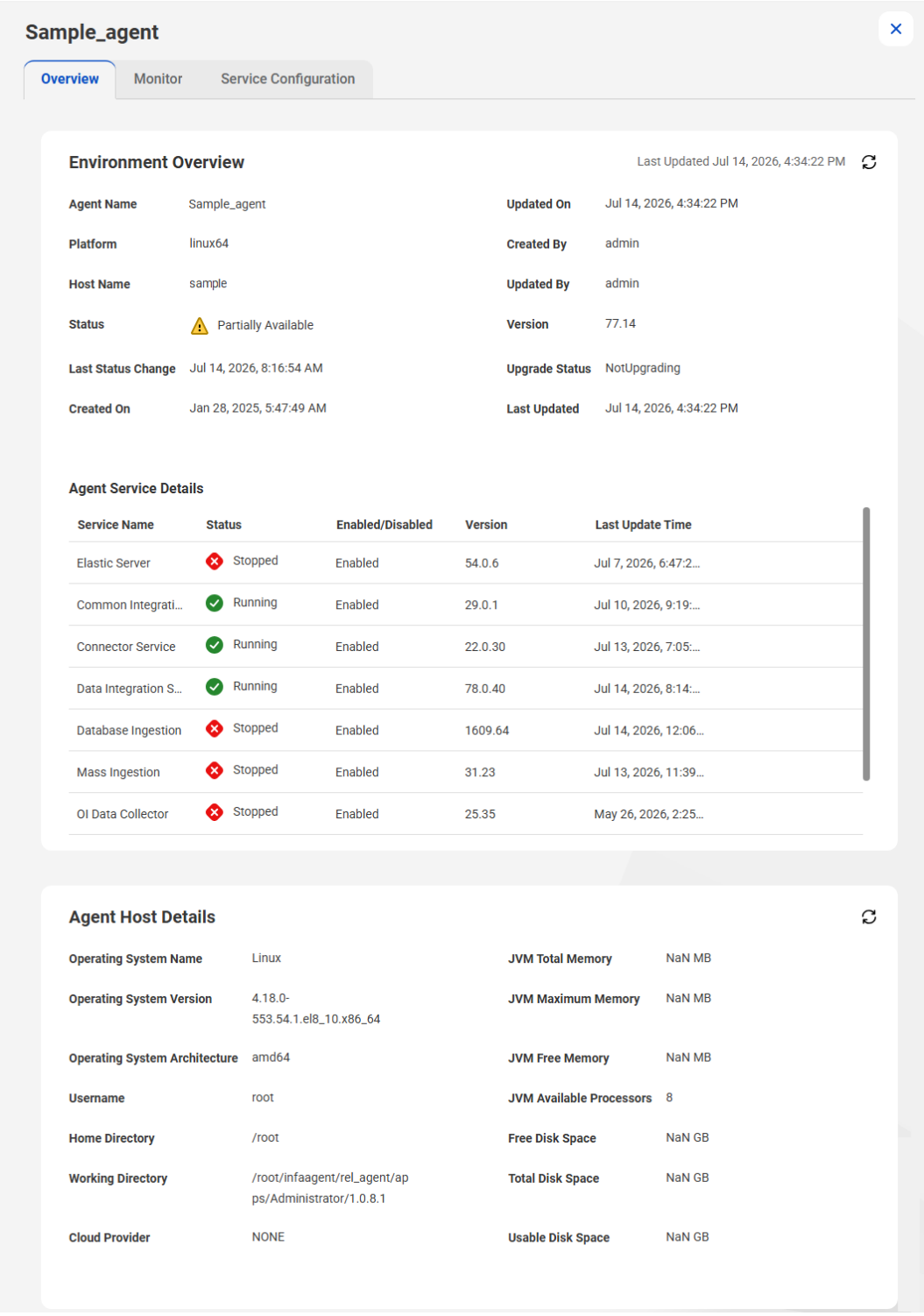
You can install and run one Secure Agent on a physical or virtual machine. After you install a Secure Agent, all users in the organization share the Secure Agent. You can configure the Secure Agent properties and move it to a different Secure Agent group. To improve scalability, you can also add multiple agents to a Secure Agent group.

Working with Secure Agents

After you create a Secure Agent, you might need to perform management tasks such as viewing and configuring agent properties, checking the host information, viewing audit logs, or refreshing the agent status. You can also delete a Secure Agent if it is no longer used.

You perform most management tasks for Secure Agents on the agent details page. To access the agent details page, click a Secure Agent on the **Runtime Environments** page.

The following image shows the agent details page:



You can perform the following tasks on the agent details page:

- View the Secure Agent details.
- View the Secure Agent host properties.

- Refresh the Secure Agent status.
- View the Secure Agent service details.
- View the Secure Agent service package details.
- View and edit Secure Agent service properties.
- View the audit log.

View the Secure Agent details

The **Environment Overview** section on the Overview tab of the agent details page displays details such as the host name, the current status, the last date and time that the agent was updated, and the agent version.

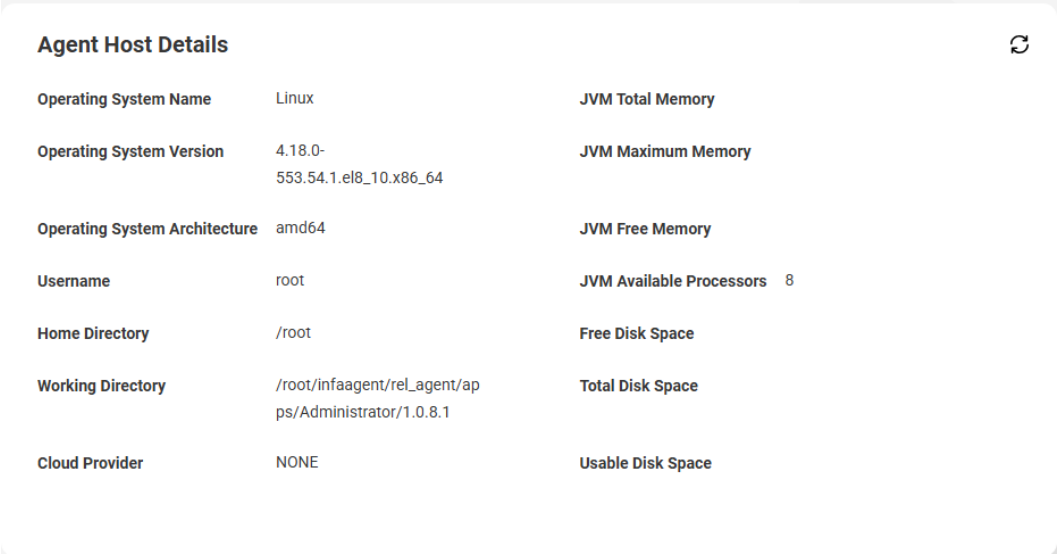
The Secure Agent can have any of the following statuses:

Status	Description
Agent Core is not running	Secure Agent is not available, but one or more of the services is running.
Not all the services are running	Secure Agent is available, but one or more of the services is not available.
Agent Core Upgrading	Secure Agent is upgrading to a new version.
Stopped	Secure Agent is not available.
Up and Running	Secure Agent and all of the services that the agent runs are available.

View the Secure Agent host properties

The **Agent Host Details** section in the **Overview** tab of the agent details page displays information about the machine that hosts the Secure Agent. For example, you can view the machine name, operating system, and available disk space.

The following image shows an example of the Secure Agent host details:



All information shown here is display only and can't be changed.

Refresh the Secure Agent status

Refresh the Secure Agent status to view the most up to date information.

To refresh the status of the Secure Agent, click the refresh icon in the upper right corner of any of the tabs.

You can also view the status on Linux by performing the following steps:

1. Navigate to the following directory: `<Secure Agent installation directory>/apps/agentcore`
2. Run one of the following commands:

```
./consoleAgentManager.sh getstatus
./consoleAgentManager.sh updatestatus
```

View the Secure Agent service details

In the **Agent Service Details** section of the **Overview** tab on the agent details page, you can view details for Secure Agent services that run on the Secure Agent such as the service name, status, version, and last update time.

A Secure Agent service can have any of the following statuses:

Status	Description
Error	The process failed.
Restarting Due to Error	The service is starting due to a failure.
Shutting Down	The service is shutting down.
Standby	The service is running, but it is not compatible with IDMC.
Starting Up	The service is starting up.
Stopped	The service is not available.
Running	The service is running.
User Stopped	The service was stopped by a user.
Warning	The service is running, but it cannot accept work.

The version number changes each time you modify the service. The Secure Agent retains the directories for the old version of the service for seven days. For example, if you update the NetworkTimeoutPeriod for version 55.0.2 of the Data Integration Server, the agent increments the version number to 55.0.3 and creates the following directory:

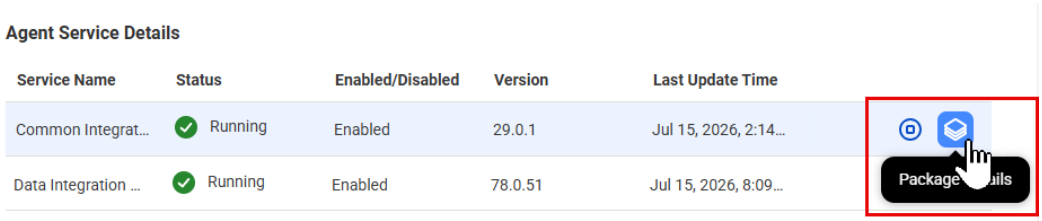
`<Secure Agent installation directory>/apps/Data_Integration_Server/55.0.3.1`

It deletes the `<Secure Agent installation directory>/apps/Data_Integration_Server/55.0.2.x` directories after seven days.

View the Secure Agent service package details

You can see the name and version number for the packages in each service that runs on the Secure Agent. You can filter the packages by service.

To see package details for a service, on the **Overview** tab, click the package icon in the row that contains the service. The following image shows the location of the icon:



View and edit Secure Agent service properties

You can details about the service properties for all the services that run on the Secure Agent. You can filter the properties by service and type.

To see package details for a service, on the **Overview** tab, click the package icon in the row that contains the service. To view or edit a property, on the **Service Configuration** tab, click the edit icon in the row that contains the service.

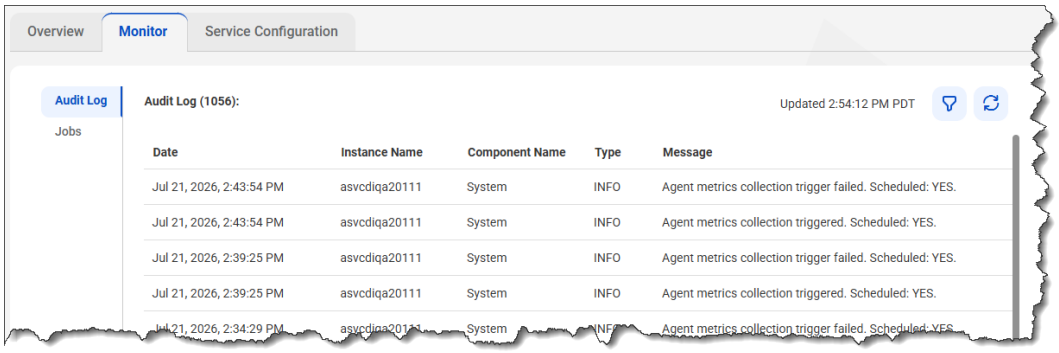
You can configure properties for each service that runs on the Secure Agent. You can also add and remove custom properties, which are used by connectors. These custom properties appear on the **Custom Configuration** tab.

For more information about Secure Agent services and service properties, see *Secure Agent Services*. For more information about custom properties, see the help for the appropriate connector.

View the audit log

You can view audit information such as start and stop times, server connections, and upgrade messages for a Secure Agent.

To view audit information, select the **Monitor** tab as shown in the following image:



Stopping and restarting the Secure Agent on Windows

The Secure Agent Manager displays the Secure Agent status. You can use the Secure Agent Manager to stop or restart the Secure Agent.

Launch the Secure Agent Manager from the Windows **Start** menu. If the Secure Agent Manager is active, you can click the Informatica Cloud Secure Agent Manager icon in the Windows taskbar notification area to open the Secure Agent Manager.

To stop the Secure Agent from the Secure Agent Manager, click **Stop**. To restart the Secure Agent, click **Restart**. The Secure Agent Manager displays a message when the action is complete.

When you close the Secure Agent Manager, it minimizes to the Windows taskbar notification tray. Closing the Secure Agent Manager does not stop the Secure Agent.

Stopping and restarting the Secure Agent on Linux

After you download the Secure Agent program files on a Linux machine, you can run the Secure Agent as a Linux process. Manually start the Secure Agent process on Linux.

1. From the command line, navigate to the following directory:

   ```
   <Secure Agent installation directory>/apps/agentcore
   ```

2. To start the Secure Agent, enter the following command:

   ```
   ./infaagent.sh startup
   ```

3. To stop the Secure Agent, enter the following command:

   ```
   ./infaagent.sh shutdown
   ```

You can view the Secure Agent status from IDMC or from a Linux command line.

Note: Effective in the July 2024 release, the older `infaagent startup` and `infaagent shutdown` commands will no longer work.

Deprecated functionality is supported, but Informatica intends to drop support in a future release. Informatica requests that you transition to different functionality before the functionality is dropped.

Stopping and starting services on a Secure Agent

By default, each Secure Agent in an organization runs all microservices that are used for data processing in the organization. You can stop and start the microservices to perform troubleshooting, optimize resources on the agent machine, or make configuration changes. When you stop or start a Secure Agent microservice, other microservices that run on the agent are not affected.

The microservices that you stop and start on a Secure Agent are the Secure Agent services, which are different from the IDMC. For example, if you want to stop the services associated with Operational Insights, you must stop the OI Data Collector service on the agent. For more information about Secure Agent services, see *Secure Agent Services*.

You might need to stop and restart a Secure Agent service in the following circumstances:

You need to troubleshoot issues with a specific Secure Agent service.

If a Secure Agent service shows an error state, you can stop the service, troubleshoot the problem, and then restart the service.

You are running memory or CPU intensive jobs, and you want to optimize computing resources on the Secure Agent machine.

For example, your organization runs Data Integration and Application Integration jobs. You want to optimize computing resources so that the Data Integration jobs run during the day and the Application Integration jobs run at night. To do this, stop Process Server during the day and restart it in the evening, and stop the Data Integration Server at night and restart it in the morning.

You update service configuration properties for the File Integration Service.

After you change configuration properties for the File Integration Service, you must restart the service. If the Secure Agent runs other services, you can stop and restart the File Integration Service without affecting the other services.

To start or stop a service on a Secure Agent, you must have update permission on the Secure Agent.

If you are the administrator of a sub-organization, you can start and stop services on the agents in the sub-organization. However, you cannot start and stop services on a Secure Agent that is in a shared Secure Agent group.

Each time you start and restart a service, the Secure Agent creates a new subdirectory for the service-related files. For example, if the Secure Agent uses version 64.0.38 of the Data Integration Server, the Secure Agent installation directory contains the following subdirectory:

```
<Secure Agent installation directory>/apps/Data_Integration_Server/64.0.38.1
```

When you stop and restart the Data Integration Server, the Secure Agent creates the following directory:

```
<Secure Agent installation directory>/apps/Data_Integration_Server/64.0.38.2
```

The Secure Agent does not delete the `.../64.0.38.1` directory.

Example

Your organization uses Data Integration and has licenses for Enterprise Data Catalog integration, file integration, and mass ingestion.

Your Secure Agent runs the following Secure Agent services:

- Data Integration Server
- EDC Search Agent
- File Integration Service
- Mass Ingestion

If you have issues with Enterprise Data Catalog search, you can stop the EDC Search Agent service while you perform troubleshooting. When you stop the EDC Search Agent service, you cannot perform data catalog discovery in Data Integration. However, jobs processed by the other services on this agent such as mappings, tasks, taskflows, and AS2 file transfers continue to run.

Guidelines for stopping and starting Secure Agent services

Use the following guidelines when you stop and start services on a Secure Agent:

- Use caution when you stop Secure Agent services because this can cause job failures.

  When you stop a Secure Agent service, any job that requires the service and is currently running on the agent stops. If there are no other agents in the group, the job can no longer run. If there are other agents in the group, you can restart the job and it will run on a different agent.
- Do not stop the Data Integration Server on an agent if you store connection properties with the agent.

  If you store connection properties with a local Secure Agent and you stop the Data Integration Server on the agent, users will not be able to access any connections or run tasks in the organization. Any job that is currently running on the agent also fails.
- Do not stop and start services to reserve a Secure Agent group for certain types of jobs.

  If you want to reserve a Secure Agent group for certain types of jobs, you can enable the required services for the Secure Agent group and disable other services. For more information about enabling and disabling services for a Secure Agent group, see "Service and connector assignment for Secure Agent groups" on page 12.

Stopping a Secure Agent service

You can stop a Secure Agent service that is in the "Running" or "Error" states to perform troubleshooting, optimize resources on the agent machine, or make service configuration changes. Stopping a Secure Agent service stops all versions of the service that are running, but doesn't affect any other service on the agent.

Note: If you stop a Secure Agent service and then restart the Secure Agent, the service remains stopped until you start it.

1. In Administrator, select **Runtime Environments**.
2. On the **Runtime Environments** page, click the name of the Secure Agent.

   Note: You might have to expand the Secure Agent group to see the list of Secure Agents within the group.
3. To stop a service, on the **Overview** tab, click the stop icon in the row that contains the service.

The Secure Agent service stops, and IDMC adds an entry in the audit log indicating that the service was stopped by a user.

Starting a Secure Agent service

You can start a Secure Agent service that is in the "Stopped" state. Starting a Secure Agent service starts the latest version of the service.

1. In Administrator, select **Runtime Environments**.
2. On the **Runtime Environments** page, click the name of the Secure Agent.

   Note: You might have to expand the Secure Agent group to see the list of Secure Agents within the group.
3. To start a service, on the **Overview** tab, click the start icon in the row that contains the service.

IDMC attempts to start the Secure Agent service. After the service starts, the status changes to "Running." If the Secure Agent service fails to start, check the audit log to find the cause of the error.

Configuring agent blackout periods

You can configure blackout periods for a Secure Agent. Blackout periods prevent data integration jobs from running on the agent during a certain period. Configure an agent blackout period to configure specific hours, days, or intervals in which no data integration jobs can run on the agent.

Agent blackout periods stop the Data Integration Server service from running jobs on a Secure Agent during the blackout period. They do not prevent other types of jobs from running on the agent. Configure an agent blackout period in the following circumstances:

- The Data Integration Server is the only service enabled on the agent and you want to stop all data integration jobs from running during a certain period.
- The Secure Agent runs multiple services and you want to stop only the data integration jobs from running during a certain period.

Note: The agent blackout period is different than the schedule blackout period for the organization. During an organization's schedule blackout period, no jobs can run on any agent. For more information about schedule blackout periods, see *Organization Administration*.

To configure a blackout period on a Secure Agent, you must create a blackout file. The blackout file is an XML file that specifies the repeat frequency, start date, and end date for each blackout period.

For example, the following blackout file contains two blackout periods: one blackout period from July 27, 2021, 5:00 AM through July 28, 2021, 11:00 PM and a second blackout period that repeats on Fridays from 2:00-4:00 PM:

```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<BlackoutWindows>
    <BlackoutWindow>
        <RepeatFrequency>OneTime</RepeatFrequency>
        <Start>2021-07-27 5:00:00</Start>
        <End>2021-07-28 23:00:00</End>
    </BlackoutWindow>
    <BlackoutWindow>
        <RepeatFrequency>Friday</RepeatFrequency>
        <Start>14:00:00</Start>
        <End>16:00:00</End>
    </BlackoutWindow>
</BlackoutWindows>
```

To configure one or more blackout periods, create a file named "blackoutWindows.dat" in the following directory on the Secure Agent machine:

```
<Secure Agent Installation Directory>\apps\Data_Integration_Server\conf\
```

If the Secure Agent is in a Secure Agent group, copy the blackout file to the `...\conf\` directory on each agent machine in the group.

If you want to use a different file name and directory, you can override the file name and file path.

After you create a blackout file, restart the Data Integration Server service on the Secure Agent so that the blackout periods take effect.

Overriding the blackout file name and directory

You can override the blackout file name and directory by setting the BlackoutWindowsFile Tomcat custom property for the Data Integration Server.

Set the following custom property for the Data Integration Server on the agent details page:

Service	Type	Name	Value
Data Integration Server	Tomcat	BlackoutWindowsFile	File path and file name for the blackout file. For example: `C:/AgentBlackouts/Agent001Blackouts.dat` **Note:** Use forward slashes (/) in the file path on both Windows and UNIX machines because the Secure Agent interprets backslashes (\) as escape characters. The file path must be accessible by the Secure Agent.

For more information about configuring custom properties for a Secure Agent service, see *Secure Agent Services*.

Blackout file structure

The blackout file is an XML file that contains elements that define each blackout period and the frequency, start time, and end time for each blackout period.

The blackout file has the following structure:

```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<BlackoutWindows>
    <BlackoutWindow>
        <RepeatFrequency></RepeatFrequency>
        <Start></Start>
        <End></End>
    </BlackoutWindow>
    <BlackoutWindow>
        <RepeatFrequency></RepeatFrequency>
        <Start></Start>
        <End></End>
    </BlackoutWindow>
    …
</BlackoutWindows>
```

The file contains the following elements:

Element	Required/ Optional	Description
BlackoutWindows	Required	Contains a BlackoutWindow element for each blackout period. Must contain one or more BlackoutWindow elements.
BlackoutWindow	Required	Defines one blackout period. Must contain one RepeatFrequency element, one Start element, and one End element.

Element	Required/ Optional	Description
RepeatFrequency	Required	Repeat frequency for the blackout period. Must contain one of the following values: - OneTime - Daily - Weekdays - Sunday - Monday - Tuesday - Wednesday - Thursday - Friday - Saturday
Start	Required	Blackout period start time in the format yyyy-mm-dd hh24:mi:ss. For example, 2019-07-25 10:26:55. The time zone is the Secure Agent time zone.
End	Required	Blackout period end time in the format yyyy-mm-dd hh24:mi:ss. For example, 2019-07-26 11:45:00. The time zone is the Secure Agent time zone.

Do not enclose element values in quotation marks.

Renaming a Secure Agent

By default, the name of a Secure Agent is the same as the name of the machine where you installed the agent. You can change the agent name.

1. On the **Runtime Environments** page, select **Edit Secure Agent Properties** from the Action menu of the Secure Agent to edit.

   Note: You might have to expand the Secure Agent group to see the list of Secure Agents within the group.
2. Edit the Secure Agent name and then click **OK**.

   Note: If you use multi-byte characters in the Secure Agent name and the agent is in a cloud-hosted environment, verify that the environment also supports these characters.

Deleting a Secure Agent

Delete a Secure Agent if you no longer need it to run tasks. Delete a Secure Agent on the **Runtime Environments** page.

Note: You cannot delete a Secure Agent if it is used in a connection or a task. For example, if the Secure Agent is the only agent in a group, and the group is used as the runtime environment for a connection or task, you cannot delete the agent.

1. In Administrator, select **Runtime Environments**.
2. Expand the Actions menu for the Secure Agent and select **Delete Secure Agent**.

   If the Secure Agent is running, a warning message appears. Stopping an active Secure Agent prevents scheduled tasks associated with the Secure Agent from running. Ignore the warning if you do not need the Secure Agent.

If you no longer need the Secure Agent, uninstall the Secure Agent after you delete it.

Upgrading a Secure Agent

The Secure Agent upgrades automatically the first time that you access a new IDMC release. The upgrade process installs a new version of the Secure Agent, updates connector packages, and applies configuration changes for the services that run on the agent. You do not need to upgrade the Secure Agent manually.

However, to prepare for an upgrade, you might need to perform tasks such as ensuring that each Secure Agent machine has enough disk space available for the upgrade. For more information about preparing for an upgrade, see *Administrator What's New*.

Note: A Secure Agent can upgrade only if the agent version is the current major release and the agent is running. For example, the new major release upgrades the Secure Agent to version 77.x. To be upgraded, an agent must be version 76.x and must be running.

For a list of the version numbers for the Secure Agent and its services, see *Secure Agent changes* in the *What's New*.

Migrating a Secure Agent

You might want to migrate a Secure Agent from one machine to another, for example, when the Secure Agent runs on a virtual machine and you want to clone the machine. To migrate a Secure Agent, download and install the Secure Agent onto the new machine, and then add the new agent to the original Secure Agent group.

Warning: Don't start and run an existing Secure Agent on a cloned machine. If you do this, the agent might not start after a product upgrade.

1. Verify that the original Secure Agent is up and running.
2. Download and install the Secure Agent onto the new machine.
3. Remove the new Secure Agent from its Secure Agent group.

   When you do this, the agent is added to a group named "Unassigned Agents." You can add any unassigned agent to a Secure Agent group.

4. Add the new Secure Agent to the original Secure Agent group.

   This ensures that connections and jobs that use the original Secure Agent group as the runtime environment continue to run after migration.

   For more information about adding and removing agents from a Secure Agent group, see "Working with Secure Agent groups" on page 16.
5. If you store connection properties on the original Secure Agent, update the connection credentials on the **Organization** page and select the new Secure Agent.
6. Optionally, remove the original Secure Agent from its Secure Agent group, and then stop the agent on the original machine and uninstall it.

Secure Agent Manager

When you install the Secure Agent on Windows, you also install the Informatica Cloud Secure Agent Manager. The Secure Agent runs as a Windows service. You can launch the Secure Agent Manager from the Windows Start menu or the desktop icon.

Use the Secure Agent Manager to perform the following tasks:

- View the status of the Secure Agent and the services that the Secure Agent runs.
- Stop and restart the Secure Agent.
- Configure Windows settings such as proxy settings and a Windows Secure Agent service login.

The Secure Agent Manager displays the status of the Secure Agent and the services that the Secure Agent runs. If the Secure Agent or one of the services that the Secure Agent runs is not starting or not running, the Secure Agent Manager displays an alert message and a link that you can click to view details.

When you close the Secure Agent Manager, it minimizes to the Windows taskbar for quick access. Closing the Secure Agent Manager does not stop the Secure Agent. When the Secure Agent Manager is minimized, you can view the Secure Agent status by hovering over the Secure Agent Manager icon.

Secure Agent logs

The different Secure Agent services generate their own logs. Use these logs to help you troubleshoot issues.

Tip: In the file paths below, "<install>" represents the directory in which the Secure Agent is installed.

General

The following table describes the general Secure Agent log files and shows the where you can find the files.

File	Description
<install>\apps\agentcore\agentcore.log	Tracks all operations managed by the Secure Agent for agent applications such as upgrade or opening tunnels for communications.
<install>\apps\agentcore\infaagent.log	Contains logs related to bootstrapping the agent.

File	Description
<install>\apps\agentcore\agentUpgrLog.txt	Keeps agent core upgrade timestamps.
<install>\apps\agentcore \consoleAgentManager.log	Contains logs related to agent registrations.
<install>\apps\Administrator\logs\tomcat \tomcat.log	Contains logs related to pre-downloading packages.

B2B Processor

The following table describes the B2B Processor log file and shows where you can find the file:

File	Description
<install>\apps\B2BProcessor \<version>\logs\b2b-aaap.log	Contains information about main execution entry points, failure messages and B2B Processor service status.

CIHProcessor

The CIHProcessor logs are generated when you use a private repository with Cloud Integration Hub.

File	Description
<install>\apps \CIHProcessor \<version>\logs\cih-aaap.log	Contains information about the following items: - Logs on the start and stop of the Cloud Integration Hub agent application. - Logs about CRUD runtime operations on the private staging database. For example, logging create, read, update, and delete actions for CIH topics. Also, logging read or write operations for Cloud Integration Hub publications and subscriptions.
<install>\apps \CIHProcessor \<version>\log\prs-audit.log	Contains entry and exit access logs of the APIs exposed by the CIHProcessor.

Common Integration Components

The following table describes the Common Integration Components log files and shows where you can find the files:

File	Description
<install>\apps\Common_Integration_Components\logs \<version>\app.log	Contains information about the Common Integration Components services.
<install>\apps\Common_Integration_Components\logs \<version>\tomcat.out	Contains log information about the Common Integration Components service stdout.

Data Access Management Proxy service

The following table describes the Data Access Management Proxy service log file and shows where you can find the file:

File	Description
<install>\apps\Data_Access_Management_Proxy \<version>\log\casapp_lcm_<version>.log	Contains a log of the lifecycle commands invoked by the Secure Agent framework.

Data Integration Server

The following table describes the Data Integration Server log files and shows where you can find the files:

File	Description
<install>\apps\agentcore	See the log files described in the General section.
<install>\apps\Data_Integration_Server \logs\tomcat	Contains information about the Tomcat server.
<install>\apps\Data_Integration_Server \logs\tomcat\tomcat<version>.log	Contains information about the Data Integration Server service.
<install>\apps\Data_Integration_Server \<version>\tomcat.out	Contains information about the Data Integration Server stdout logs.
<install>\apps\Data_Integration_Server \<version>\scripts.log	Contains information about the Data Integration Server life cycle scripts, for example: deploy, start, status, and stop scripts.
<install>\apps\Data_Integration_Server \logs\session logs	Contains information about the Data Integration Server task session.

File Integration service

The following table describes the File Integration Service log file and shows where you can find the file:

File	Description
<install>\apps \FileIntegrationService\logs \FileIntegrationService.log	Contains logs related to the File Integration service, which includes the following information: - Logs related to the file servers, such as AS2, HTTPs, SFTP, MLLP - Transfer tasks such as encrypt, decrypt, compress, decompress. - API-triggered jobs, such as sendFiles to SFTP/AS2/FTPs. - Filer server users - Connectivity to fis-proxy

Mass Ingestion runtime

The following table describes the Mass Ingestion Runtime log file and shows where you can find the file:

File	Description
<install>\apps\MassIngestionRuntime\logs\informaticamft.log	Contains log information related to File Mass Ingestion and File Listener logs running the MassIngestionRuntime service. Also contains logs related to the startup and status of this service.

Operational Insights Data Collector

The following table describes the Operational Insights Data Collector log files and shows where you can find the files:

File	Description
<install>\apps\OpsInsightsDataCollector\logs\App.log	Contains startup information for the Operational Insights Data Collector (also known as the "OI Data Collector").
<install>\apps\OpsInsightsDataCollector\logs\datacollector.log	Contains information about on-premises and cloud data collections. Also contains information about the publishing performed by the Operational Insights Data Collector service.

Process Server

The following table describes the Process Server log files and shows where you can find the files:

File	Description
<install>\apps\process-engine\logs\catalina.log	Contains logging information from the Process Server. This is useful for diagnosing runtime errors.
<install>\apps\process-engine\logs\scripts.log	Contains logging from scripts executed by agentcore to start or stop the components of the Process Server, including the database and server itself.
<install>\apps\process-engine\logs\localhost-access.log	Contains information associated with a request, for example: IP address, time, request method such as GET or POST, and the resource from which the request is coming.
<install>\apps\process-engine\logs\PostGreSql\upgrade.log	Contains logging that captures information about if or when the database version is upgraded.
<install>\apps\process-engine\logs\PostGreSql\postgresql.log	Contains logs for the PostgreSQL database, which is useful for diagnosing database-related issues.

Secure Agent Java version

The Secure Agent uses an embedded Java instance and does not rely on external Java installations outside of the Secure Agent installation directory.

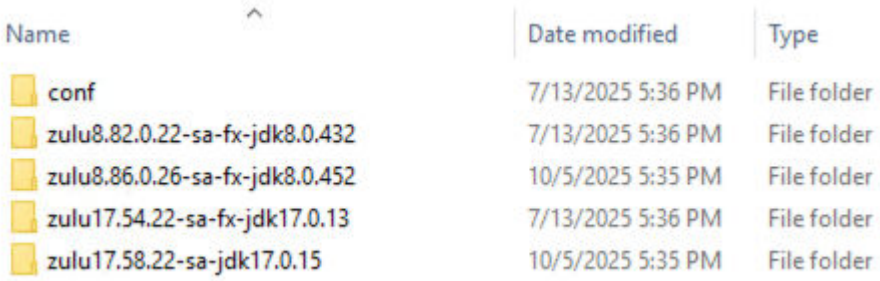
When the Secure Agent is installed initially, the embedded Java instance is located in the following folder:

```
<Secure Agent installation directory>/jdk
```

When new Java versions are installed during upgrades, their packages are stored in the following folder:

```
<Secure Agent installation directory>/apps/jdk
```

The following image shows an example of multiple Java versions:



You can find the current versions of Java in use by examining the files in the following folder:

```
<Secure Agent installation directory>/apps/jdk/conf
```

In the `conf` folder you see the files:

- `jdk17_version`, which lists the version of JDK 17.x in use.
- `version`, which lists the version of JDK 8.x in use.

Troubleshooting a Secure Agent

The Secure Agent did not install or start correctly.

If the Secure Agent does not install or start correctly, complete the following tasks:

1. Review the following logs:

File	Description
<Secure Agent installation directory>\apps\agentcore \infaagent.log	Contains startup and shutdown information about the Secure Agent.
<Secure Agent installation directory>\apps\agentcore \agentcore.log	Contains information about the activities related to the Secure Agent, including details of all the services enabled for the agent.

File	Description
<Secure Agent installation directory>\apps \Data_Integration_Server\logs\tomcat \tomcat<version>.log	Contains details related to the Tomcat process for the Data Integration Server. It records the task execution details, including the start time, end time, and statistics for the tasks. Also includes design time and metadata-related activities for the Data Integration assets.
<Secure Agent installation directory>\apps \Data_Integration_Server\<version>\tomcat.out	Contains basic information about the Tomcat process for the Data Integration Server, including when the Tomcat process started and stopped. Also includes Secure Agent connection details, such as certificate and SSL-related information.
<Secure Agent installation directory>\apps \Data_Integration_Server\<version>\scripts.log	Contains information about the Data Integration Server, including details about how the script used by the Data Integration Server was run. Use this log if there is an issue with the Data Integration Server.

2. For Secure Agents that run on Windows, view the application logs in the Windows Event Viewer.

I started the Secure Agent, but the status is inactive.

The Secure Agent might take a few minutes to start. The status refreshes every 5 seconds. If the Secure Agent does not become active, complete the following tasks:

- If your organization uses a proxy server to access the internet, verify that the proxy settings are set correctly.
- View the details in infaagent.log in the directory where you installed the Secure Agent.

I installed the Secure Agent, but I want to install another on a different machine. How do I do that?

On the new machine, use your login to connect to IDMC. Then, download and install the Secure Agent.

One of my services shows an error status after I restarted the service successfully.

If a service fails with an error status, the error status for the service might continue to display in the Agent Service Details after the service starts up successfully. The error stays on the page until an internal job that cleans up obsolete messages runs. You can ignore the error.

I am trying to uninstall the Secure Agent, but the Secure Agent status still shows "Up and Running."

When you uninstall the Secure Agent without first stopping the Secure Agent, the Agent Core and other services might continue to run for several minutes. To avoid this issue, stop the Secure Agent before you uninstall it.

Why does my Secure Agent always display "Agent Core Upgrading" in Administrator?

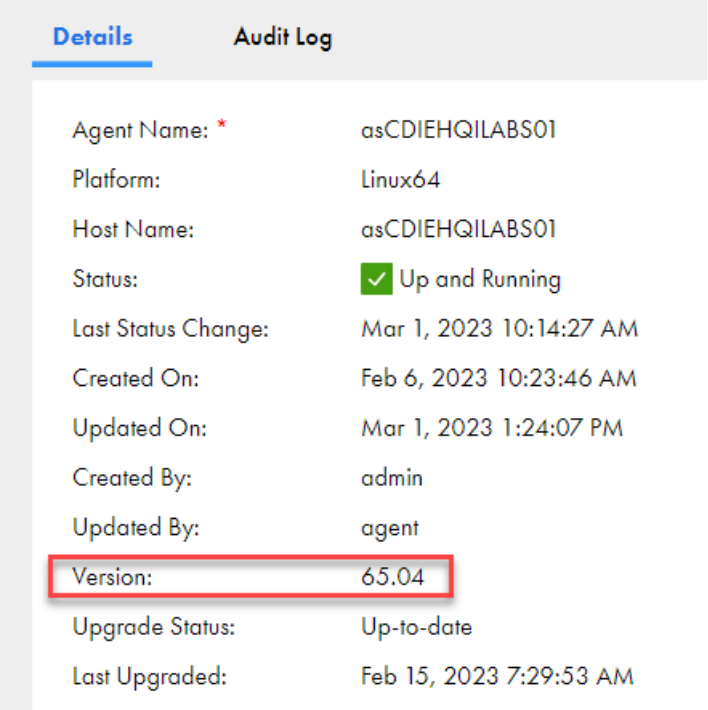
On the **Runtime Environments** page in Administrator, the status of an agent always displays "Agent Core Upgrading". You see the following message in the `agentcore.log` file:

```
2022-10-11 17:02:57,560 GMT tid="21" tn="Agent Core State Machine Thread" ERROR
[com.informatica.saas.infaagent.agentcore.AgentCoreStateMachine] - Authentication failed
due to IO error: [cannot decrypt null or empty string].
```

This issue occurs when the agent missed one or more previous major upgrades. For example, you stopped an agent that was on version 62.x and the current version is 65.x when you restart it. The automatic upgrade only supports upgrading from the previous major version, 64.x. Since your version is older than version 64.x, the automatic upgrade fails.

To resolve the issue, either reregister or reinstall the Secure Agent.

You can see the agent version in the Details tab of a Secure Agent:



CHAPTER 7

Secure Agent installation

You can install the Secure Agent in a number of different ways, including to a cloud environment or local environment.

You can install a Secure Agent in the following ways:

On an AWS, Google Cloud, or Microsoft Azure cloud environment

On AWS, you are redirected to the AWS Marketplace to continue the installation.

On Google Cloud, you first log in using your Google Cloud credentials and then you enter details about the Secure Agent and virtual network. The online wizard completes the installation for you.

On Azure, you first log in using your Azure credentials and then you enter details about the Secure Agent and virtual network. The online wizard completes the installation for you.

When installing in these environments, you do not need to download the Secure Agent installer separately.

In different cloud environment

If you are not using AWS, Google Cloud, or Azure, you first set up a VM in your cloud environment and then follow the appropriates steps for installing the Secure Agent on Windows or Linux.

On a local machine or VM running Windows

On Windows, the Secure Agent runs as a Windows service. Ensure that your Windows environment meets the Secure Agent requirements before downloading the Secure Agent installer.

On a local machine or VM running Linux

On Linux, the Secure Agent runs as a process. Ensure that your Linux environment meets the Secure Agent requirements before downloading the Secure Agent installer.

Installing on AWS

The Secure Agent installer can help you create a runtime environment in supported regions on Amazon Web Services (AWS). The runtime environment you create is a Secure Agent group that contains one Secure Agent.

You can create a runtime environment in the following regions:

- us-east-1
- us-west-2
- ca-central-1

- eu-central-1
- eu-west-1
- eu-west-2
- ap-northeast-1
- ap-southeast-2

When you create a runtime environment on AWS, you create a new stack where the Secure Agent is deployed. You can create the stack in a new or existing virtual private cloud (VPC). The installer creates an Amazon Elastic Compute Cloud (EC2) instance within the VPC.

To create a runtime environment, you must have a subscription with AWS that includes create, modify, and delete privileges for the following resource types:

- EC2 instances
- Elastic IP addresses
- Elastic network interfaces
- Internet gateways
- Route tables
- Security groups
- Subnets
- VPCs

You must also have read and launch permissions for machine images.

Note: Exclude the following directory from security scans: `<Secure Agent installation directory>/apps`. Scanning files in this directory can interfere with the operation of the Secure Agent.

1. In Administrator, select **Runtime Environments**.
2. On the **Runtime Environments** page, click **Manage Cloud Secure Agents**.
3. Click **New Cloud Secure Agent**.
4. Select **Amazon Web Services**.
5. Click **Next**.
6. On the **Environment Configuration** page, copy the install token.

   The install token is valid for 24 hours and can't be reused.
7. Choose whether to create the runtime environment on an existing or new VPC.
8. Click **Continue Configuration in AWS**.

   The AWS **Sign in** screen opens in a new browser tab.
9. Sign in to your AWS account.

   The **Quick create stack** page opens.
10. In the **Stack name** area, enter a stack name.
11. In the **Parameters** area, under **Network Configuration**, configure the following properties based on whether you're using an existing VPC or a new VPC.

- For an existing VPC, configure the following properties:

Property	Value
VPC ID	Select the ID for the VPC where you want to deploy the Secure Agent.
Subnet ID	Enter or select a subnet within the VPC.
Allowed Remote Access CIDR	Enter the CIDR block that specifies the IP addresses where the Secure Agent can be installed. CIDR (Classless Inter-Domain Routing) is a method for allocating IP addresses. It configures a network rule to allow remote access to the Secure Agent. The "/x" portion of the address determines how many IP addresses are available in the subnet, for example: 108.124.81.10/32

- For a new VPC, configure the following properties:

Property	Value
Availability Zones	Select the availability zone for your region.
VPC CIDR	Enter the CIDR block that specifies the IP addresses where you want to create the VPC.
Subnet CIDR	Enter the CIDR block that specifies the IP addresses for the subnet in the availability zone that you selected.
Allowed Remote Access CIDR	Enter the CIDR block that specifies the IP addresses where the Secure Agent can be installed.

12. Under **Amazon EC2 Configuration**, configure the following properties:

Property	Value
Key Pair Name	Enter the name of an existing EC2 key pair to enable external access to the EC2 instance. Corresponding key pair files are required for SSH access to the server.
Instance Type	Select the instance type for the EC2 instance or accept the default. Default is m5.xlarge.
Enable Elastic IP Addressing	Choose whether to assign elastic IP addresses to the EC2 instance or accept the default. Default is no.

13. Under **Informatica Intelligent Data Management Cloud (IDMC) Account Details**, configure the following properties:

Property	Value
IDMC POD Master URL	Accept the default value for the IDMC POD Master URL. This is the URL that you use to access IDMC. **Warning:** Changing this URL can result in stack deployment failure.
IDMC User Name	Enter your IDMC user name.
IDMC User Token	Paste the install token that you copied. If you forgot to copy the install token, you can switch back to IDMC and generate a new one.
Secure Agent Group Name	Accept the default value for the Secure Agent group name. This is the name of the runtime environment that you're creating.

14. Click **Create stack**.

    It takes a few minutes to create the stack. Be sure to monitor the stack creation and address any issues that might occur.

    When the stack is created successfully, the EC2 Instance status changes from CREATE_IN_PROGRESS to CREATE_COMPLETE.

15. In IDMC, on the **Environment Configuration** page, click **Finish**.

    IDMC creates your runtime environment and displays it on the **Runtime Environments** page.

    Tip: To see the progress of your pending Secure Agents, click **Manage Cloud Secure Agents** on the **Runtime Environments** page. The status appears at the top of the page.

It takes a few minutes for the Secure Agent services to start. When the Secure Agent is ready to use, the status changes from "Pending Environment Set Up" to "Up and Running." You might need to refresh the page to see the updated status.

Installing on Google Cloud

The Secure Agent installer can create a runtime environment on Google Cloud for you, based on just a few properties that you enter on the configuration page. Ensure that you have a subscription with Google Cloud that includes permissions to deploy resources.

Note: Exclude the following directory from security scans: `<Secure Agent installation directory>/apps`. Scanning files in this directory can interfere with the operation of the Secure Agent.

1. In Administrator, select **Runtime Environments**.
2. On the **Runtime Environments** page, click **Manage Cloud Secure Agents**.
3. Select **Google Cloud Platform**.
4. Click **Next**.
5. Select the Google account to use.

6. Enter the following properties:

Property	Description
Project	A project defines how IDMC interacts with Google services and what resources it uses. Select your Google Cloud project from the drop-down list. **Note:** If you don't have a project, exit the installation wizard and create your project on Google Cloud. You can't create a project from within IDMC.
Secure Agent Name	Enter a name for your Secure Agent. The name needs to conform to the following rules: - The name can be up to 43 characters long, with a combination of letters, numbers, and hyphens. - The first character must be a lowercase letter. - The last character can't be a hyphen. - All letters must be lowercase. By default, the runtime environment uses the same name as the agent.
Region	Select the region to deploy the Secure Agent. Choose a region that's appropriate for your organization and your customers.
Machine Type	Select the machine type for your virtual machine. If you're not familiar with Google machine types, start with a size with at least 4 cores and 16 GB of memory.
Virtual Network	Specify whether to use an existing virtual network based on your Google subscription or create a new virtual network. A virtual network uses hardware and software to emulate a physical network.
Virtual Network Name	Select an existing virtual network or enter the name for a new virtual network.
Subnet	Select the subnet to use or enter a name for a new subnet.
Subnet Address	Select the subnet address that includes all the resources or enter a new subnet address. Subnet addressing allows a system made up of multiple networks to share the same Internet address.

7. Select the **I acknowledge this action will incur costs on Google Cloud Platform** check box to acknowledge that costs will be incurred on your Google account.
8. Click **Create**.
   IDMC creates your runtime environment and displays it on the **Runtime Environments** page.

Troubleshooting connection issues on Google Cloud

The firewall on Google Cloud can block access to your VM. If this occurs, add a firewall rule to allow RDP and SSH access to your VM instances.

When Google Cloud blocks access, the runtime environment fails to start with the following error:

```
Connection Failed. We are unable to connect to the VM on port 22.
```

1. On the Google Cloud console, go to the **Firewall Rules** page.
2. Click **Create firewall rule**.

3. Create a firewall rule with the following settings:

Setting	Value
Name	Enter a name for the firewall rule. For example: `allow-ingress-from-iap(<name>)`
Direction of traffic	Ingress
Action on match	allow
Target	All instances in the network
Source filter	IP ranges
Source IP ranges	35.235.240.0/20
Protocols and ports	Select **TCP** and enter `22,3389` to allow both RDP and SSH.

4. Click **Create**.

Installing on Microsoft Azure

The Secure Agent installer can configure a runtime environment on Microsoft Azure. Note that running data integration tasks on Azure incurs costs based on the workload and the VM size.

Note: Exclude the following directory from security scans: `<Secure Agent installation directory>/apps`. Scanning files in this directory can interfere with the operation of the Secure Agent.

Before you proceed, ensure that you have a Microsoft Azure subscription with permissions that allow you to deploy resources. If admin consent is enabled at your organization, reach out to the Azure administrator for app consent approvals. For more information about admin consent requests, see the Microsoft documentation.

1. In Administrator, select **Runtime Environments**.
2. On the **Runtime Environments** page, click **Manage Cloud Secure Agents**.
3. Click **New Cloud Secure Agent**.
4. Select **Microsoft Azure**.
5. Click **Next**.
6. Select the Microsoft account to use.

7. Enter the following properties:

Property	Description
Subscription	Select your Microsoft Azure subscription. The subscription must include permissions to deploy the following resources: - Network security group - Virtual network (including subnet) - Network interface - Public IP address - OS disk - Virtual machine Be sure to grant permission to the Hyperscalar Azure Integration App when prompted. **Note:** If you do not have an Azure subscription, exit the installer and sign up for one with Microsoft. You cannot sign up from within IDMC.
Resource Group	A resource group is a container that holds related resources for your runtime environment. IDMC uses one resource group for each Secure Agent to simplify management of the VM resources for that agent. You typically create new resource groups, but you can use any existing group that is empty. **Tip:** Use the same or similar name as the Secure Agent to more easily identify which resource group belongs with each agent.
Resource Group Name	Name of the resource group. Enter the name of a new group or select an existing group. Ensure that any existing resource group is empty, otherwise this message appears*: "API Input validation failed."*
Location	Select the region to deploy the Secure Agent. Choose the Azure region that's appropriate for your organization and your customers. Not every resource is available in every region.
VM Name	Enter a name for the virtual machine (VM) that will be created.
VM User Name	Enter your name as the virtual machine user.
VM Password	Enter a password to access the virtual machine.
Secure Agent Name	Enter a name for your Secure Agent. By default, the runtime environment has the same name as the agent. **Tip:** Use the same or similar name as the resource group, to more easily identify which resource group belongs with each agent.
VM Size	Select a size for your virtual machine. If you are unfamiliar with Azure image sizing, start with a size with at least 4 cores and 16 GB of memory. Note that your Azure hourly charges are affected by the VM size.
Virtual Network	Select an existing virtual network based on your Microsoft Azure subscription and location or create a new virtual network.
Virtual Network Name	Select an existing virtual network or enter the name for a new virtual network. When you select an existing virtual network, this associates the newly created VM with the existing VNet.
Virtual Network Address	Select an existing virtual network address or enter a new address.

Property	Description
Subnet Name	Select the subnet to use or enter a name for a new subnet. The subnet holds all the Azure resources that are deployed to the virtual network.
Subnet Address	Select the subnet address that includes all the resources or enter a new subnet address. Subnet addressing allows a system made up of multiple networks to share the same Internet address.
CIDR IP Address Range	Enter the CIDR IP address range. CIDR (Classless Inter-Domain Routing) is a method for allocating IP addresses. It configures a network rule to allow remote access to the Secure Agent. The "/x" portion of the address determines how many IP addresses are available in the subnet, for example: 108.124.81.10/32

Tip: For more information, refer to "Explore Azure Virtual Networks" in the Microsoft documentation.

8. Click **Create**. Administrator creates your runtime environment and displays it on the **Runtime Environments** page.

   Tip: To see the progress of your pending Secure Agents, click **Manage Cloud Secure Agents** on the **Runtime Environments** page. The status appears at the top of the page.

Installing on Windows

On Windows, the Secure Agent runs as a Windows service. When you install the Secure Agent, you also install the Informatica Cloud Secure Agent Manager.

By default, the Secure Agent starts when you start Windows. You can stop and restart the Secure Agent using the Secure Agent Manager or Windows Services. If you install the Secure Agent on a different volume than you use to run the installation program, you must start and stop the Secure Agent from Windows Services.

You can also use the Secure Agent Manager to check the Secure Agent status and configure proxy information. The Secure Agent works with BASIC, DIGEST, and NTLMv2 proxy authentication.

You can launch the Secure Agent Manager from the Start menu or desktop icon. When you close the Secure Agent Manager, it minimizes to the Windows taskbar notification area for quick access.

When you install a Secure Agent, you perform the following tasks:

1. Verify that the machine meets the minimum requirements.
2. Download the Secure Agent installer files.
3. Install and register the Secure Agent.

Secure Agent requirements on Windows

You can install the Secure Agent on any machine that has internet connectivity and can access IDMC.

Verify the following requirements before you install the Secure Agent on Windows:

- The Secure Agent machine uses a supported operating system. For the list of supported operating systems for the Secure Agent, see the Product Availability Matrix (PAM) for Intelligent Data Management Cloud on the Knowledge Base.

- The Secure Agent machine has the following attributes at a minimum:
  - x86 64-bit architecture
  - 4 CPU cores
  - 16 GB RAM
  - 5 GB free disk space
- The Secure Agent machine is on a volume with at least 250GB disk space, with at least 5 GB free space or three times the size of the Secure Agent installation, whichever is greater.
- The account you use to install the Secure Agent has access to all remote directories that contain flat source or target files.
- No other Secure Agent is installed on the machine. If another Secure Agent is installed on the machine, uninstall it first.

For more information, see this KB article:
Minimum requirements and best practices when installing Informatica Cloud Secure Agent.

Configuring the firewall

If your organization uses a protective firewall, include the IDMC domain name or IP address ranges in the list of approved domain names or IP addresses. To ensure that the Secure Agent can perform all necessary tasks through the firewall, enable the port that the Secure Agent uses.

The Secure Agent uses port 443 (HTTPS) to connect to the internet. Configure your firewall to allow traffic to pass over port 443. Also exclude the following directory from security scans: `<Secure Agent installation directory>/apps`. Scanning files in this directory can interfere with the operation of the Secure Agent.

The list of allowed domains and IP addresses can vary according to your POD (Point of Deployment). You can identify your POD through the URL that appears when you open any service. The first few characters of the URL string identify the POD. For example, if the URL starts with `usw3.dm-us.informaticacloud.com`, your POD is USW3.

For the list of all the domains and IP addresses that you must allow for the different PODs, see Pod Availability and Networking.

Setting Secure Agent permissions on Windows

A Secure Agent requires certain permissions to transfer data between sources and targets.

When you install a Secure Agent on Windows, the Secure Agent must be part of the local Administrators group.

Configuring Windows settings

Before you use the Secure Agent on Windows, configure proxy settings and a Windows Secure Agent service login.

You can configure proxy settings in Secure Agent Manager. Configure a login for the Windows Secure Agent service on Windows.

Note: If you use the Secure Agent for Informatica Cloud Data Wizard, you do not need to configure proxy settings or a Windows service login for the Secure Agent.

Downloading and installing the Secure Agent on Windows

To install the Secure Agent on a Windows machine, download and run the Secure Agent installer and then register the agent.

Before you download and install the Secure Agent, verify that no other Secure Agent is installed on the machine. If any other Secure Agent exists, you must uninstall it first.

Tip: To verify the checksum of the Secure Agent installation program, use the agent REST API version 2 resource. For more information about the agent resource, see *REST API Reference*.

1. Open Administrator and select **Runtime Environments**.
2. On the **Runtime Environments** page, click **Download Runtime Installer**.
3. Select **Secure Agent** as the environment type.
4. Select **Windows 64** as the platform.
5. Click **Copy** to copy the install token.

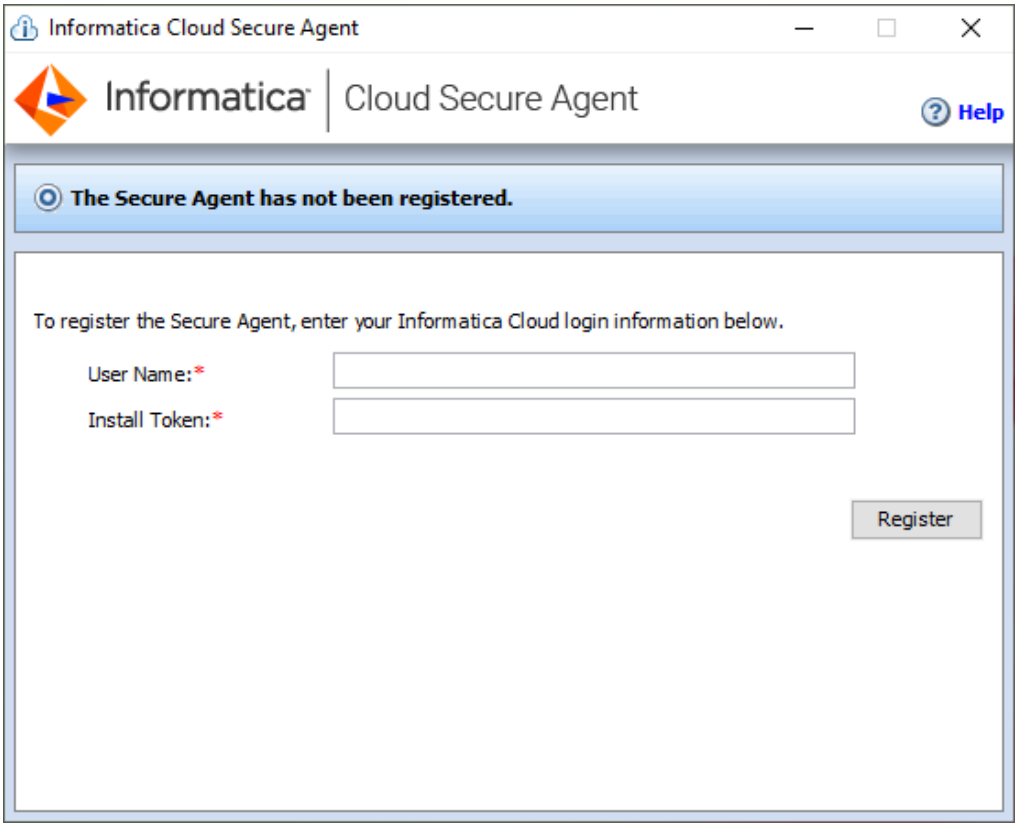
   You will need this token to install the Secure Agent. The token expires after 24 hours. If the token expires before you install the Secure Agent, repeat steps 1 to 5 to obtain a new token. You don't need to download the installer again.
6. Click **Download**.

   The installer is downloaded to your machine. The name of the installer is `agent64_install_ng_ext.<agent core version>.exe`.
7. Run the installer as an Administrator and complete the following steps:
   a. Specify the Secure Agent installation directory, and click **Next**.

      Note: Ensure that the file path doesn't contain multibyte characters. If the path contains multibyte characters, the Secure Agent might not start.

   b. Click **Install** to install the agent.

The **Cloud Secure Agent** dialog box opens and prompts you to register the agent as shown in the following image:



8. In the Secure Agent Manager, enter the following information, and then click **Register**:

Option	Description
User Name	User name that you use to access IDMC.
Install Token	Token that you copied when you downloaded the runtime installer.

The Secure Agent Manager displays the status of the Secure Agent. It takes a minute for all of the services to start.

Tip: When you register the agent, it is added to its own Secure Agent group by default. You can add the agent to a different Secure Agent group if necessary.

9. If your organization uses an outgoing proxy server to connect to the internet, enter the proxy server information.
10. Close the Secure Agent Manager.

    The Secure Agent Manager minimizes to the taskbar and continues to run as a service until stopped.

Configuring the proxy settings on Windows

If your organization uses an outgoing proxy server to connect to the internet, the Secure Agent connects to IDMC through the proxy server.

Important: Activating proxy settings might expose sensitive data to any user with Secure Agent installation privileges. This sensitive data can include connector configurations and credentials. To mitigate the risk of sensitive data leakage, you must strictly limit edit and update permissions on all connections to authorized personnel only. For more information about role privileges, see *User Administration*.

The Secure Agent installer configures default proxy server settings for the Secure Agent based on settings configured in the browser. The proxy settings are stored in the proxy configuration file, `proxy.ini`, in the following directory:

```
<Secure Agent installation directory>/apps/agentcore/conf
```

The following code shows the default contents of the proxy configuration file:

```
InfaAgent.ProxyPassword=ZU8KjIzgtVrVmfRMUPzPMw\=\=
InfaAgent.ProxyNtDomain=
InfaAgent.ProxyHost=foo.bar.com
InfaAgent.ProxyPasswordEncrypted=true
InfaAgent.NonProxyHost=localhost|127.*|[\:\:1]
InfaAgent.ProxyUser=
InfaAgent.ProxyPort=12345
InfaAgent.AuthenticationOrder=
```

To update proxy hosts and ports, as well as the username and password used to connect to the proxy server, use the Secure Agent Manager. The Secure Agent Manager validates the values that you enter. For more information, see "Updating proxy settings through the Secure Agent Manager" on page 141.

To update other properties, edit the proxy configuration file directly. When you edit the proxy configuration file, ensure that you use the correct formats to avoid errors. For more information, see "Updating proxy settings in the proxy configuration file" on page 142.

To allow the Secure Agent to connect to data sources and targets, configure proxy settings through JVM Options in addition to configuring the proxy configuration file. For more information, see the help for the appropriate connector.

Some use cases require the Secure Agent to bypass the proxy server and communicate directly with the instance metadata service (IMDS), which stores temporary security credentials. For example, bypassing the proxy server is required if you use an Azure managed identity to access sources and targets in a mapping in advanced mode and if you enable the **Use EC2 Role to Assume Role** property in an Amazon S3 V2 connection. To allow the agent to communicate directly with the IMDS, edit the proxy configuration file and set the IP address of the IMDS, 169.254.169.254, in the value for the property *InfaAgent.NonProxyHost*.

Contact your network administrator for the correct proxy settings.

Updating proxy settings through the Secure Agent Manager

To update proxy hosts, proxy ports, and the username and password used to connect to the proxy server, edit the proxy settings through the Secure Agent Manager.

1. In the Secure Agent Manager, click **Proxy**.
2. To enter proxy server settings, click **Use a Proxy Server**.

3. Enter the following information:

Field	Description
Proxy Host	Required. Host name of the outgoing proxy server that the Secure Agent uses.
Proxy Port	Required. Port number of the outgoing proxy server.
User Name	User name to connect to the outgoing proxy server.
Password	Password to connect to the outgoing proxy server.

4. Click **OK**.

   The Secure Agent Manager restarts the Secure Agent to apply the settings.

Updating proxy settings in the proxy configuration file

To update proxy settings that you can't update through the Secure Agent Manager, edit the proxy configuration file directly.

1. Open the `proxy.ini` file.
2. Update the property values as required. Use the pipe character (|) as a delimiter to combine a list of host names and IP addresses. You can enter a wildcard to the left for host names or to the right for IP addresses.

   For example, the following value uses two formats to exclude cluster IP addresses in the CIDR block 172.16.0.0/16:

   ```
   InfaAgent.NonProxyHost=localhost|127.|[\:\:1]|123.432.|172.16.*|172.16.0.0/16
   ```
3. Restart the Secure Agent for the changes to take effect.

The proxy details appear on the Secure Agent Manager settings page for the proxy server.

Configuring a login for a Windows Secure Agent service

On Windows, configure a network login for the Secure Agent service. The Secure Agent can access the network with the privileges and permissions associated with the login.

Configure a login for the Secure Agent machine to allow the Secure Agent to access directories to configure and run tasks. When you configure connections, configure tasks, and run tasks that use flat file or FTP/SFTP connection types, the Secure Agent might require read and write permissions on the related directories.

For example, to browse to a directory to configure a flat file or FTP/SFTP connection, the Secure Agent login might require permission to access the directory. Without a Secure Agent login with the appropriate permissions, IDMC cannot display the directory in the **Browse for Directory** dialog box.

1. Go to the **Services** window from the Windows Administrative tools.
2. In the **Services** window, right-click the Informatica Cloud Secure Agent service and choose **Properties**.
3. In the **Properties** dialog box, click the **Log On** tab.
4. To configure a login, select **This Account**.
5. Enter an account and password.

   Use an account with the required privileges and permissions for the network security defined for the domain. By default, the account format is <domain name>\<user name>.

6. Click **OK**.
7. In the **Services** window, restart the Secure Agent service for the changes to take effect.

Uninstalling the Secure Agent on Windows

You can uninstall the Secure Agent. You might uninstall the Secure Agent if you no longer want to run the Secure Agent on the machine or if you want to reinstall the Secure Agent.

Before you uninstall the Secure Agent, verify that no connection or task is configured to use it.

1. Click **Start** > **All Programs** > **Informatica Cloud Secure Agent** > **Uninstall Informatica Cloud Secure Agent**.

   The Secure Agent uninstaller launches.
2. Click **Uninstall**.
3. When the uninstall completes, click **Done**.
4. Delete any remaining files in the installation directory.

   After you uninstall the Secure Agent, delete all files and directories associated with the Secure Agent installation.

Note: Uninstalling the Secure Agent does not delete log files from the Secure Agent directory. If you want to reinstall a Secure Agent on the machine, you must delete all files and directories associated with the Secure Agent installation or reinstallation will fail. If you want to save the log files, copy them to a different directory, and then delete the Secure Agent installation directory.

Installing on Linux

On Linux, the Secure Agent runs as a process. You can use a shell command line to install, register, start, stop, and uninstall the Secure Agent.

You can also use the shell command line to check the Secure Agent status.

When you install a Secure Agent, you perform the following tasks:

1. Verify that the machine meets the minimum requirements.
2. Download the Secure Agent installer files.
3. Install and register the Secure Agent.

Consider the following guidelines:

- Create a specific user profile to install the Secure Agent with full access to all folders from the Secure Agent installation directory. Don't install the Secure Agent as the root user.
- You can't install more than one Secure Agent on the same machine under the same user account. Multiple agents may exist under different user accounts.
- Don't install the Secure Agent on any node within the Informatica domain.

Secure Agent requirements on Linux

You can install the Secure Agent on any machine that has internet connectivity and can access IDMC. Before you install the Secure Agent on Linux, verify the system requirements.

Verify the following requirements before you install the Secure Agent on Linux:

- Verify that the Secure Agent machine uses a supported operating system. For the list of supported operating systems for the Secure Agent, see the Product Availability Matrix (PAM) for Intelligent Data Management Cloud on the Knowledge Base.
- Verify that the Secure Agent machine is running the x86 64-bit architecture, with at least 11 GB free disk space.
- Verify that the `libidn.x86_64` package is installed.
  If the package isn't present, install it using the following command: `sudo yum install libidn.x86_64`

  Note: The command to install the package might vary based on your Linux distribution.
- Verify that the `libidn.so.*` libraries are installed.
  If the libraries aren't present, run the following commands:
  1. Change to the appropriate directory on the Secure Agent machine.
     - For 64-bit systems: `cd /usr/lib/x86_64-linux-gnu`
     - For 32-bit systems: `cd /usr/lib/i386-linux-gnu`
  2. Create a symbolic link using the following command:

     ```
     sudo ln -s libidn.so.12 libidn.so.11
     ```

  If you are installing the Secure Agent on RHEL 9, create an additional symbolic link using the following command:

  ```
  sudo ln -s libidn2.so.0 libidn.so.11
  ```
- The account that you use to install the Secure Agent must have access to all remote directories that contain flat source or target files.
- If you use PowerCenter, install the Secure Agent using a different user account than the account you used to install PowerCenter.
  IDMC and PowerCenter use some common environment variables. If the environment variables are not set correctly for IDMC, your jobs might fail at run time.

For more information, see this KB article:
Minimum requirements and best practices when installing Informatica Cloud Secure Agent.

Configuring the firewall

If your organization uses a protective firewall, include the IDMC domain name or IP address ranges in the list of approved domain names or IP addresses. To ensure that the Secure Agent can perform all necessary tasks through the firewall, enable the port that the Secure Agent uses.

The Secure Agent uses port 443 (HTTPS) to connect to the internet. Configure your firewall to allow traffic to pass over port 443. Also exclude the following directory from security scans: `<Secure Agent installation directory>/apps`. Scanning files in this directory can interfere with the operation of the Secure Agent.

The list of allowed domains and IP addresses can vary according to your POD (Point of Deployment). You can identify your POD through the URL that appears when you open any service. The first few characters of the URL string identify the POD. For example, if the URL starts with `usw3.dm-us.informaticacloud.com`, your POD is USW3.

For the list of all the domains and IP addresses that you must allow for the different PODs, see Pod Availability and Networking.

Setting Secure Agent permissions on Linux

A Secure Agent requires certain permissions to transfer data between sources and targets.

When you install a Secure Agent on Linux, the Secure Agent must have read/write/execute permissions for the installation directory.

Downloading and installing the Secure Agent on Linux

To install the Secure Agent on a Linux machine, download and run the Secure Agent installer and then register the agent.

Before you download and install the Secure Agent, verify that no other Secure Agent is installed on the machine using the same Linux user account. If any other Secure Agent exists, you must uninstall it first.

Tip: To verify the checksum of the Secure Agent installation program, use the agent REST API version 2 resource. For more information about the agent resource, see *REST API Reference*.

1. Open Administrator and select **Runtime Environments**.
2. On the **Runtime Environments** page, click **Download Runtime Installer**.
3. Select **Secure Agent** as the Environment Type.
4. Select **Linux 64** as the Platform.
5. Click **Copy** to copy the install token.

   You will need this token to install the Secure Agent. The token expires after 24 hours. If the token expires before you install the Secure Agent, repeat steps 1 to 5 to obtain a new token. You don't need to download the installer again.
6. Click **Download**.

   The installer is downloaded to your machine. The name of the installer is `agent64_install_ng_ext.<agent core version>.bin`.
7. Save the installer to a directory on the machine where you want to run the Secure Agent.

   Note: Ensure that the file path doesn't contain spaces or multibyte characters. If the file path contains spaces, the installation might fail. If the path contains multibyte characters, the Secure Agent might not start.
8. From a shell command line, navigate to the directory where you downloaded the installation program and enter the following command:

   ```
   ./agent64_install_ng_ext.bin -i console
   ```
9. When the installer completes, navigate to the following directory:

   ```
   <Secure Agent installation directory>/apps/agentcore
   ```
10. To start the Secure Agent, enter the following command:

    ```
    ./infaagent.sh startup
    ```

    The Secure Agent Manager starts. Register the agent using the user name that you use to access IDMC and also supply the install token.
11. To register the agent, in the `<Secure Agent installation directory>/apps/agentcore` directory, enter one of the following commands using your IDMC user name and the token that you copied:

    - To add the agent to its own Secure Agent group, use the following command:

      ```
      ./consoleAgentManager.sh configureToken <user name> <install token>
      ```

- To add the agent to an existing Secure Agent group, use the following command:

```
./consoleAgentManager.sh configureTokenWithRuntime <user name> <install token>
<Secure Agent group name>
```

Note: If the command includes a Secure Agent group name that doesn't exist, the Secure Agent is not assigned to a group. Be sure to use a valid Secure Agent group name.

The following table lists the command options:

Option	Description
User Name	Required. IDMC user name of the user installing the Secure Agent.
Install Token	Required. The install token that you copied when you downloaded the runtime installer.
Secure Agent group name	Optional. Include a group name when you want to add the agent to an existing Secure Agent group instead. If this option isn't included in the command, the agent will be in its own Secure Agent group.

You can check the registration status of a Secure Agent using the following command:

```
./consoleAgentManager.sh isConfigured
```

Configuring the proxy settings on Linux

If your organization uses an outgoing proxy server to connect to the internet, the Secure Agent connects to IDMC through the proxy server.

Important: Activating proxy settings might expose sensitive data to any user with Secure Agent installation privileges. This sensitive data can include connector configurations and credentials. To mitigate the risk of sensitive data leakage, you must strictly limit edit and update permissions on all connections to authorized personnel only. For more information about role privileges, see *User Administration*.

The Secure Agent installer configures default proxy server settings for the Secure Agent based on settings configured in the browser. The proxy settings are stored in the proxy configuration file, `proxy.ini`, in the following directory:

```
<Secure Agent installation directory>/apps/agentcore/conf
```

The following code shows the default contents of the proxy configuration file:

```
InfaAgent.ProxyPassword=ZU8KjIzgtVrVmfRMUPzPMw\=\=
InfaAgent.ProxyNtDomain=
InfaAgent.ProxyHost=foo.bar.com
InfaAgent.ProxyPasswordEncrypted=true
InfaAgent.NonProxyHost=localhost|127.*|[\:\:1]
InfaAgent.ProxyUser=
InfaAgent.ProxyPort=12345
InfaAgent.AuthenticationOrder=
```

To update proxy hosts and ports, as well as the username and password to connect to the proxy server, use the command line. The command line validates the values that you enter. For more information, see "Updating proxy settings through the command line" on page 147.

To update other properties, edit the proxy configuration file directly. When you edit the proxy configuration file, ensure that you use the correct formats to avoid errors. For more information, see "Updating proxy settings in the proxy configuration file" on page 147.

To allow the Secure Agent to connect to data sources and targets, configure proxy settings through JVM Options in addition to configuring the proxy configuration file. For more information, see the help for the appropriate connector.

Some use cases require the Secure Agent to bypass the proxy server and communicate directly with the instance metadata service (IMDS), which stores temporary security credentials. For example, bypassing the proxy server is required if you use an Azure managed identity to access sources and targets in a mapping in advanced mode and if you enable the **Use EC2 Role to Assume Role** property in an Amazon S3 V2 connection. To allow the agent to communicate directly with the IMDS, edit the proxy configuration file and set the IP address of the IMDS, 169.254.169.254, in the value for the property *InfaAgent.NonProxyHost*.

Contact your network administrator for the correct proxy settings.

Updating proxy settings through the command line

To update proxy hosts, proxy ports, and the username and password to connect to the proxy server, use the command line to update the proxy configuration file.

1. Navigate to the following directory:

   ```
   <Secure Agent installation directory>/apps/agentcore/conf
   ```

2. To update the proxy configuration file, `proxy.ini`, run the following command:

   ```
   ./consoleAgentManager.sh configureProxy <proxy host> <proxy port> <proxy user name>
   <proxy password>
   ```

3. Restart the Secure Agent.

Updating proxy settings in the proxy configuration file

To update proxy settings that you can't update through the command line, edit the proxy configuration file directly.

1. Open the `proxy.ini` file.
2. Update the property values as required. Use the pipe character (|) as a delimiter to combine a list of host names and IP addresses. You can enter a wildcard to the left for host names or to the right for IP addresses.

   For example, the following value uses two formats to exclude cluster IP addresses in the CIDR block 172.16.0.0/16:

   ```
   InfaAgent.NonProxyHost=localhost|127.|[\:\:1]|123.432.|172.16.*|172.16.0.0/16
   ```

3. Restart the Secure Agent for the changes to take effect.

Uninstalling the Secure Agent on Linux

You can uninstall the Secure Agent. You might uninstall the Secure Agent if you no longer want to run the Secure Agent on the machine or if you want to reinstall the Secure Agent.

Before you uninstall the Secure Agent, verify that no connection or task is configured to use it.

1. From the command line, navigate to the following directory:

   ```
   <Secure Agent installation directory>/apps/agentcore
   ```

2. Stop the Secure Agent Linux process by entering the following command:

   ```
   ./infaagent.sh shutdown
   ```

3. To uninstall the Secure Agent, run rm -rf on the directory where you installed the Secure Agent to remove Secure Agent files.

Troubleshooting a Secure Agent installation

The Secure Agent download failed.

If you receive a download failure after clicking **Download Runtime Installer**, allow the following domain through your firewall and try downloading again:

```
https://global-package.dm.informaticacloud.com
```

For more information, see Change in Package Dependency Manager IP Addresses and Domain for IDMC Knowledge article.

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